

Post Mining Land Use | Rezoning | Macquarie Coal Complex

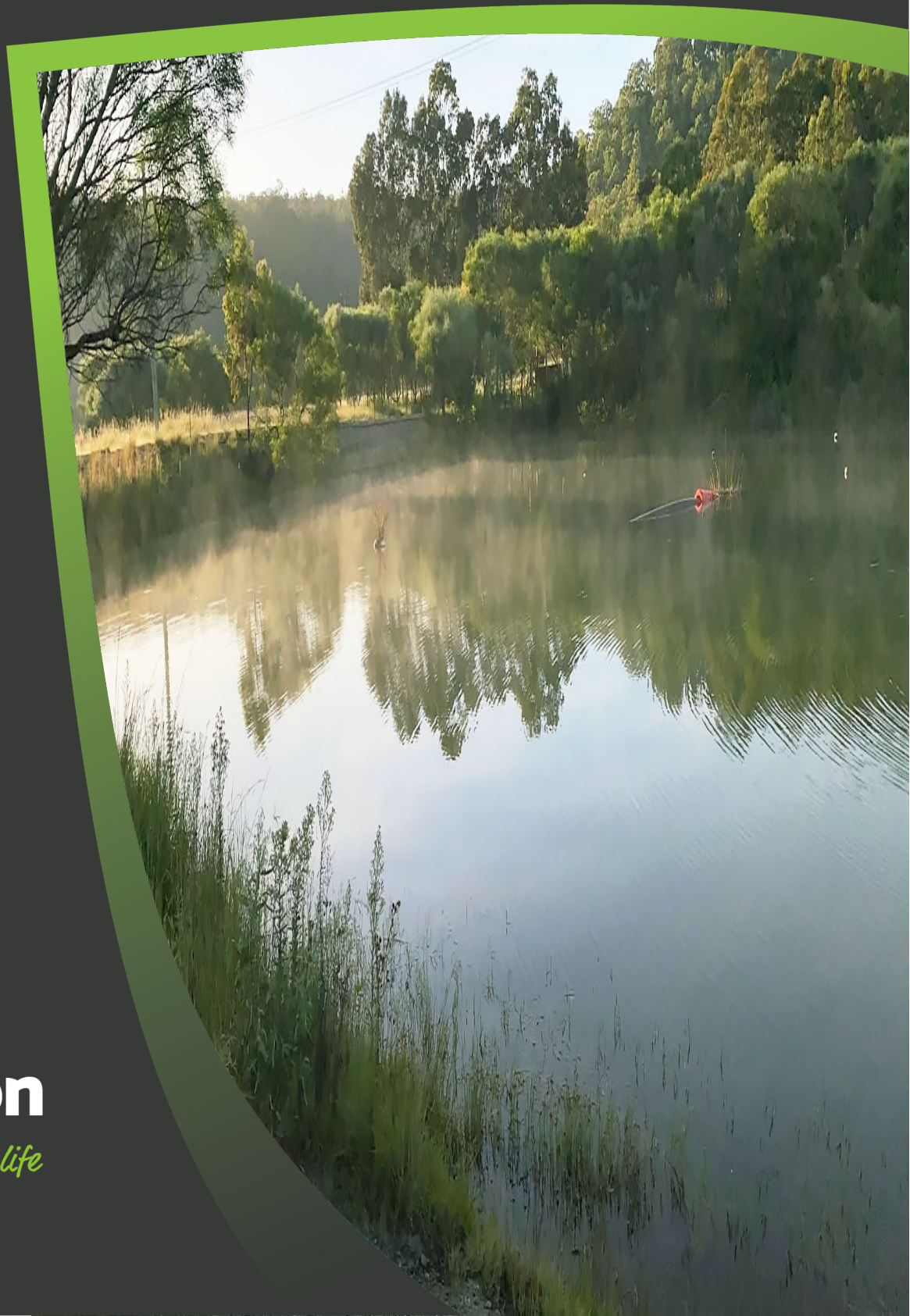
Flooding and Water Cycle Management Report

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Executive summary

This Flooding and Water Cycle Management Report (This Report) presents a strategic flooding and water cycle management (WCM) assessment for the Macquarie Coal Complex Precinct (the Precinct) and Teralba Southgate to support a state-led rezoning and Master Plan. The assessment confirms that the Precinct can be developed with a coordinated flooding and water cycle management strategy that supports future land use while protecting people, property, infrastructure and the surrounding environment.

The assessment has been undertaken at a master planning scale using available flood studies, water cycle data, site investigations and topographic analysis. It focuses on identifying key constraints, opportunities and system behaviours rather than detailed design. On this basis, it demonstrates that flooding, stormwater, water quality and emergency management outcomes are practically achievable across the Precinct through an integrated, precinct-scale approach.

Flooding Strategy and Precinct Response

Flooding is a defining characteristic of the Precinct, driven predominantly by regional flooding from Cockle Creek, supported by local overland flow processes and downstream tidal influences. These mechanisms establish the spatial framework for development by identifying where land should be protected, where development can occur, and where mitigation is required.

The Master Plan adopts a fundamental strategy of avoidance first, followed by selective, managed encroachment in lower impact areas where development can be supported without increasing risk. This approach preserves key floodplain functions while maximising viable development land.

The recommended flooding strategy is based on several core principles:

- Preservation of primary floodways and conveyance corridors
- Retention of flood storage areas and natural drainage pathways
- Direction of development toward areas of lower flood risk
- Integration of flood mitigation measures to support safe development

This approach ensures that development responds to the floodplain rather than working against it, maintaining the hydraulic integrity of the system while enabling growth.

Protection of Surrounding Environment and Property

A key outcome of the assessment is that the proposed strategy protects both the environmental values of the Precinct and the surrounding catchment, while avoiding adverse impacts on adjacent landholdings.

The Precinct drains to sensitive receiving environments including Cockle Creek, Burkes Creek, Diega Creek and Lake Macquarie, which contain important ecological features such as riparian corridors, wetlands and key fish habitat.

The master plan integrates these systems into the development framework by:

- Maintaining riparian corridors as functional flood conveyance and ecological buffers
- Protecting water quality through integrated treatment and management measures
- Avoiding increases in downstream flood risk or changes to flow behaviour
- Ensuring no material transfer of flood or drainage impacts to neighbouring land

This ensures that future development does not degrade environmental value or increase risk beyond the site, a critical requirement for strategic planning and approval.

Deliverability of Mitigation Within the Precinct

A significant finding of the assessment is that the majority of required mitigation measures can be accommodated within the Precinct footprint. This includes:

- Floodway corridors and overland flow paths
- Regional and local detention systems
- Stormwater conveyance networks
- Water quality treatment infrastructure
- Landform adjustments to manage flow and storage

These measures can be embedded into the structure of future development, allowing flood risk and water cycle objectives to be addressed without reliance on extensive off-site works or unacceptable external impacts.

This outcome demonstrates that the Precinct is not constrained in a way that precludes development, but instead requires a coordinated and integrated design response at the master planning level.

Master Plan Intent and Flexibility

The master plan establishes a clear, achievable and flexible framework for future development. It defines the spatial logic, performance objectives and key principles required to guide design and approvals, rather than prescribing a single fixed engineering solution.

Importantly:

- The strategy is intended to be implemented progressively
- Future proponents can refine and adapt solutions
- Any modifications must remain consistent with the overall objectives
- All outcomes must comply with relevant legislation, policy and performance criteria

This ensures the plan remains practical and adaptable to future land use changes, evolving design approaches and updated technical standards, while maintaining a consistent approach to flood risk and water management.

Demonstrated Feasibility at Strategic Level

Based on the site review and strategic assessment, all key aspects of flooding and water cycle management are considered feasible across the Precinct.

This includes:

- Flood risk management and land use compatibility
- Stormwater collection, conveyance and discharge
- Water quality management and protection of receiving environments
- Emergency access and evacuation planning

While further modelling and design will be required at later stages, the assessment confirms that there is a credible and workable pathway to achieve these outcomes within future development.

Future Opportunities and Sustainable Water Strategies

In addition to conventional infrastructure, the Precinct presents opportunities to adopt more sustainable and integrated water cycle approaches that reduce reliance on traditional systems.

Potential strategies include:

- Stormwater harvesting for non-potable reuse at lot or precinct scale
- Rainwater capture systems to reduce runoff volumes
- Passive irrigation of landscapes and green infrastructure such as through the use of street trees
- Distributed bioretention, detention and infiltration, where ground conditions allow
- Integration of water sensitive urban design (WSUD) across the development

These strategies can:

- Reduce demand on downstream drainage infrastructure
- Improve water quality outcomes
- Enhance resilience to climate variability
- Support more efficient and sustainable long-term servicing

Collectively, they represent a pathway toward a lower impact, more adaptive water management system for the Precinct.

Conclusion

Overall, the flooding and water cycle management assessment supports a master plan that is practical, flexible and environmentally responsible. It demonstrates that development within the Precinct can proceed, provided that key floodplain functions are maintained and an integrated strategy is implemented.

The proposed approach protects surrounding communities and environmental systems, accommodates required mitigation within the site, and establishes a robust framework to guide future planning, design and delivery.

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Main report

Flooding and WCM Report



1 Introduction

This report confirms that the Precinct can be planned with a practical and achievable flooding and water cycle management strategy that supports future development while protecting people, property, infrastructure and the surrounding environment. The assessment has been undertaken at a strategic master planning level and is based on a review of the site, available flood studies, water cycle management information, topography, waterways, receiving environments and existing infrastructure. On that basis, the overall conclusion is that the key flooding, stormwater, water quality and evacuation outcomes for the Precinct are capable of being achieved through a coordinated precinct-scale approach, subject to detailed design in later phases.

Flooding is a defining consideration for the Precinct, particularly in relation to Cockle Creek, its tributaries, local overland flow paths and downstream tailwater influences. However, the report demonstrates that these constraints do not preclude future land use planning. Rather, they establish a clear framework for where development should occur, where flood conveyance and storage should be maintained, and where supporting mitigation is required. The proposed flooding strategy is built around preserving key floodways, accommodating regional flow paths, providing detention and storage where necessary, and locating future development in areas that can achieve an acceptable level of flood risk management. This creates a structure for growth that responds to the floodplain rather than working against it.

A key message of the assessment is that the proposed strategy protects the environmental values of the site and the surrounding catchment, while also avoiding the transfer of flood impacts to neighbouring landholdings. The Precinct contains and drains to sensitive receiving environments, including Cockle Creek, Burkes Creek, Diega Creek and Lake Macquarie. It also contains riparian corridors, wetlands, key fish habitat and other ecological values that require protection. The recommended approach maintains these systems as important components of the masterplan, allowing them to continue functioning as flood conveyance corridors, ecological buffers and water quality protection features. In this way, the strategy is not only about making development possible, but about ensuring that the surrounding environment and adjoining properties remain protected from increased flood risk, water quality decline or inappropriate drainage outcomes.

Importantly, the report finds that many of the mitigation measures needed to support future development can be accommodated within the Precinct footprint. This includes the retention of floodway corridors, provision of regional or local detention, integration of stormwater conveyance pathways, on-site detention, water quality treatment systems and landform responses that work with the site's natural drainage patterns. While detailed sizing and alignment of infrastructure will need to be resolved by future proponents, the strategic assessment demonstrates that sufficient opportunities exist within the Precinct to embed these measures into future development layouts. This is a significant outcome because it shows that the masterplan can establish a viable pathway forward without relying on unreasonable off-site intervention or unacceptable impacts to external land.

The intent of the masterplan is therefore to set an achievable and defensible strategy at a precinct scale, rather than prescribe a single fixed engineering outcome. The report makes clear that future proponents may modify elements of the ultimate flooding and water cycle response as individual development proposals evolve, provided the overall objectives of the masterplan are maintained and all relevant legislation, policy and performance criteria are met. This is an important distinction. The report does not seek to lock in detailed design at rezoning stage. Instead, it establishes the spatial logic, technical principles and performance expectations needed to guide future planning, design and approvals in a consistent way.

Based on the site review and high-level strategic assessment completed for this report, all core aspects of stormwater management, flooding response, water quality management and emergency evacuation are considered practically achievable across the Precinct. This does not remove the need for future modelling, detailed design, staging reviews and approval pathways. However, it does confirm that there is a credible and workable route to resolving these matters through later project phases. In particular, the report shows that flood-compatible land use planning, structured drainage networks, protection of overland flow paths, water quality treatment trains and future emergency access planning can all be integrated into the development of the Precinct in a practical manner.

Looking ahead, there is also an opportunity to reduce demand on conventional water cycle management infrastructure by exploring more sustainable and integrated servicing strategies as the Precinct develops. These

may include stormwater harvesting for non-potable reuse, rainwater capture at lot or precinct scale, reuse of treated water for landscape irrigation or industrial purposes, passive irrigation of open space, distributed retention and infiltration where ground conditions allow, and broader water sensitive urban design initiatives that reduce runoff volumes while improving water quality outcomes. Such measures could lessen the burden on downstream conveyance and treatment systems, improve resilience and create a more efficient long-term water servicing model for the Precinct. Overall, the report supports a masterplan that is practical, flexible and environmentally responsible, and which provides a sound basis for future proponents to progress detailed design while meeting flood, water cycle and planning objectives.

1.1 Site location and context

Located on Awabakal Country within the Lake Macquarie Local Government Area, the Precinct comprises of approximately 1160 hectares of land across the suburbs of Killingworth, Wakefield, Barnsley and Teralba Southgate. The Precinct forms part of the broader Lower Hunter region and is strategically positioned within an established industrial and infrastructure corridor, with proximity to key regional centres. Figure 1-1 illustrates the regional context of the Precinct. Macquarie Coal Complex is a former mining operation comprising West Wallsend Colliery (underground mine), Westside Mine (open cut), the Macquarie Coal Preparation Plant (MCP), Tailings Storage Facility (TSF) and associated infrastructure. Coal mining has been a defining land use in the area since operations commenced in 1969, shaping both the physical landscape and the economic profile of surrounding communities. The site is currently closed and undergoing rehabilitation, presenting a large, consolidated landholding with significant potential for transition to new employment-generating uses.

At a regional scale, the Precinct is well positioned to leverage existing transport and infrastructure networks that have historically supported mining operations, including arterial road connections linking to the Hunter Expressway, M1 Pacific Motorway and broader freight networks connecting to the Port of Newcastle. This strategic connectivity supports the potential for the site to accommodate a range of employment uses, including advanced manufacturing, logistics and emerging industries, consistent with regional economic transition objectives.

The broader land use context includes a mix of urban residential communities, industrial lands and environmental corridors. Similar to other legacy industrial sites transitioning within NSW, the Macquarie Coal Complex site represents a significant interface between former heavy industry and surrounding communities, requiring careful planning to manage land use compatibility, environmental constraints and future development outcomes.

The scale, single ownership (Glencore) and relative separation from sensitive urban uses provide a unique opportunity to deliver a coordinated, master-planned transition of the site. Large tracts of underutilised and rehabilitated land offer flexibility to accommodate new industry typologies, while enabling the retention and enhancement of ecological areas and landscape buffers.

Transition of the Macquarie Coal Complex aligns with the strategic directions of the Hunter Regional Plan 2041, which seeks to diversify the regional economy, leverage existing infrastructure and repurpose former mining land to support long-term employment growth. At a local level, the Precinct supports the Lake Macquarie Local Strategic Planning Statement 2025–2045, particularly in relation to growing and diversifying the economy, protecting environmental assets and delivering infrastructure that supports productivity and sustainable growth.

As with other large-scale industrial transition projects, the evolution of the Precinct from mining to future uses represents a complex, long-term process. It requires coordination of mine closure and rehabilitation activities, consideration of environmental and physical constraints and alignment of multiple statutory planning pathways. The master planning process provides a framework to manage this transition holistically, balancing economic, environmental and community outcomes while enabling the site's progressive transformation into a new employment precinct.

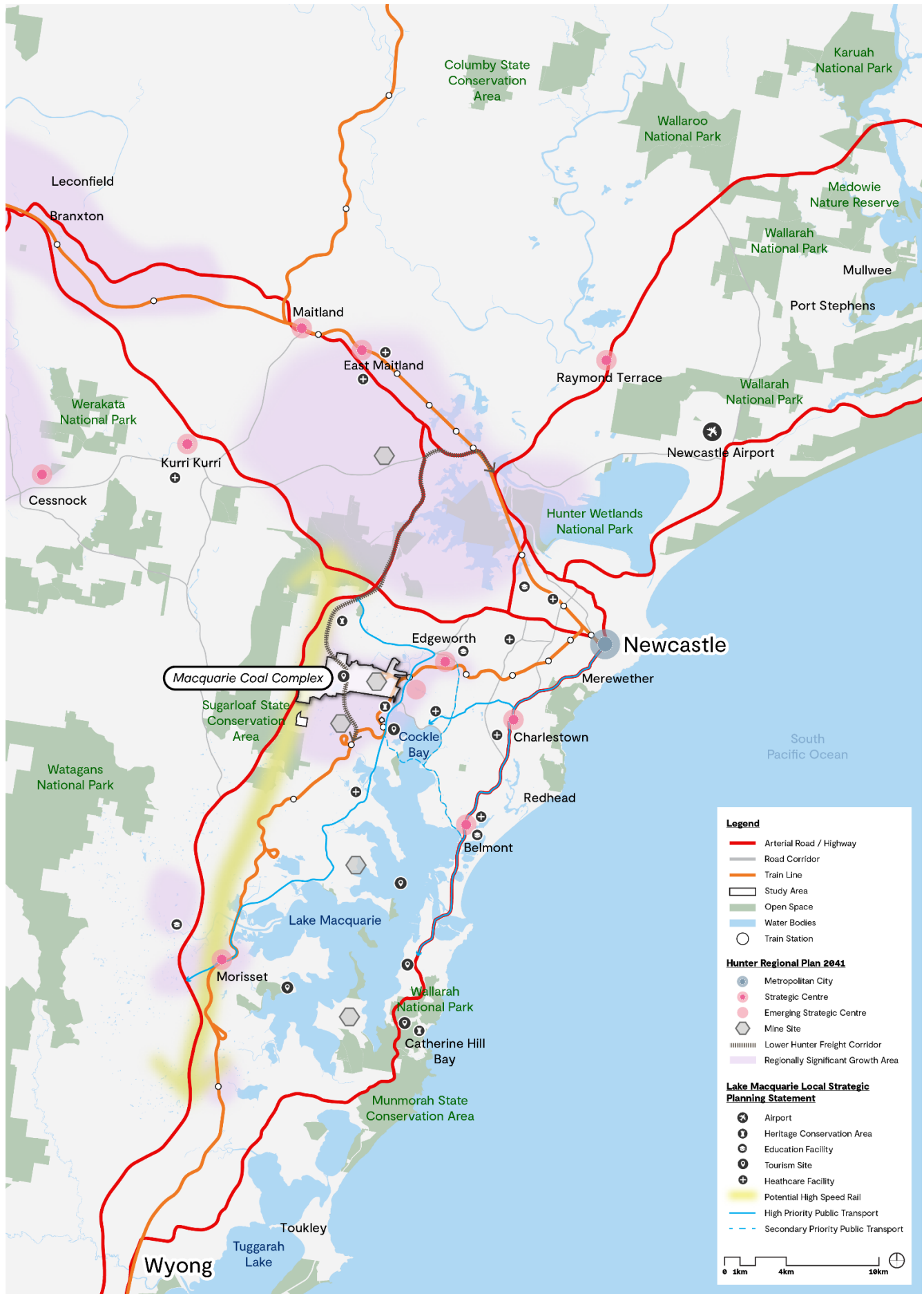


Figure 1-1 Regional context and connectivity from GHD Macquarie Coal Complex Pilot Project Master Plan 2026

1.2 Purpose of this Report

This assessment has been prepared to provide a strategic flooding and drainage constraints review to support preparation of the Master Plan and future rezoning investigations.

The assessment is intended to:

- provide an overview of the existing site and water context
- identify flooding and drainage constraints that influence land use allocation;
- identify areas potentially suitable for future development and areas that should remain constrained;
- identify strategic flood-related risks affecting future access, servicing and staging;
- identify key information gaps and future investigation requirements;
- provide flood-related objectives and performance criteria suitable for incorporation into the Master Plan; and
- provide an overview of water cycle management strategies and opportunities suitable for the Precinct

This assessment has been prepared at a strategic planning level and relies primarily upon existing flood studies and publicly available information.

1.3 Scope

This assessment has been prepared to support the Master Plan by identifying flooding and Water Cycle Management (WCM) constraints and opportunities that may influence future land use planning and development.

The scope of the assessment includes:

- Desktop review of existing flooding and WCM information.
- Strategic assessment of flooding, drainage and WCM constraints across the Precinct.
- Identification of flood risk, drainage and emergency management considerations relevant to future development.
- High-level review of flood mitigation and stormwater management opportunities.
- Development of mapping, planning recommendations and Master Plan objectives and performance criteria.

The following items are outside the scope of this assessment and should be addressed during future planning and design stages:

- Detailed flood modelling and drainage design.
- Detailed water cycle management design and infrastructure sizing.
- Detailed emergency management and evacuation planning.
- Development application assessments and regulatory approvals.

2 Information Sources, Assumptions and Limitations

2.1 Data Sources Reviewed

2.1.1 Documents and Local Study Review

Documents, developments and local studies interacting or impacting the Precinct have been collected for review and consideration in the construction of the Precinct structure plan. The documents reviewed are presented in Table 2-1. Where applicable, references have been made throughout the report where the relevant studies have been referenced.

Table 2-1 Documents, Developments and Local Study Review

Document	Description
Lake Macquarie City Council Development Control Plan 2014	This document the development control plan for the developments within the Lake Macquarie Council Local Government Area (LGA).
Discharge Water Management Strategy Macquarie Coal Preparation Plant SLR Consulting, 2018	This strategy addresses the Environment Protection Licence (EPL) 1360 notice of variation which requires a Discharge Water Management Study (DWMS) for the Macquarie Coal Preparation Plant (MCP).
Glencore Macquarie Coal Annual Review 2008 to 2024	These reports provide an ‘environmental audit’ for the purposes of section 122B (2) of the Environmental Planning and Assessment Act 1979. They summarise operations and environmental management on site of various parameters including water.
Upper Cockle Creek Floodplain Risk Management Study and Plan 2019	This study reviews and updates previous flood investigations for the Upper Cockle Creek catchment to define existing flood behaviour and risk. It incorporates PMF and dam failure scenarios and considers potential climate change impacts. The findings inform land use planning, development controls and future flood risk management.
Lake Macquarie Waterway Flood Risk Management Study and Plan 2012	This document establishes Council’s adopted framework for managing flood risk associated with lake-based inundation, catchment rainfall and elevated ocean conditions. Prepared in accordance with the NSW Floodplain Development Manual and incorporating sea level rise and climate change benchmarks, the Plan informs strategic land use planning, flood planning levels and long-term adaptation pathways. Note that this study
Lake Macquarie Coastal Management Program (CMP, 2023)	The Lake Macquarie Coastal Management Program identifies coastal flood mapping and inundation risks and recommends management and adaptation actions to address these risks; however, any regulatory controls are implemented through separate statutory planning instruments.
Winding Creek and Lower Cockle Creek Flood Study 2017	This Study and Plan reassess flood behaviour for Winding Creek and Lower Cockle Creek using updated hydrologic and 2D hydraulic modelling calibrated to major historical events. Design flood levels, flows and hazards were defined across a range of AEP events and the PMF, including sensitivity, blockage and climate change scenarios.
Lake Macquarie Waterway – Flood Risk Management Study and Plan (WMAwater, 2012)	The Lake Macquarie Waterway – Flood Risk Management Study and Plan (WMAwater, 2012) establish Council’s adopted framework for managing flood risk associated with lake-based inundation, oceanic influences, and catchment inflows. The study incorporates hydrologic and hydraulic modelling to define flood behaviour under existing and future conditions, including the effects of tidal variation, storm surge, and sea level rise. The plan informs strategic land use planning, flood planning levels, and long-term flood risk management measures across the Lake Macquarie waterway and surrounding foreshore areas.

2.1.2 Data Collection

Data used and collected for the report has been summarised in Table 2-2.

Table 2-2 Summary of the data supplied for review

Title	Data type	Source / Comment	Date Received
LiDAR in 5 m resolution	GEOTIFF	Elvis (Geoscience Australia)	05/01/2026
Flood models	Upper Cockle Creek Flood Study (2019): - Hydrology; Hydraulic Models; GIS data files - Winding Creek and Lower Cockle Creek Flood Study (2018): - PDF; GIS Layers; Models - Lt Creek Floodplain Risk Management Study and Plan (2018): - Results; Hydrologic Data; Flood Damages; GIS Data; Hydraulics - Stony Creek Flood Study (2005): - Sobek Results; Sobek Models	NSW SES Flood Data Portal Access to the flood data was not provided in sufficient time to review and incorporate the GIS flood information into this assessment. Flood information shown in figures were derived by tracing the flood mapping from the reports and indicative only.	Access to flood data provided 10/06/2026
NSW Estuary Health Risk Dataset	Shapefile	Data available online at NSW estuary health risk dataset. NSW Department of Climate Change, Energy, the Environment and Water, 2025	Sourced from Data.NSW
Council	Spreadsheet	Council's estuary health water quality monitoring (carried out by DCCEEW on behalf of council) undertakes monthly monitoring in Cockle Creek and at Speers Point	3/06/2026
MUSIC modelling Rainfall Data	MUSIC file	MUSIC Link – Lake Macquarie City Council	Sourced from MUSIC on May 2026
MUSIC modelling hydrological and pollutant parameters	MUSIC file	MUSIC Link – Lake Macquarie City Council	Sourced from MUSIC on May 2026

2.2 Assumptions, Limitations and Dependencies

This assessment has been undertaken as a strategic assessment to inform the Master Plan. It is based on existing available information at the time of the assessment. The assumptions, limitations and dependencies of this study are shown in Table 2-3.

Table 2-3: Summary of assumptions, limitations and dependencies

Item	Type	Description
Existing Flood Studies	Dependency	The assessment relies on the Upper Cockle Creek Floodplain Risk Management Study and Plan (2019), Winding Creek and Lower Cockle Creek Flood Study (2017) and Lake Macquarie Waterway Flood Risk Management Study and Plan (2012) as the primary sources of flood information.
Existing Flood Studies	Limitation	The Lake Macquarie Waterway Flood Risk Management Study and Plan (2012) is based on modelling that adopted ARR 1987 rainfall and methodologies. The flood levels and advice have not been updated to reflect the latest ARR 2019 Version 4.2, however is understood to still be aligned with Council's current flood planning advice.
Strategic Level Assessment	Limitation	The assessment has been prepared to support Master Plan development and is not intended to replace detailed flood impact assessments, drainage design or development application investigations.
Existing Flood Models	Dependency	Flood extents, hazard mapping and planning outcomes are dependent on the continued suitability of the adopted Council flood models and their underlying assumptions.
Climate Change Allowances	Limitation	Existing flood studies predate the latest ARR2019 Version 4.2 guidance relating to PMF and 1% AEP climate change impacts. Future modelling updates may alter the extent of flood constraints and developable land. Additionally, MUSIC modelling is based on historical data that does not account for climate change.
PMF Mapping	Limitation	PMF flood risk has been assessed using existing study outputs and available mapping. Updated modelling may modify PMF extents, hazard classifications and emergency response considerations.
Flood Emergency Classification	Assumption	Emergency response classifications have been derived from visual interpretation of available PMF mapping and topographic information and should be confirmed through detailed assessment during future planning stages.
Road Network	Assumption	Existing and future road access arrangements have been considered conceptually only. Future road alignments, flood immunity and evacuation performance may alter development suitability and emergency response classifications.
Future Landform	Assumption	The assessment assumes broadly existing landforms. Future filling, excavation or post-mining landform modification may alter local flood behaviour and drainage characteristics and may change the flood risk and consequently impact conclusions made and advice provided.
Drainage Assessment	Limitation	No detailed stormwater modelling or drainage network design has been undertaken. Drainage feasibility has been assessed qualitatively based on topography, floodplain characteristics and receiving water constraints.
Overland Flow Paths	Limitation	Existing overland flow paths have been identified from available mapping and topographic interpretation only. Detailed drainage investigations may identify additional local flow paths requiring protection or management.
Development Staging	Dependency	The identification of lower constraint land does not confirm development suitability and remains dependent on detailed flood, drainage, servicing and emergency management investigations.
Tidal Inundation	Dependency	Coastal inundation risk and downstream drainage performance are dependent on adopted sea level rise projections and future Lake Macquarie water level assumptions used for planning purposes. The levels adopted from Lake Macquarie Waterway Flood Risk Management Study and Plan (2012) are: 1.0 m AHD for the year 2100 sea level rise scenario 2.32 m AHD for the year 2100, 1% AEP coastal inundation

Item	Type	Description
Authority Requirements	Dependency	Future development remains subject to the requirements of Lake Macquarie City Council and relevant State agencies, including any updates to flood planning controls, climate change guidance and emergency management requirements applicable at the time of assessment.
Groundwater	Limitation	Groundwater and hydrogeology information from the site is limited and sparse. Geotechnical investigations are needed to understand this further.
WSUD Elements	Assumption	Sizing of WSUD infrastructure is based on simple modelling and considered preliminary only. No detailed modelling has been undertaken and sizes have been provided as a ballpark figure. Further refinement will be required to determine sizing and locations of WSUD assets.

3 Site and Water Context

3.1 Relevant Legislations, Policy and Guidelines

Legislation relevant to the project for this assessment is contained in Table 3-1 whilst relevant policies and guidelines are summarised in Table 3-2.

Table 3-1 Relevant legislation

Legislation	Relevance to project
Water Management Act 2000 (NSW) and Water Management (General) Regulation 2025	The Water Management Act 2000 (NSW) (WM Act) is administered by the NSW Department of Planning and Environment (DPE) (formerly NSW Department of Planning, Industry and Environment) and is intended to ensure that water resources are conserved and properly managed for sustainable use benefitting both present and future generations. The WM Act is also intended to provide a formal means for the protection and enhancement of the environmental qualities of waterways and their instream uses as well as provide for protection of catchment conditions. The intent and objectives of the WM Act have been considered as part of this assessment. Provisions of the WM Act require the development of management plans to deal with flooding regimes and the way they are managed in relation to risks to property and life and to ecological impacts. The WM Act also defines approvals required for carrying out works situated near a river or floodplain via flood work approvals or drainage work approvals. The Water Management (General) Regulation 2025 replaced the 2018 version. The regulation specifies important procedural and technical matters related to the administration of the Act and also specifies exemptions from licence and approval requirements under the Act. The regulation provides for Strahler stream order to be determined for the purposes of the regulation by reference to a stream order database (published on the department's website).
Water Sharing Plans	Water sharing plans (WSPs) are defined under the WM Act to protect water sources and their dependent ecosystems and basic landholder rights. They are also required to promote sustainable use of water resources and define the rules of how water will be allocated and traded in NSW. The purpose of a water sharing plan is to: ■ protect the fundamental environmental health of the water source ■ ensure the water source is sustainable in the long-term

Legislation	Relevance to project
	- provide all water users with a clear picture of when and how water will be available for extraction. The following Water Sharing Plans apply to the Macquarie Coal Complex: - Hunter Unregulated and Alluvial Water Sources 2022 - Lake Macquarie Coastal Floodplain Alluvial Groundwater Source - Unnamed Upriver Alluvium in WSP in the North Lake Macquarie Water Source - North Coast Fractured and Porous Rock Groundwater Sources 2016 - Sydney Basin-North Coast Groundwater Source
State Environmental Planning Policy (Resilience and Hazards) 2021	The State Environmental Planning Policy (Resilience and Hazards) 2021 provides the primary statutory framework for the consideration of flood risk in land use planning across New South Wales. The policy requires that development proposals on flood-prone land are compatible with the flood hazard, do not adversely affect flood behaviour, and incorporate appropriate risk management measures. It also ensures that development does not result in increased flood impacts on surrounding properties and that flood storage and conveyance functions are maintained. This policy underpins the assessment of rezoning and subdivision proposals within the Precinct.
Fisheries Management Act 1994 (FM Act)	The FM Act aims “to conserve, develop and share the fishery resources of the State for the benefit of present and future generations”. To meet the aims, Part 7 of the FM Act outlines legislative provisions to protect fish habitat and Part 7A outlines provisions to conserve threatened species of fish and marine vegetation and their habitat. Key Fish Habitat (KFH) are defined under the FM Act (NSW). The presence of KFH is an indication of ecological value of the watercourse. There is mapped KFH (shown on the Fisheries NSW Spatial Data Portal) associated with the following waterways: - Cockle Creek - Burkes Creek - Diega Creek
Protection of the Environment Operations Act 1997 (POEO Act)	The NSW POEO Act establishes the procedures for issuing licences for environmental protection and is administered by the NSW Environment Protection Authority (EPA). An Environment Protection Licence (EPL) is required under Chapter 3 of POEO Act to undertake a scheduled activity or scheduled development work. Under Part 5.3 of the POEO Act, it is an offence to pollute waters unless an EPL is held and the conditions of any discharge in the EPL are met. Schedule 1 of the POEO Act provides a list of activities that require an EPL. EPL 1360 is an environmental license to Oceanic Coal Australia Pty Limited (subsidiary of Glencore Australia). It governs the environmental operations, water discharge, and noise limits for the West Wallsend Colliery and the Macquarie Coal Preparation Plant (MCP) in New South Wales.

Table 3-2 Relevant policies and guidelines

Document	Relevance to project
Australia Rainfall and Runoff (ARR) 2019 and 1987 Editions	The Australian Rainfall and Runoff (ARR) guideline is a governing document for hydrological and hydraulic analysis. It provides designers and analysts with tools, information and data for the assessment of design flood estimation in Australia. ARR 1987 has been a long-standing guideline to flood estimation in the industry. This has now been updated to ARR 2019 which brings with it updated rainfall data, patterns and methodologies.

Document	Relevance to project
	The principles and recommendations of ARR have indirectly informed the strategic assessment and should be adopted for future detailed flood modelling and development assessments within the Precinct, including the consideration of climate change in accordance with the latest published guidance.
Australian Institute of Disaster Resilience (AIDR) Handbook	Generally, provides guidance on national principles and practices for disaster resilience. More specifically, the handbook provides the definition on the required flood emergency response classification of the floodplain (in accordance with AIDR Guide 7-2 (2014)) and flood planning constraint categories (FPCCs 1 to 4 in accordance with AIDR Guide 7-5 (2017)). This will assist in informing the developmental constraints of flooding across the Lake Macquarie SAP.
Lake Macquarie Coastal Management Program 2023	Provides an update of the Coastal Zone Management Plan (CZMP) 2015 and responds to the objects of the Coastal Management Act 2016.
Lake Macquarie Development Control Plan 2014	Outlines the development control plan for the developments within the Lake Macquarie Council Local Government Area (LGA). Part 5 <i>Development in Industrial Business Park</i> , outlines the applicable flood levels for development and different land uses. This will form the development constraint for the proposed development. Table 2 of this DCP outlines Flood Planning levels as they relate to Industrial Developments.
Lake Macquarie Local Environmental Plan 2014	Outlines how land can be used and developed within the Lake Macquarie Council Local Government Area (LGA).
Lake Macquarie Waterway Flooding and Tidal Inundation Policy (2020)	This policy sets a planning level benchmark for floor levels for buildings to 2050 and 2100 based on the 1:100-year lake flood plus a sea level rise of 0.4m to 2050 and of 0.9m by 2100. Council will consider the effect of climate change to inform planning decisions and when determining applications. Each DA or planning decision is to be considered on a case-by-case basis. For habitable dwellings, the flood planning levels are: ■ Planning level to 2050: 2.36m AHD (includes 0.5m freeboard) ■ Planning level to 2100: 2.82m AHD (includes 0.5m freeboard) The planning benchmarks will be used as the basis for Council staff and the community to proceed with risk assessment, policy development, community empowerment, and planning and development decisions. The current risk will be reviewed in 10 years to ensure development of the most suitable solutions, based on current advice from the Intergovernmental Panel for Climate Change. Planning and development assessment guidelines will include criteria for the three risk areas identified in the Lake Macquarie Waterway Flood Risk Management Study and Plan (2012): ■ “high lake hazard” area below 1.0 metre AHD ■ “high flood hazard” area below 1.5 metres AHD ■ “low flood hazard” area for flood-affected properties above 1.5 metres AHD and below 3.0 metres AHD.
Managing Urban Stormwater – Soils and Construction, Volume 1 (Landcom, 2004) and Volume 2A (DECC, 2008a) and Volume 2C (DECC, 2008b)	Managing Urban Stormwater – Soils and Construction, commonly known as the ‘Blue Book’, guides local councils and the development industry on stormwater management, particularly erosion and sediment control during the construction phase of the development. Better stormwater management will: ■ reduce pollution to downstream areas and receiving waterways ■ reduce land degradation

Document	Relevance to project
	- raise awareness and application of ecologically sustainable development principles - improve health, ecology and amenity of urban waterways. Consideration of erosion and sediment control mechanisms is needed during the construction phase of the project to reduce the impacts of land disturbance. These are considered in the performance criteria of the Master Plan in Section 10.2.
National Water Quality Management Strategy (WQA, 2018)	The National Water Quality Management Strategy is a nationally agreed upon strategy and framework to enable effective water quality management for the delivery of fit-for-purpose water that supports community values. This strategy contains guidelines and policies relevant to the project including: - The NSW Water Quality and River Flow Objectives (DECCW, 2006) - The Australian and New Zealand Guidelines for Fresh and Marine Water Quality (ANZG, 2018). - The project considers water quality management strategies such that the environmental values of the sensitive receiving watercourses are not adversely impacted.
NSW Aquifer Interference Policy	The purpose of this policy is to clarify the water licensing and assessment processes for aquifer interference activities under the Water Management Act 2000 and other relevant legislative frameworks. More specifically the policy: - clarifies the requirements for obtaining water licenses for aquifer interference activities under NSW water legislation; and - establishes and objectively defines considerations in assessing and providing advice on whether more than minimal impacts might occur to a key water-dependent asset as a result of aquifer interference. The project area contains GDEs which are discussed further in Section 3.2.8.
NSW Estuary Health Risk Dataset	The NSW Estuary Health Risk Dataset is designed to support councils in Stage 1 of preparing coastal management programs (CMPs) under the NSW Coastal Management Manual (2018). The dataset identifies areas in a catchment that pose relatively higher risks to the health of estuaries and areas where land use intensification is best avoided and/or more stringent management controls are needed. The dataset can be used to help map where further studies and/or management actions in a catchment would contribute to achieving some of the management objectives for coastal environment areas and coastal wetlands and littoral rainforests areas, specified in the Coastal Management Act 2016. The use of the dataset is further described in Section 3.1.1 Risk-Based Water Quality Criteria & NSW Estuary Health Risk Dataset.
NSW Flood Prone Land Package (2021)	Provides the NSW strategic planning framework for flood-prone land. The package adopts a risk-based approach to floodplain management, considering flood hazard, flood function, emergency management and climate change to inform land use planning, development suitability and flood resilience. The principles have been applied to identify flood constraints and guide future planning within the Precinct.
NSW Flood Risk Management Manual (NSW Government 2023)	Provides the current best practice framework for flood risk management in NSW, replacing and updating elements of the 2005 Floodplain Development Manual. It outlines a risk-based approach to flood assessment, incorporating consideration of climate change, emergency response, land use planning, and community vulnerability. The guideline forms the basis of which the management of flooding and development should be considered in this project. It also informs the definition of hazard categories that will assist in defining suitable developable land.

Document	Relevance to project
NSW Groundwater Dependent Ecosystems Policy (DLWC, 2002)	The NSW state groundwater dependent ecosystems policy is a framework to manage the state's groundwater resources to sustain environmental, social and economic uses for the people of NSW / prepared by the Department of Land & Water Conservation. This policy identifies and describes what groundwater and inflow dependent ecosystems are. The project area contains GDEs which are discussed further in Section 3.2.8.
NSW State Groundwater Quality Protection Policy (DLWC, 1998)	The NSW state groundwater quality protection policy is a framework to protect groundwater resources against pollution and to slow/halt or reverse any degradation of groundwater resources, ensure sustainability of groundwater dependent ecosystems, maintain the full range of beneficial uses to these resources and maximise economic benefit to the region, state and nation. As such, this framework focuses on three pillars which include: groundwater quality protection, groundwater quality management and groundwater dependent ecosystems. The project area contains GDEs which are discussed further in Section 3.2.8.
NSW Water Quality and River Flow Objectives	The NSW Water Quality and River Flow Objectives are the agreed environmental values and long-term goals for NSW's surface waters (NSW Government, 2006). The NSW Water Quality Objectives (WQOs) set out: ■ The community's values and uses for our rivers, creeks, estuaries and lakes (i.e. healthy aquatic life, water suitable for recreational activities like swimming and boating, and drinking water) (NSW Government, 2006); and ■ A range of water quality indicators to help us assess whether the current condition of our waterways supports those values and uses (NSW Government, 2006). The NSW WQOs provide environmental values for NSW waters and the ANZECC 2000 / ANZG (2018) Guidelines provide the technical guidance to assess the water quality needed to protect those values (NSW Government, 2006). The values include protection of aquatic ecosystems, visual amenity, secondary and primary recreation, livestock watering, irrigation water, homestead water, drinking water and water quality suitable for growing aquatic foods (NSW Government, 2006). The project falls within the Lake Macquarie & Tuggerah Lakes WQO Zone. ■ Cockle Creek, Diega Creek and Burkes Creek are defined as "waterways affected by urban development". ■ Lake Macquarie is defined as an estuary Further discussion of the NSW Water Quality and River Flow Objectives is in Section 0.
Policy and Guidelines for Fish Habitat Conservation and Management (DPI Fisheries, 2013)	The policy provides guidance on designs of instream structures to maintain fish passage and is applicable to planning and development proposals and other activities that affect freshwater ecosystems in NSW. There are mapped KFHs located within and around the project boundary, refer to Section 3.2.8.

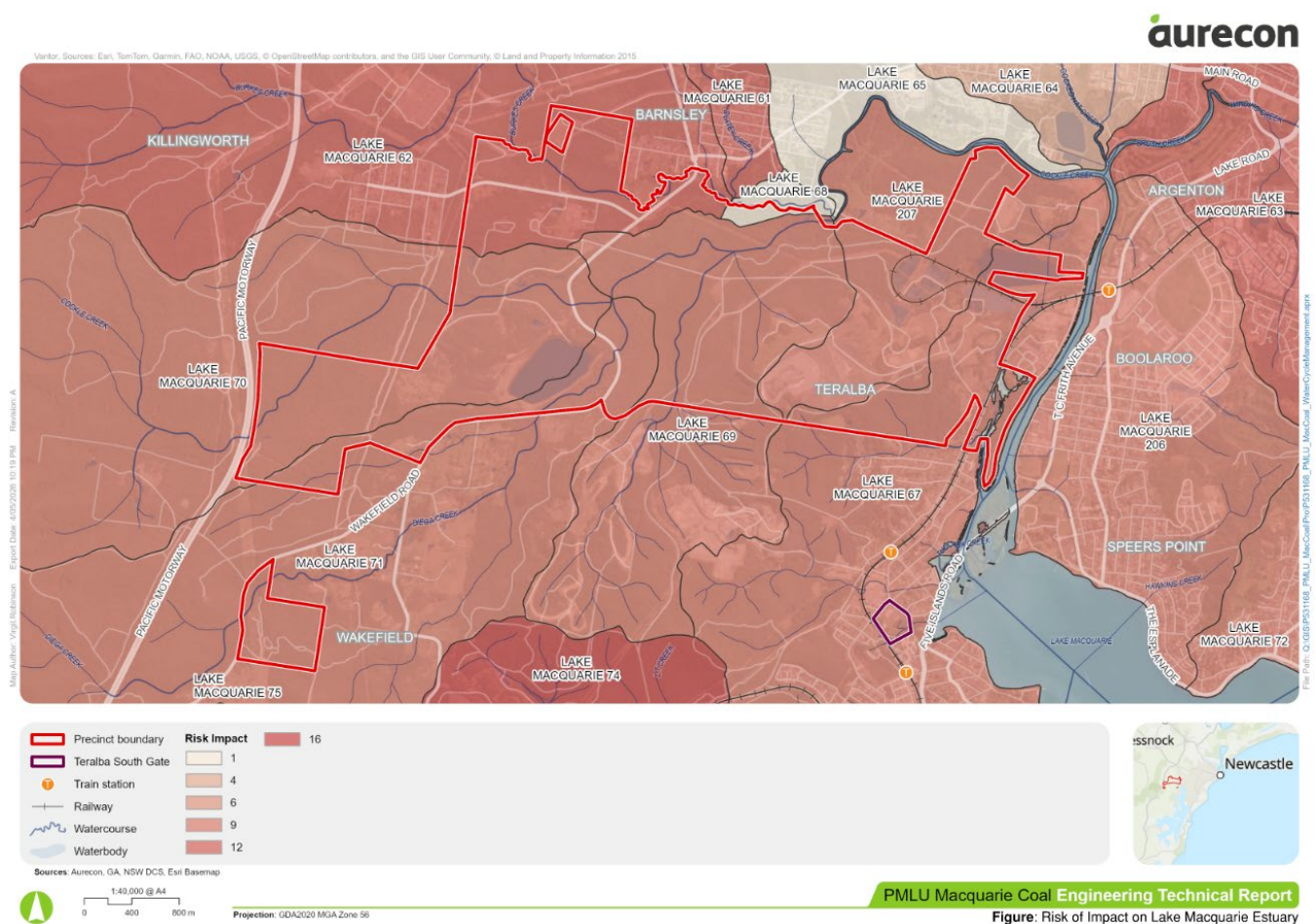
3.1.1 Water Quality Policy and Guidelines

The receiving waterways for the Precinct are within the Lake Macquarie catchment. Lake Macquarie is an estuary which has high environmental and recreational value associated. The estuary includes seagrass meadows, mangrove and oyster reefs and stormwater runoff associated with development has the potential to degrade water quality and threaten the ecological and recreational values of the Estuary. Given the high value of the estuary, there are numerous relevant guidelines and controls described below aimed to mitigate impacts to water quality from development. Development within the Precinct would be required to address the key criteria listed in the sections below.

Risk-Based Water Quality Criteria & NSW Estuary Health Risk Dataset

Given the high ecological value of the downstream Lake Macquarie Estuary, the NSW Governments policy on establishing stormwater quality targets for sensitive receptors; *Risk-based Framework for Considering Waterway Health Outcomes in Strategic Land-use Planning Decisions* was considered for nutrient loading limit targets for Precinct development. The former Office of Environment and Heritage (OEH) applied the first two steps of the Risk-based Framework and produced the NSW Estuary Health Risk dataset. The aim of the dataset is to spatially prioritise where further studies and/or management actions in a catchment would help achieve outcomes for coastal environment areas, coastal wetlands and littoral rainforests areas specified in the Coastal Management Act 2016. The dataset identifies areas in a catchment that pose relatively higher risks to the health of estuaries and areas where land use intensification is best avoided and/or more stringent management controls are needed.

The risk of impact on the Lake Macquarie Estuary has been spatially represented in Figure 3-1. The nutrient loading limits for the Lake Macquarie Estuary catchment applied at different zones intersecting with the project boundary have been summarised in Table 3-3.



ZONE ID	TN (kg/ha)	TP (kg/ha)	TSS (kg/ha)
LAKE MACQUARIE 71	2.67	0.28	149.8
LAKE MACQUARIE 75	1.42	0.14	59.14
LAKE MACQUARIE 207	7.75	0.71	534.51

Council Water Quality Controls

Lake Macquarie City Council DCP has water quality targets adopted based for specific development types. For urban areas targets are based on best practice load-based reduction targets when compared to an unmitigated developed scenario. For greenfield sites (rural and forested sites), current site performance must be modelled, and these will be used as the target for any further development i.e. NorBE. The targets set by Council have been summarised in Table 3-4.

Table 3-4 Lake Macquarie City Council DCP stormwater management targets

Pollutant	Urban areas (residential, commercial, industrial)	Greenfield (NorBE)
Total nitrogen	45%	No net increase
Total phosphorus	45%	No net increase
Total suspended solids	80%	No net increase
Gross pollutants	No target	No net increase

NSW Water Quality Objectives

The NSW Water Quality and flow objectives define the existing community uses of the existing watercourses and groundwater environment. The Precinct is within the Lake Macquarie and Tuggerah Lakes catchment. An extract of the NSW WQO for the Lake Macquarie and Tuggerah Lakes is provided in Table 3-5 and the map is provided in Figure 3-2.

The *environmental values*, their *indicators* and their *guideline levels* are listed in Table 3-5.

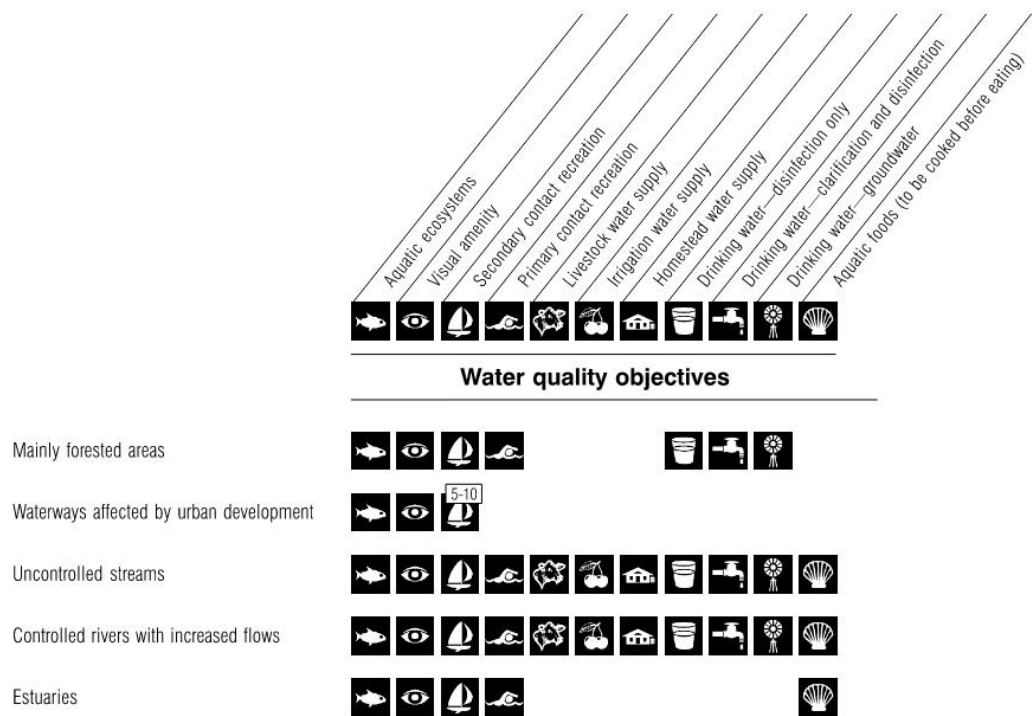


Figure 3-2 NSW Water Quality Objectives – Lake Macquarie and Tuggerah Lakes

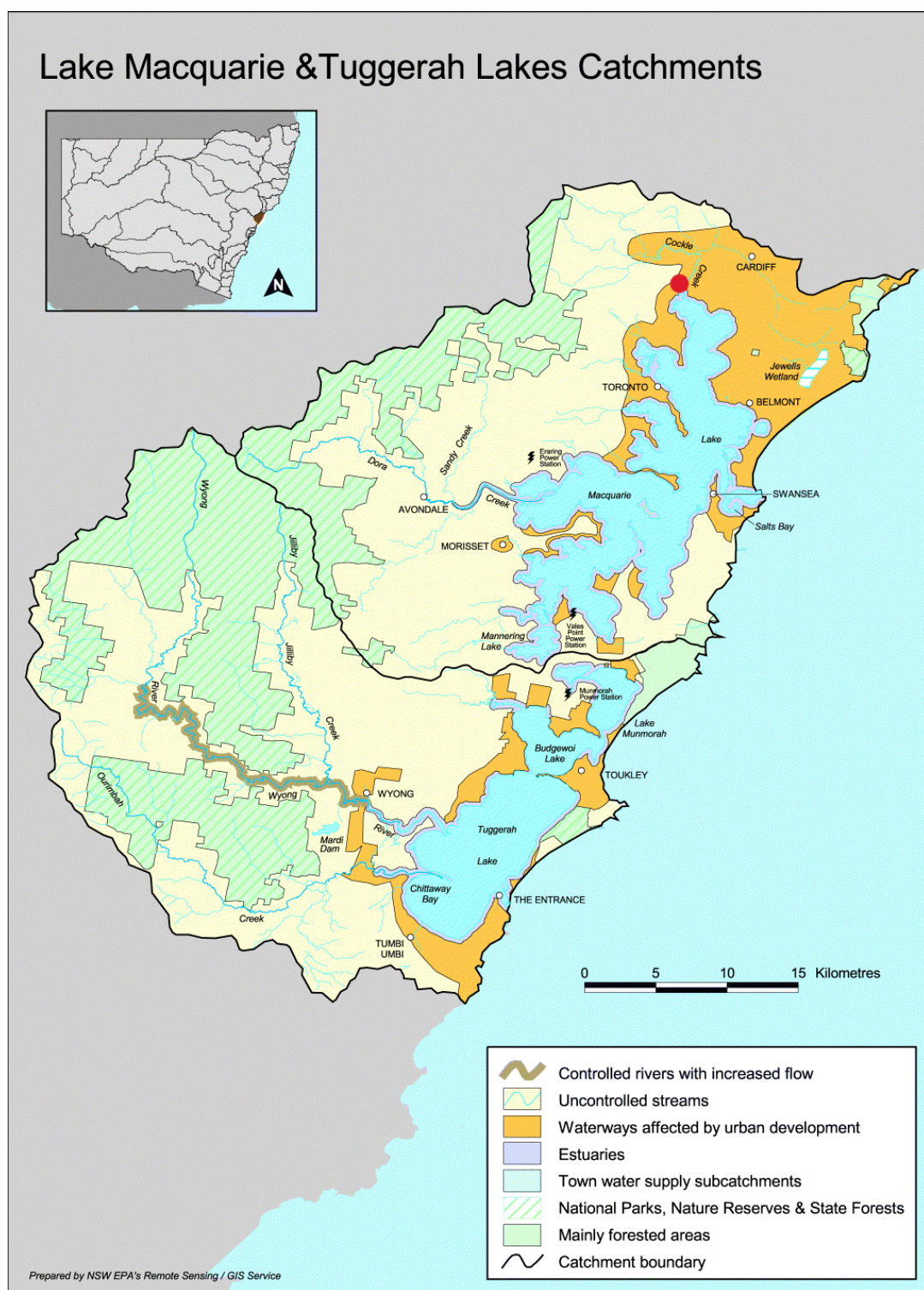


Figure 3-3 NSW Water Quality Objective Zones for Precinct (indicative location marked in red)

Table 3-5 Water quality objectives for the Lake Macquarie and Tuggerah Lakes catchment

Environmental value	Indicator	Units	Guideline value
Aquatic ecosystems	Chlorophyll a (Chl a)	µg/L	- Upland rivers: not applicable - Lowland rivers: 5 µg/L - Lakes & reservoirs: 5 µg/L. - Estuaries: 4 µg/L

Environmental value	Indicator	Units	Guideline value
	Total Phosphorus (TP)	µg/L	- Upland rivers: 20 µg/L - Lowland rivers: 25 µg/L for rivers flowing to the coast; 50 µg/L for rivers in the Murray-Darling Basin - Lakes & reservoirs: 10 µg/L - Estuaries: 30 µg/L
	Total Nitrogen (TN)	µg/L	- Upland rivers: 250 µg/L - Lowland rivers: 350 µg/L for rivers flowing to the coast; 500 µg/L for rivers in the Murray-Darling Basin - Lakes & reservoirs: 350 µg/L - Estuaries: 300µg/L
	Temperature	*C	See ANZECC 2000 Guidelines, table 3.3.1.
	Dissolved oxygen (DO) (% saturation)	Lower and upper limit	- Upland rivers: 90–110% - Lowland rivers: 85–110% - Freshwater lakes & reservoirs: 90–110% - Estuaries: 80–110%
	pH	Lower and upper limit	- Upland rivers: 6.5–8.0 - Lowland rivers: 6.5–8.5 - Freshwater lakes & reservoirs: 6.5–8.0 - Estuaries: 7.0–8.5
	Salinity	µS/cm	- Upland rivers: 30–350 µS/cm - Lowland rivers: 125–2200 µS/cm
	Turbidity	NTU	- Upland rivers: 2–25 NTU - Lowland rivers: 6–50 NTU - Lakes & reservoirs: 1–20 NTU - Estuaries: 0.5–10 NTU
	Toxicants		As per ANZG 2018
Visual use	Visual clarity and colour	To protect the aesthetic quality of a waterbody: - the natural visual clarity should not be reduced by more than 20%; - the natural hue of the water should not be changed by more than 10 points on the Munsell Scale; - the natural reflectance of the water should not be changed by more than 50%.	
	Surface films and debris	Oil and petrochemicals should not be noticeable as a visible film on the water nor should they be detectable by odour. Waters should be free from floating debris and litter.	
	Nuisance organisms	Macrophytes, phytoplankton scums, filamentous algal mats, blue-green algae, sewage fungus and leeches should not be present in unsightly amounts.	

Environmental value	Indicator	Units	Guideline value
Secondary Recreation ¹	Faecal coliforms		Median bacterial content in fresh and marine waters of < 1000 faecal coliforms per 100 mL, with 4 out of 5 samples < 4000/100 mL (minimum of 5 samples taken at regular intervals not exceeding one month)
	Enterococci		Median bacterial content in fresh and marine waters of < 230 enterococci per 100 mL (maximum number in any one sample: 450-700 organisms/100 mL).
	Algae and blue-green algae		< 15 000 cells/mL
	Nuisance organisms		Use visual amenity guidelines. Large numbers of midges and aquatic worms are undesirable.
	Chemical contaminants		Waters containing chemicals that are either toxic or irritating to the skin or mucous membranes are unsuitable for recreation. Toxic substances should not exceed values in tables 5.2.3 and 5.2.4 of the ANZECC 2000 Guidelines.
	Visual clarity and colour		Refer to visual use guidelines above.
	Surface films		Refer to visual use guidelines above.
Primary Recreation ²	Turbidity		A 200 mm diameter black disc should be able to be sighted horizontally from a distance of more than 1.6 m (approximately 6 NTU).
	Faecal coliforms		Beachwatch considers waters are unsuitable for swimming if: - the median faecal coliform density exceeds 150 colony forming units per 100 millilitres (cfu/100mL) for five samples taken at regular intervals not exceeding one month, or - the second highest sample contains equal to or greater than 600 cfu/100mL (faecal coliforms) for five samples taken at regular intervals not exceeding one month. ANZECC 2000 Guidelines recommend: - Median over bathing season of < 150 faecal coliforms per 100 mL, with 4 out of 5 samples < 600/100 mL (minimum of 5 samples taken at regular intervals not exceeding one month).
	Enterococci		Beachwatch considers waters are unsuitable for swimming if: - the median enterococci density exceeds 35 cfu/100mL for five samples taken at regular intervals not exceeding one month, or - the second highest sample contains equal to or greater than 100 cfu/100mL (enterococci) for five samples taken at regular intervals not exceeding one month. ANZECC 2000 Guidelines recommend: - Median over bathing season of < 35 enterococci per 100 mL (maximum number in any one sample: 60-100 organisms/100 mL).
	Protozoans		Pathogenic free-living protozoans should be absent from bodies of fresh water. (Note, it is not necessary to analyse water for these pathogens unless temperature is greater than 24 degrees Celsius).
	Algae & blue-green algae		< 15 000 cells/mL

Environmental value	Indicator	Units	Guideline value
	Nuisance organisms	Use visual amenity guidelines. Large numbers of midges and aquatic worms are undesirable.	
	pH	5.0-9.0	
	Temperature (prolonged exposure)	°C	15°-35°C
	Chemical contaminants	Waters containing chemicals that are either toxic or irritating to the skin or mucus membranes are unsuitable for recreation. Toxic substances should not exceed the concentrations provided in tables 5.2.3 and 5.2.4 of the ANZECC 2000 Guidelines 2000.	
	Visual clarity and colour	Refer to visual use guidelines above	
	Surface films	Refer to visual use guidelines above.	

3.1.2 Council Onsite Detention Requirements

The LMCC DCP states that on site detention (OSD) may be required for residential developments larger than a dual occupancy and for all commercial and industrial developments. OSD is the temporary capture and storage of stormwater to ensure that peak runoff from a development site does not exceed existing conditions. It is designed to prevent localised flooding and any adverse impacts to downstream development/properties. There are a number of OSD systems that Council consider including:

- In-Ground Tanks
- Above Ground Storages
- Rainwater Tanks

The DCP provides design guidelines for OSD systems with storage volumes less than 50 m³. If a storage volume greater than 50 m³ is required, a recognised routing method must be used.

3.1.3 Controlled Activities on Waterfront Land

These guidelines are important to understand the stream order and permissible activities on or near waterways within the Precinct. These guidelines can be used to determine if a proposed activity is on waterfront land and whether it requires a controlled activity approval under the WM Act.

Local Watercourse Orders

Stream order is used to describe the hierarchy of streams from the top to the bottom of a catchment. There are different methods for describing stream order. The Water Management (General) Regulation 2025 uses the Strahler system as shown in Figure 3-4. The stream order is important for determining riparian corridor extent and waterway crossing type.

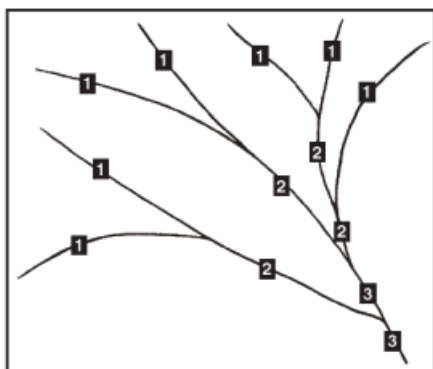


Figure 3-4 Strahler order system (DPE, 2022a)

Planning Requirements

Controlled activities carried out in, on, or under waterfront land are regulated by the Water Management Act 2000 (WM Act). The Natural Resources Access Regulator (NRAR) administers the controlled activity provisions of the WM Act which are outlined in the Guidelines for Riparian Corridors (RC) on Waterfront Land. Waterfront land includes the bed and bank of any river, lake or estuary and all land within 40 metres of the highest bank of the river, lake or estuary.

The Guidelines recommend a vegetated riparian zone (VRZ) width based on watercourse order as classified under the 'Strahler' system, see Section 3.1.3 Local Watercourse Orders. The width of the vegetated riparian zone should be measured from the top of the highest bank on both sides of the watercourse, refer Figure 3-5.

A summary of the vegetated riparian zone and total riparian corridor width requirements are presented in Table 3-6. These riparian corridor widths are also stipulated in the Lake Macquarie City Council DCP.

The Guidelines also outline certain works and activities that can occur on waterfront land and in riparian corridors through the riparian corridor matrix, see Table 3-7.

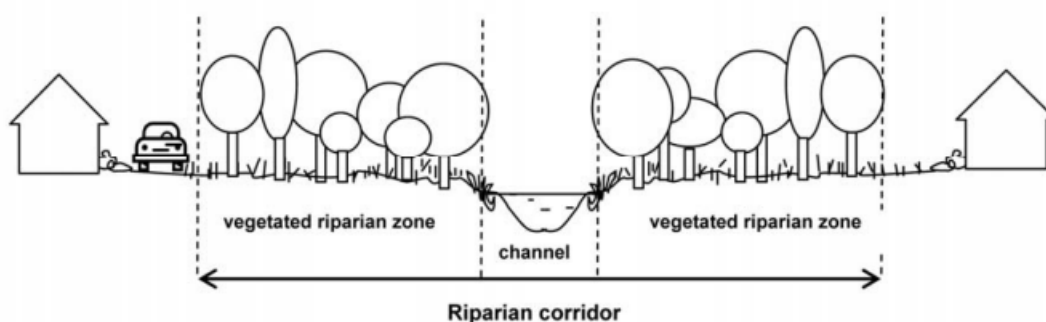


Figure 3-5 Riparian corridor (DPE, 2022b)

Table 3-6 Riparian corridor widths (DPE, 2022b)

Watercourse type	VRZ width (each side of watercourse)	Total RC width
1 st order	10 metres	20 m + channel width
2 nd order	20 metres	40 m + channel width
3 rd order	30 metres	60 m + channel width
4 th order and greater (includes estuaries, wetlands and any parts of rivers influenced by tidal waters)	40 metres	80 m + channel width

Table 3-7 Riparian corridor matrix (DPE, 2022b)

Stream order	VRZ	RC offsetting for non-RC uses	Cycleways and paths	Detention basins		Stormwater outlet structures and essential services	Stream realignment	Road crossings		
				Only within 50% outer VRZ	Online			Any	Culvert	Bridge
1 st	10m	X	X	X	X	X	X	X		
2 nd	20m	X	X	X	X	X		X		
3 rd	30m	X	X	X		X			X	X
4 th	40m	X	X	X		X			X	X

3.2 Catchment, Waterways and Receiving Environments

3.2.1 Climate and Rainfall

Rainfall and evaporation data were assessed to understand the distribution of water across the Precinct and to provide a contextual overview of the localised climate. Rainfall data was available from the BoM database (BoM, 2025a) at station Bolton Point (The Ridge Way) (#061133) which is located four kilometres from the Teralba Southgate sub-precinct of the project area. Data was summarised from this station between 2006 to 2026. The evaporation and temperature data was available from the BoM Newcastle weather station (#061055), from 2010 to 2026.. Data analysis was undertaken for each calendar month of the rainfall and evaporation datasets and is summarised in Table 3-8, Figure 3-6 and Figure 3-7.

The small variation in rainfall indicates a low level of seasonality within the Precinct. The lowest months of rainfall occurred between July and September and the highest monthly rainfall occurred between January and June.

The evaporation data indicates lowest rates of evaporation are between April and August and the highest evaporation rates are between October and February. Based on the comparison in rainfall and evaporation historical trends, there is a deficit between September and January where evaporation is notably higher than rainfall.

The temperature data for the Precinct has been summarised in Figure 3-7. The graph shows that there is minimal variation in temperature for each month, with the largest variation for maximum temperatures in July and for minimum temperature in May.

The Precinct is characterised as having uniform rainfall (BOM, 2025b). This aligns with the locally recorded rainfall data (refer Figure 3-6) and temperature data (refer Figure 3-7).

Table 3-8 Rainfall and evaporation data statistics

Statistic	Rainfall (mm)	Evaporation (mm)
Mean	3.5	4.0
Lowest	0	0.3
25th %ile	0	2.3
75th %ile	0	5.5
Highest	189	16.6

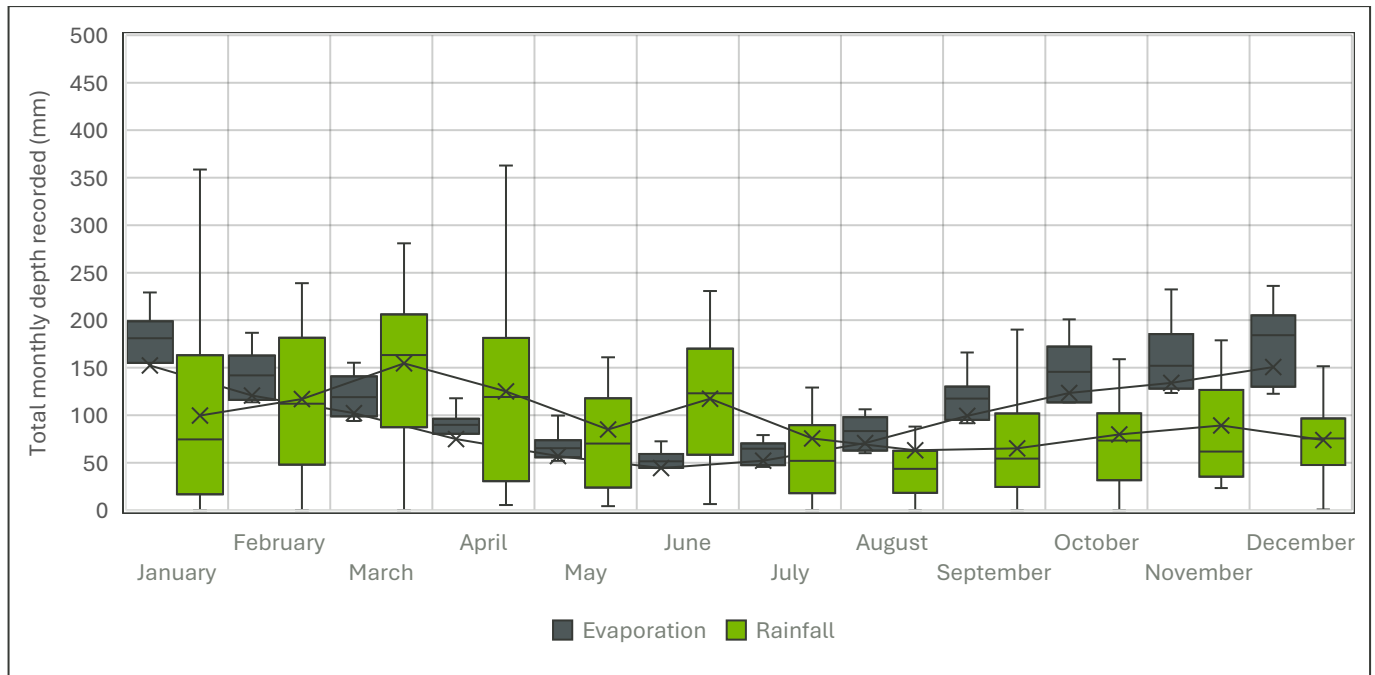


Figure 3-6 Evaporation and rainfall climate data

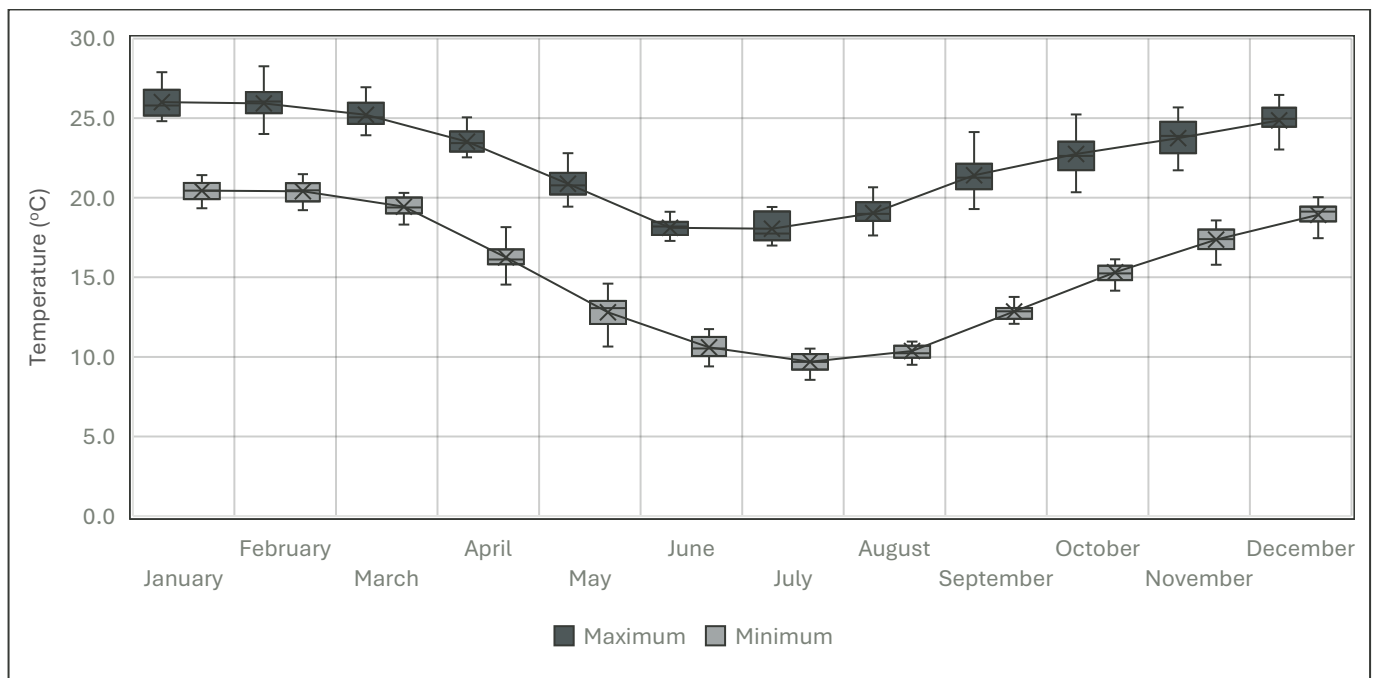
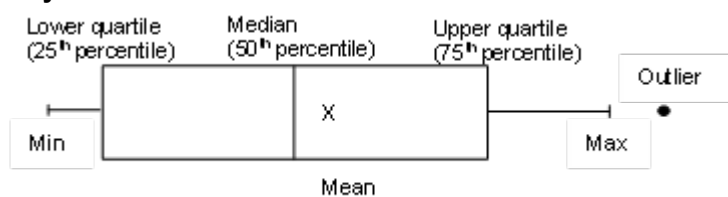


Figure 3-7 Average monthly maximum and minimum temperature data

Key:



3.2.2 Catchment and local topography

The Precinct is characterised by large areas of bushland through the western and central part of the Precinct extent. There are multiple dams scattered throughout with the Pit Void Dam located in the western part, the largest dam of the Precinct. There are sections of industrial development in the northwest section and around the Tailings Dam facility. The Teralba section to the south west consists mainly of bushland with some cleared sections in the west and the Wakefield section to the south east consists mainly of industrial development.

The Precinct is located within the Upper Cockle Creek catchment. Cockle Creek forms a major tributary of Lake Macquarie and drains a catchment of approximately 109 km². Burkes Creek contributes to approximately 15 km² of catchment area before joining Cockle Creek. The northern and eastern portions of the precinct are influenced by the Cockle Creek floodplain, while the northern boundary is influenced by Burkes Creek and associated floodplain areas. The local floodplain catchment is shown in Figure 3-8.

The Precinct flows in a north easterly direction towards Cockle Creek and Lake Macquarie Estuary with elevations typically ranging between 66 m AHD in the southwest section and 0 m AHD at Cockle Creek. Elevations rise sharply at the Tailings Storage Facility and surrounding development and north of the Pit Void Dam. The slope of the catchment and Precinct is shown in Figure 3-9.

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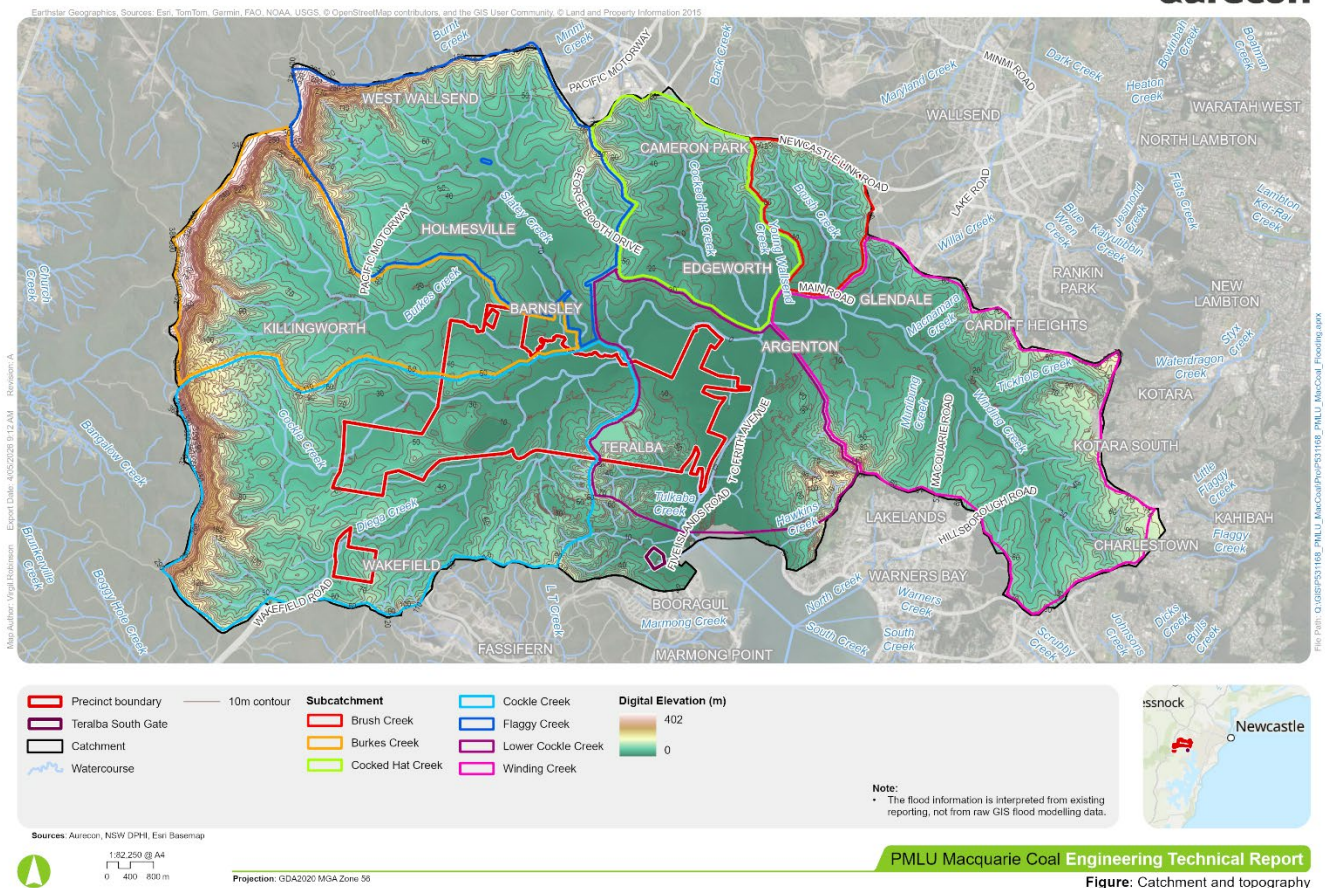


Figure 3-8: Catchment and topography

PMLU Macquarie Coal Engineering Technical Report
Figure: Catchment and topography

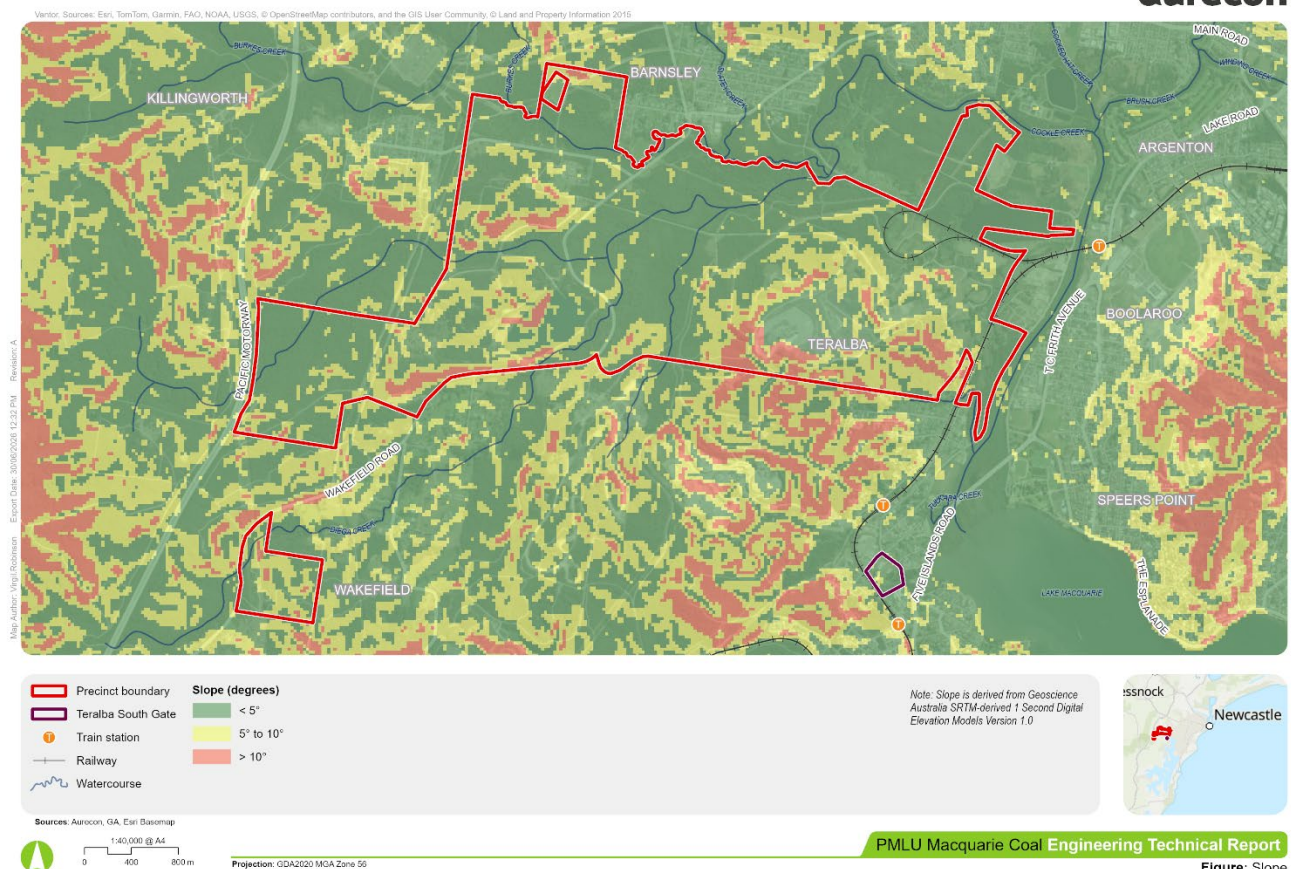


Figure 3-9 Slope of site

3.2.3 Watercourses

The major watercourses within the Precinct are Cockle Creek and two of its tributaries, Burkes Creek and Diega Creek. There are an additional 4 unnamed watercourses which flow through the Precinct.

Cockle Creek is a 3rd and 4th order waterway within the Precinct boundary and turns into a 5th order waterway just south of the Weir Road. It is a non-perennial waterway and is one of the major tributaries of Lake Macquarie. It is tidal just upstream of the Weir Road, discussed further in Section 3.2.4. Burkes Creek is located to the north of the Precinct and is a 4th order, non-perennial waterway. Diega Creek is located in the south of the Precinct and is a 4th order, non-perennial waterway before its confluence with Cockle Creek, to the north of the Final Void Pit.

The Precinct discharges to Cockle Creek to the north and east of the boundary. The lower parts of Cockle Creek can be flooded by elevated ocean levels caused by high tides and storm surge, as well as intense rainfall over the catchment (NSW Government SES, 2020).

Cockle Creek ultimately discharges to Lake Macquarie. Lake Macquarie is an estuary which has a permanently open, trained entrance. It supports diverse habitats including seagrass meadows, mangroves, saltmarsh, rocky reefs, oyster reefs, sand flats, mud flats and deep open waters and provides recreational value for water-based activities (OEHL, 2026).

Water Quality

To understand the health of receiving waterways for the project, data was assessed from sparse water quality monitoring available from Lake Macquarie Council as part of their estuary health management and Glencore annual reports as part of their environmental protection licences (EPLs) for their licensed discharge locations.

Water quality monitoring data was collected by Council at Cockle Creek and Lake Macquarie at Speers Point. This data was collected from 2019 to 2025 and has been compared with the WQOs as described in Section 3.1.1 NSW

Water Quality Objectives in Table 3-9 for key parameters. This data shows the water quality at Speers Point meets the WQOs for the key parameters. The data at Cockle Creek indicates low turbidity, a stable pH but some nutrient loading and algae samples that exceed the WQOs within the estuary. These exceedances are fairly minor, however do indicate a slightly disturbed environment.

Table 3-9 Summary of water quality for Council data

Parameter	Unit	WQO Estuary	Lake Macquarie Speers Point			Cockle Creek Estuary		
			Mean	Median	80th %ile	Mean	Median	80th %ile
TSS	mg/L	40	3	3	4	8	7	10
Turbidity (field)	NTU	0.5–10	1.3	1.1	1.7	5.5	3.7	5.4
Chlorophyll-a	µg/L	4	2	2	3	6	5	8
Total Nitrogen	µg/L	300	168	160	210	269	250	340
Total Phosphorus	µg/L	30	17	15	22	33	30	45
Ammonia as N (For 95% level of protection)	µg/L	370	4	3	5	7	5	10
pH	-	7.0–8.5	8.1	8.1	8.2	8.0	8.0	8.1
Specific Conductivity			50559	51697	53076	45497	48928	51242

* Bolded values represent when WQO has been exceeded

Anecdotal evidence from council regarding the large lagoon on Weir Road on the Macquarie Coal Complex, indicated challenging water quality issues in the past with low dissolved oxygen resulting in large fish kill incidents. Sampling of macroinvertebrates and water quality in western portion of site within Cockle Creek near the pit lake from 2023-2025 classified the system as severely impaired based on macroinvertebrate scores.

Environmental Protection Licence (EPL 1360) allows discharge to Cockle Creek and Burkes Creek (an upstream tributary). This has two licenced points which each have flow and water quality concentration limits. Surface water quality has been monitored at the two licensed points as part of the EPL 1360 with data available from 2008-2024. There has also been due diligence monitoring conducted upstream and downstream of the licensed locations within Cockle Creek and Burkes Creek which was conducted within. The testing suite required as part of this EPL licence included TSS, pH, conductivity and a suite of metals. The data for TSS, pH and conductivity has been summarised in Table 3-10 for the due diligence locations and EPL locations. Although raised, overall the WQOs are still met within Burkes Creek and Cockle Creek for the tested parameters.

Table 3-10 Summary of water quality data from Glencore monitoring

Parameter	Unit	WQO Estuary	Upstream Burkes Creek	EPL Point 2	Downstream Burkes Creek	Upstream Cockle Creek	EPL Point 1	Downstream Cockle Creek
			Mean	Mean	Mean	Mean	Mean	Mean
TSS	mg/L	40	13.4	15.2	52.8	16.1	115.6	31.5
pH	NTU	7.0–8.5	6.8	8.0	7.2	7.35	8.1	7.7
Specific Conductivity	µg/L		257.8	650	656.5	27049	960	27386

Based on the review of available data, the quality of water within Diega Creek and Cockle Creek within the Precinct is not fully known, but due to historical land use is considered a slightly to moderately disturbed environment. Downstream of the Precinct in the receiving waterways, there is more data which indicates some disturbance has

impacted the water quality, but in general the water quality is considered good. Potential contamination has not been quantified by this assessment. It is recommended that a surface water quality monitoring program be developed to further understand and quantify the water quality within the Precinct.

3.2.4 Tidal Limits

Cockle Creek is subject to tidal influence from Lake Macquarie upstream to Barnsley Weir (The Weir Bridge), located approximately 50 m upstream of The Weir Road crossing. The weir forms a significant hydraulic control within the Cockle Creek system and represents the upstream limit of tidal influence under non-flooding conditions. Based on Council survey, the weir has a crest level of about 0.44m AHD. The mapped tidal limit (Latitude: - 32.93690, Longitude: 151.59911) is shown on Figure 3-10.

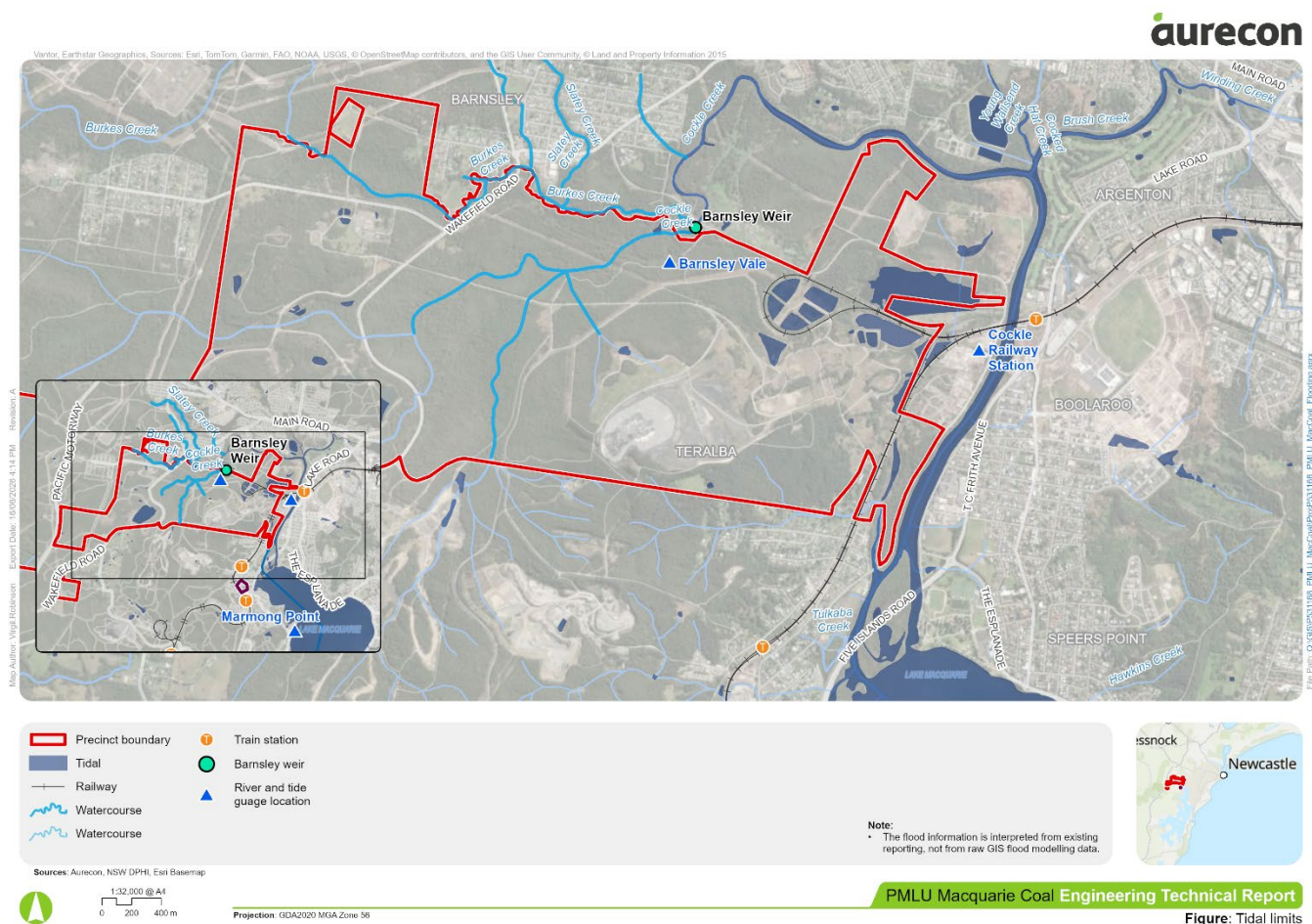


Figure 3-10 Tidal Limits

The adopted tidal planes for Lake Macquarie at Marmong Point (Gauge 211460), which informs the planning and drainage design considerations, are presented in Table 3-11.

Table 3-11 Tidal planes for Lake Macquarie at Marmong Point (211460) (MHL 2025)

Tidal Plane	Water Level (m AHD) (Annual Average)
Highest Astronomical Tide (HAT)	0.320
High Water Solstices Springs (HHWSS)	0.228
Mean High Water Springs (MHWS)	0.163
Mean High Water (MHW)	0.153

Although the Precinct is located upstream of the tidal limit, drainage infrastructure servicing the eastern portions of the Precinct may ultimately discharge to receiving systems that are hydraulically influenced by the Barnsley Weir and downstream tidal conditions within Cockle Creek and Lake Macquarie. Elevated downstream water levels associated with tidal cycles or coincident flood events have the potential to reduce gravity drainage capacity, increase upstream tailwater levels and extend local drainage ponding durations.

These downstream hydraulic controls should be considered during future design of stormwater infrastructure, particularly where drainage outlets are located at relatively low elevations downstream of the weir. The potential for backwater effects and tidal ingress should also be considered in the design of stormwater quality treatment infrastructure located near receiving waterways. Low-level wetlands, bioretention systems and other water quality treatment measures may be susceptible to prolonged inundation or reverse flows where outlet structures are influenced by elevated tailwater conditions.

Accordingly, future drainage strategies should consider the need for local drainage controls, such as flap gates or tide gates where appropriate, together with the selection of drainage infrastructure and water quality treatment components that are capable of operating effectively under periodic tailwater or tidal influence.

3.2.5 Soils

An assessment of the Hydrological Soil Group dataset (DCCEEW, 2026) indicates that the Precinct is generally underlain by low permeability soils consisting of Soil Group C. This soil group is described as:

- C - soils having **slow infiltration** rates when thoroughly wetted and consisting chiefly of soils with a layer that impedes downward movement of water, or soils with moderately fine to fine texture. These soils have a slow rate of water transmission.

Sections surrounding the Pit Void Dam and the Tailings Storage Facility to the northeastern boundary are 'not assessed' and thus the soil group is unknown. The Tailing Storage dam is to be clay capped and thus is assumed to be largely impermeable. The Hydrological Soil Groups are shown in Figure 3-11.

An assessment of the soil landscapes of the Precinct using the eSPADE spatial service (DPIE, 2026) indicates the site is dominated by Cockle Creek and Killingworth soil landscapes. Table 3-12 indicates the soil landscapes of the study area and the dominant topsoil defined under these landscapes. The site area also contains unknown soil landscapes referred to as 'disturbed terrain' around the mining activity locations. The exact dominant topsoil of these areas is not known. See Figure 3-12 for spatial representation of soil landscapes.

The NSW 1:25,000 Acid Sulfate Soil Risk Mapping, accessible via eSPADE v2.2 indicates the following risk classifications across the Precinct area:

- Around West Wallsend Colliery (WWC) / Westside Mine areas) were mapped as having no known ASS occurrence (Class N). Site topography ranges from approximately 17–21 m AHD, which is above the typical estuarine threshold for coastal ASS formation.
- Macquarie Coal Preparation Plant (MCP) areas). Parts of the MCP area are mapped with low probability of ASS occurrence at 1–3 m below ground surface (Class L2), and areas of disturbed terrain (Classes X0–X2) at elevations generally below 4 m AHD. The north-eastern corner of Catchment Dam 3 is located at the boundary of a high ASS probability area (Class H). Disturbed terrain classifications reflect historical land reclamation, mining, dam construction, and swamp filling activities, and require additional investigation to confirm ASS potential.
- Low-lying areas proximate to Cockle Creek are mapped with low to high ASS probability in areas below approximately 1–2 m AHD, consistent with estuarine depositional environments. This includes areas mapped with ASS risk that have not been field-verified.

Further detail about the presence of Acid Sulfate Soils are included in the Contamination Technical Assessment (CTA).

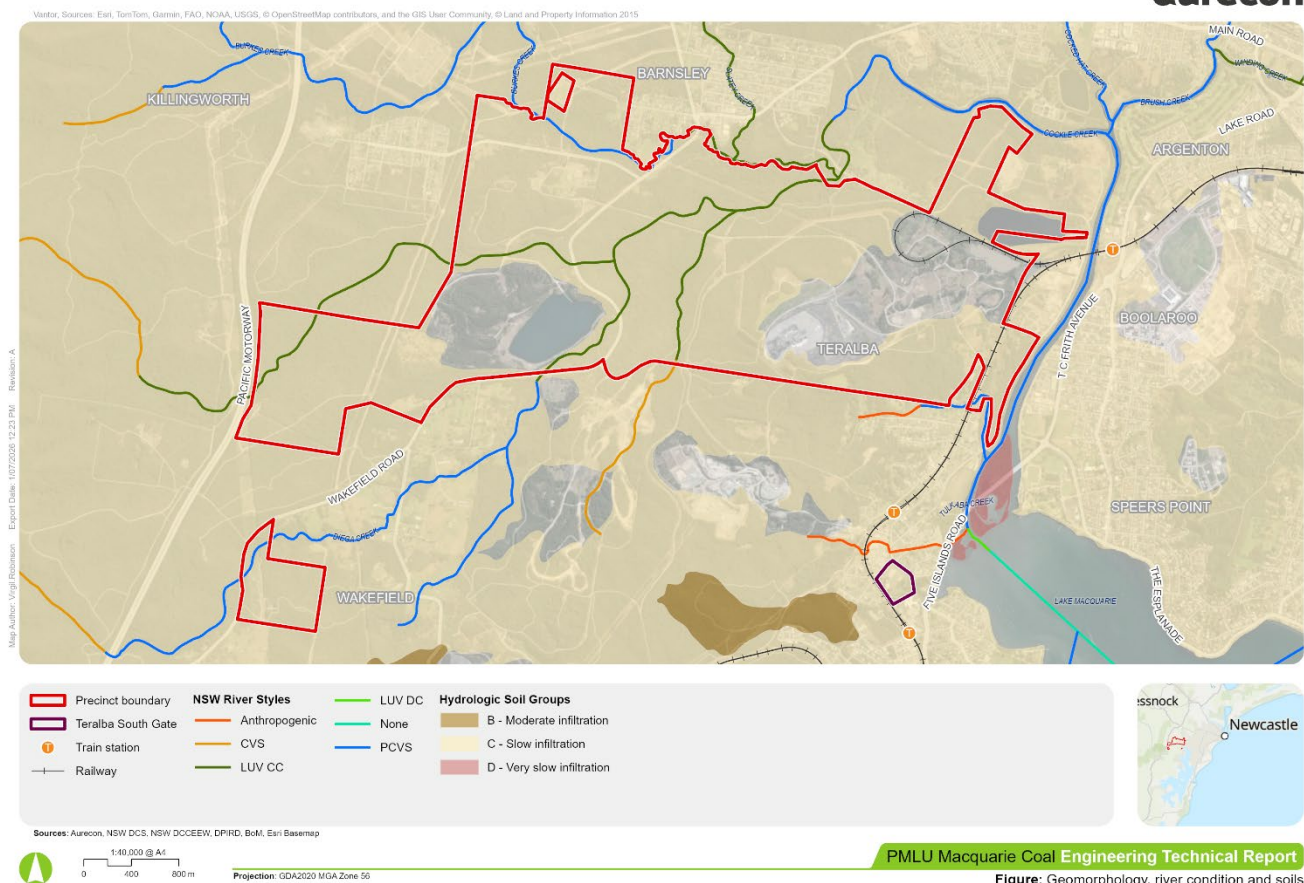


Figure 3-11 Geomorphology, river condition and soils

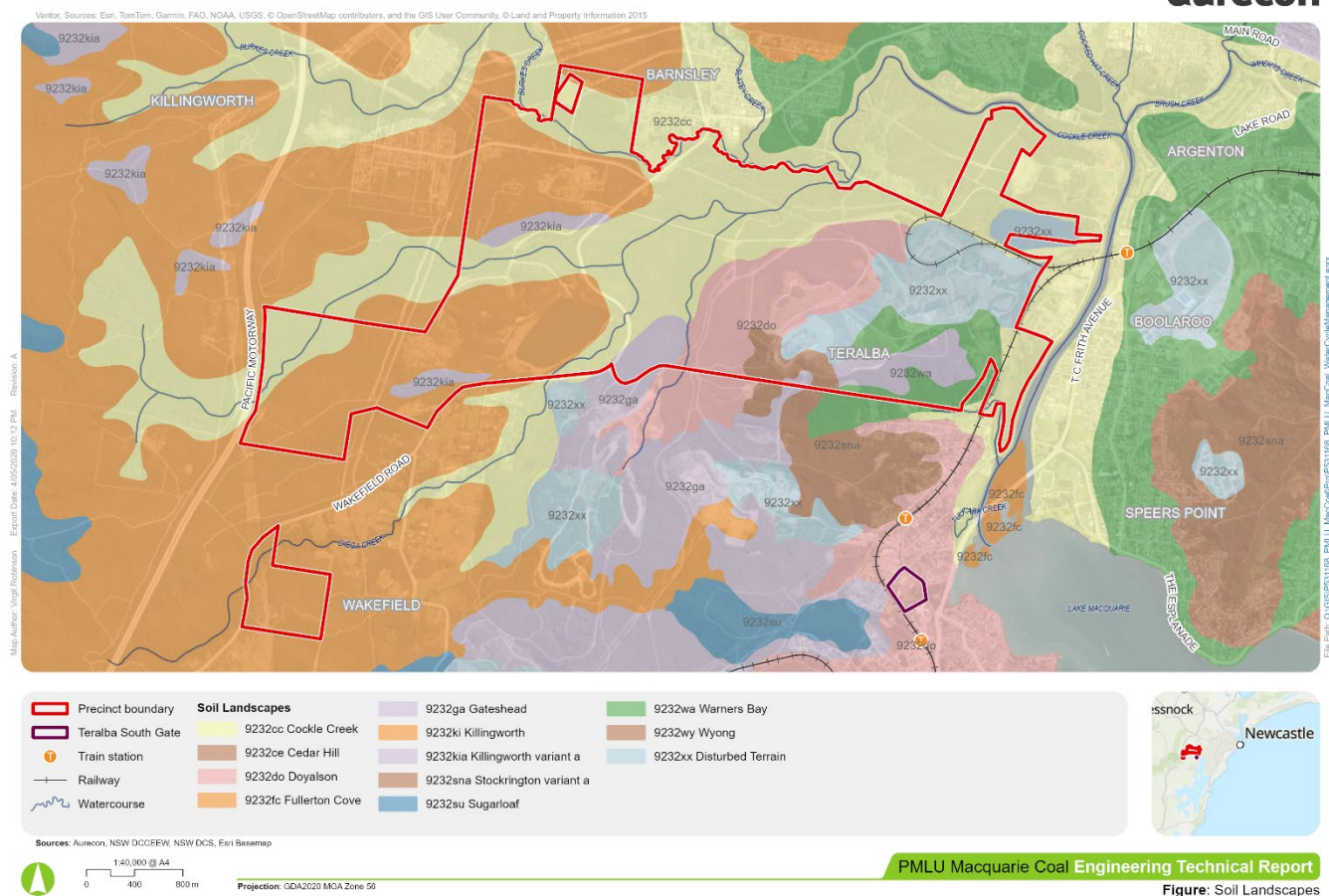


Figure 3-12 Soil landscapes

Table 3-12 Soil landscapes of the Precinct

Soil Landscape	Code	Dominant topsoil texture	% of Precinct Area
Cockle Creek	9232cc	Sandy loam to silty and sandy clay loam	35%
Doyalson	9232do	Loamy sand	12%
Gateshead	9232ga	Sandy loam to sandy clay loam, commonly light sandy clay loam	5%
Killingworth	9232ki	Commonly sandy loam or silt loam, but may vary depending on the parent material from loamy sand to clay loam	31%
Killingworth variant a	9232kia	Commonly sandy loam or silt loam, but may vary depending on the parent material from loamy sand to clay loam	1%
Warners Bay	9232wa	Silt loam through fine sandy clay loam to silty clay loam	4%
Stockrington variant a	9232sna	Sandy loam through sandy clay loam to clay loam sandy	2%

3.2.6 Groundwater

There is limited, sparse information about the hydrogeology within the Precinct. Some publicly available data was available to the north east of the project around the Cockle Creek estuary. There are 14 groundwater wells that have been identified in two clusters, to the south and northeast of the Precinct, based on MinView data (2024). Only wells within 100 m of the Precinct boundary have been considered for this assessment. Groundwater levels from available logs are generally limited and Table 3-13 includes standing groundwater depths for wells with data. See Figure 3-13 for groundwater bore locations.

Some further data was available from around Diega Creek. However, there is some uncertainty about groundwater levels from this due to conflicting information.

It is anticipated that groundwater is expected to be shallow near Cockle Creek and tributaries in low lying areas. Perched groundwater may be encountered within granular or alluvial materials over less permeable alluvium or residual soils. High elevations are expected to be well drained, but levels could be variable and recovering where mining has influenced the groundwater regime, particularly around Diega Creek.

Groundwater recharge of the alluvium is also anticipated to be slow due to the hydrological soil group, however further testing would be required to confirm and quantify the hydraulic conductivity of the Precinct and determine recharge rates in different areas.

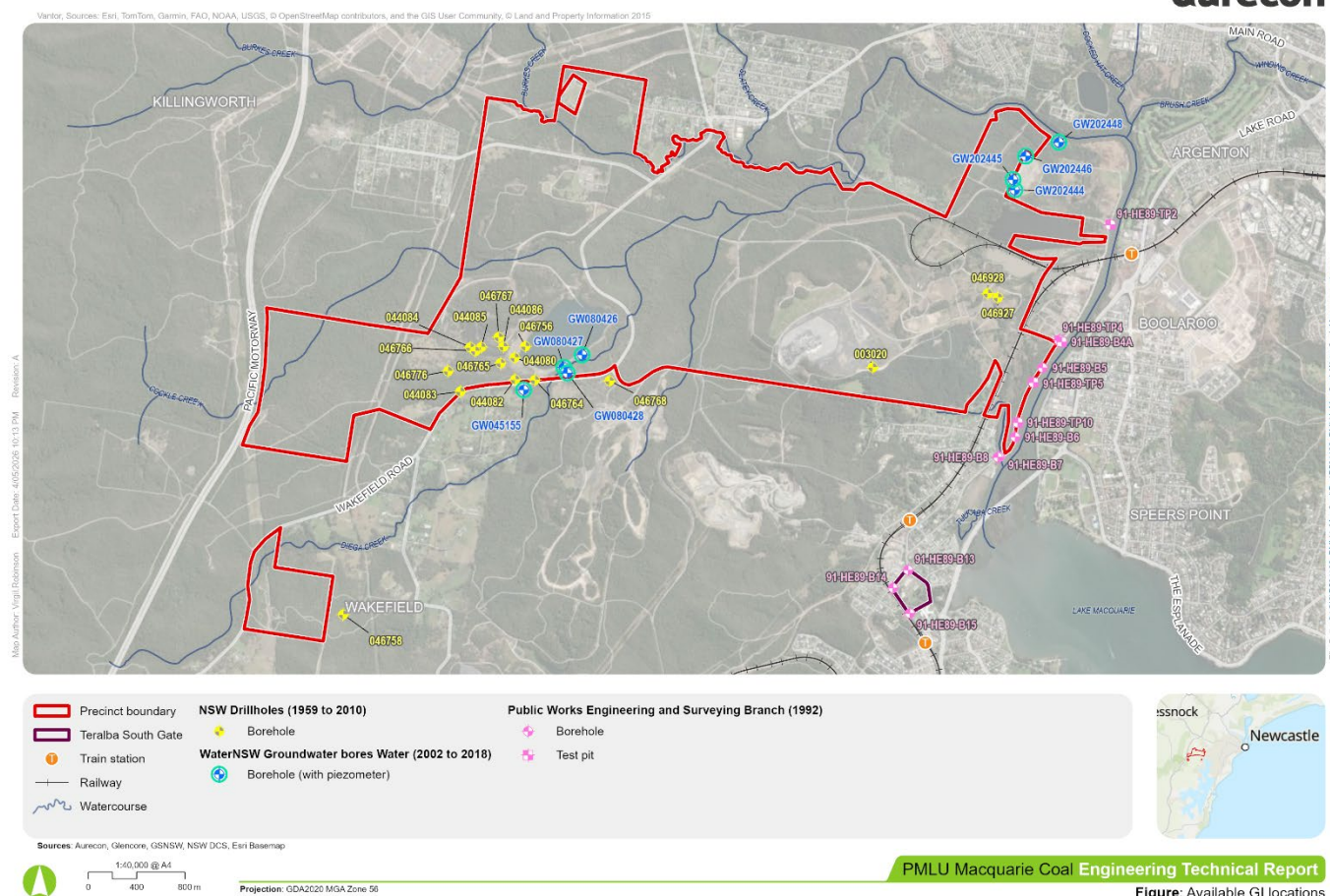


Figure 3-13 Available ground investigation locations

Table 3-13 Groundwater wells summary (WaterNSW, 2025)

Borehole ID	Cluster and approximate location	Screen depths	Standing groundwater depth	Year of completion	Top of bedrock (as logged)
		[m bgl]	[m bgl]		[m bgl]
GW202446	Northeast, north of The Weir Road	2.7 – 5.2	4.1	2010	> 5.2
GW202445		2.5 – 5.0	3.8	2010	> 5.2
GW202444		2.6 – 4.6	4.2	2010	> 5.2

3.2.7 Existing Land Zoning

The existing land use of the Precinct is predominantly zoned as C2 which is classified as environmental conservation and SP1 classified as industrial (mining). The previous mining activities on site consisted of both underground and open cut mining for black coal extraction. To the northeast there is an existing rail loop and a Tailing Storage Facility to the south of the Precinct. A Final Void Pit from previous mining activities is located to the southwest of the Precinct. There are multiple catchment dams and ponds located in the northeast of the Precinct. These surface water features and water infrastructure are discussed in Section 3.3. Wakefield Road runs north-south of the Precinct and is perpendicular to The Broadway Road which is directed towards Killingworth. The existing land zoning and infrastructure is shown in Figure 3-14.

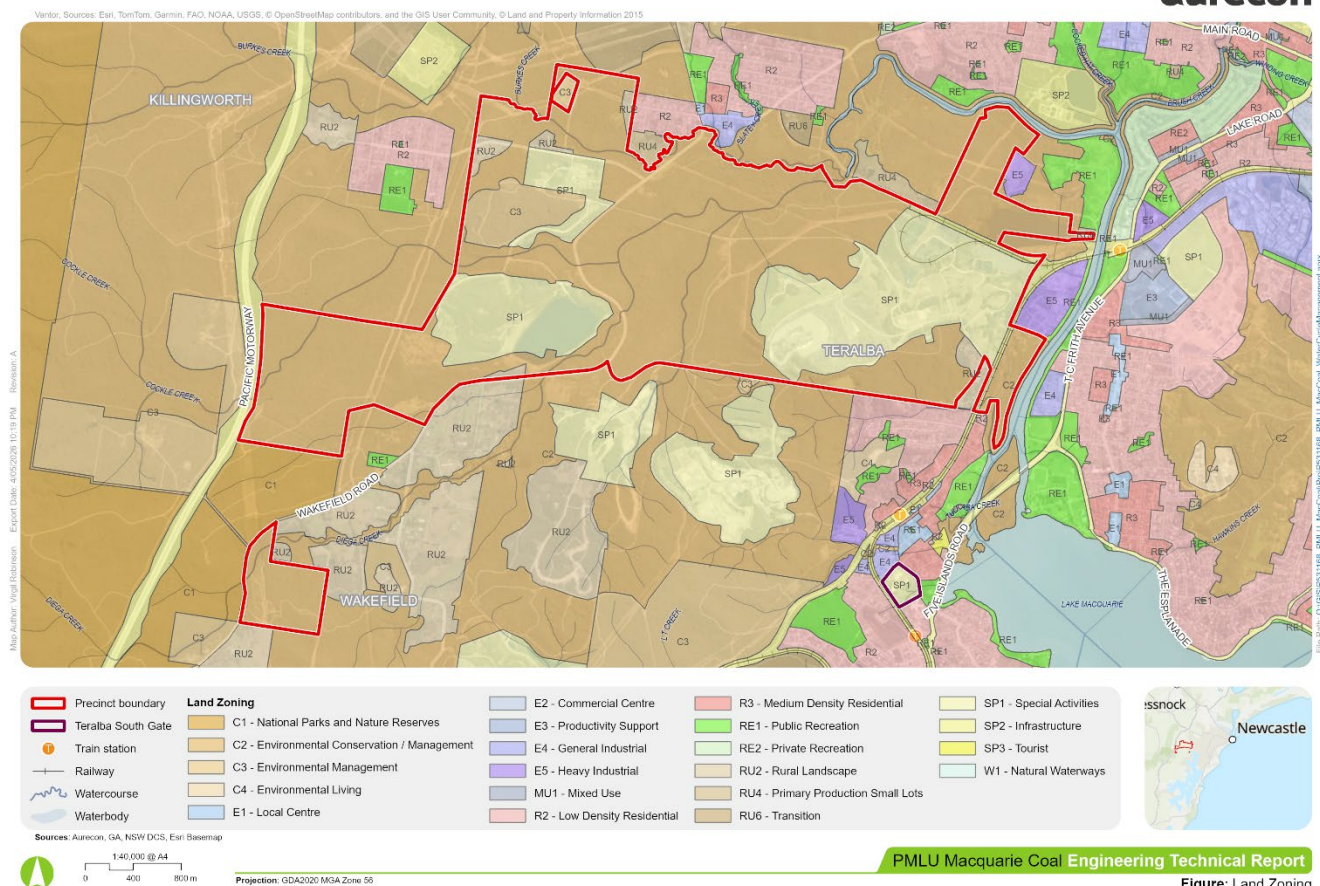


Figure 3-14 Existing land zoning

3.2.8 Sensitive Receiving Environments

The location of sensitive aquatic environments identified within the Precinct are shown on Figure 3-16 and summarised in following sections.

NSW Wetlands

Lake Macquarie and Cockle Creek are identified as wetland areas in the NSW Wetlands Dataset and are receiving waterways for this project. This dataset designates areas that may be environmentally sensitive for exempt or complying development. Lake Macquarie is listed as a “coastal lagoon and lake” and Cockle Creek as an “estuarine wetland”. The dataset also identified areas around the Tailings Storage Facility and to the northeast of the Precinct as “reservoirs”.

The locations of these wetlands are shown in Figure 3-16.

Lake Macquarie contains wetlands identified through the NSW Wetlands Dataset including Belmont Lagoon, listed as coastal vegetation and Black Neds Bay, categorised as an estuarine wetland. These are not receiving waterways but are nearby nationally important wetlands.

Coastal Management SEPP Wetlands

Lake Macquarie and Cockle Creek Estuary are also listed as Coastal environment area under the State Environmental Planning Policy (SEPP) (Resilience and Hazards) 2021. In addition, sections of Diega Creek in the southwest of the Precinct are listed under this classification. Locations classified as coastal environment area are shown in Figure 3-16.

Groundwater Dependent Ecosystems

There are known groundwater dependent ecosystems (GDEs) in the Precinct. GDEs are defined as ecosystems that rely on groundwater for some or all of their water requirements. The location of GDEs are shown in Figure 3-15 and a description of each GDE and their probability is summarised in Table 3-14.

Table 3-14 Groundwater Dependent Ecosystems in Precinct

GDE	GDE Type	Location	HEVAE Value Category	GDE Probability
Coastal Creekflat Layered Grass-Sedge Swamp Forest	Terrestrial	Around Diega Creek in the southern part of the Precinct and Burkes Creek to the north of the Precinct.	Medium to High	Not assessed
Coastal Creekline Dry Shrubby Swamp Forest	Terrestrial	Around a tributary of Diega Creek in the southwest corner of Precinct.	Medium to High	Not assessed
Estuarine Swamp Oak Twig-rush Forest	Terrestrial	In the eastern corner of the Precinct near the outlet of Cockle Creek to Lake Macquarie Estuary.	Medium to High	Not assessed
Hunter Coast Lake Flats Apple Forest	Terrestrial	In the eastern corner of the Precinct around Cockle Creek.	Medium to High	Medium
Lower Hunter Red Gum-Paperbark Riverflat Forest	Terrestrial	Around Burkes Creek in the northern part of the Precinct.	High	Medium
Lower North Creekflat Mahogany Swamp Forest	Terrestrial	Around Cockle Creek in the western section of the Precinct.	Medium to Very High	Not assessed
Lower North Riverflat Eucalypt-Paperbark Forest	Terrestrial	Around confluence of Diega and Cockle Creek in the centre of the Precinct.	Medium to Very High	Not assessed
Southern Lower Floodplain Freshwater Wetland	Terrestrial	In the eastern corner of the Precinct near the outlet of Cockle Creek to Lake Macquarie Estuary.	Medium to High	Not assessed

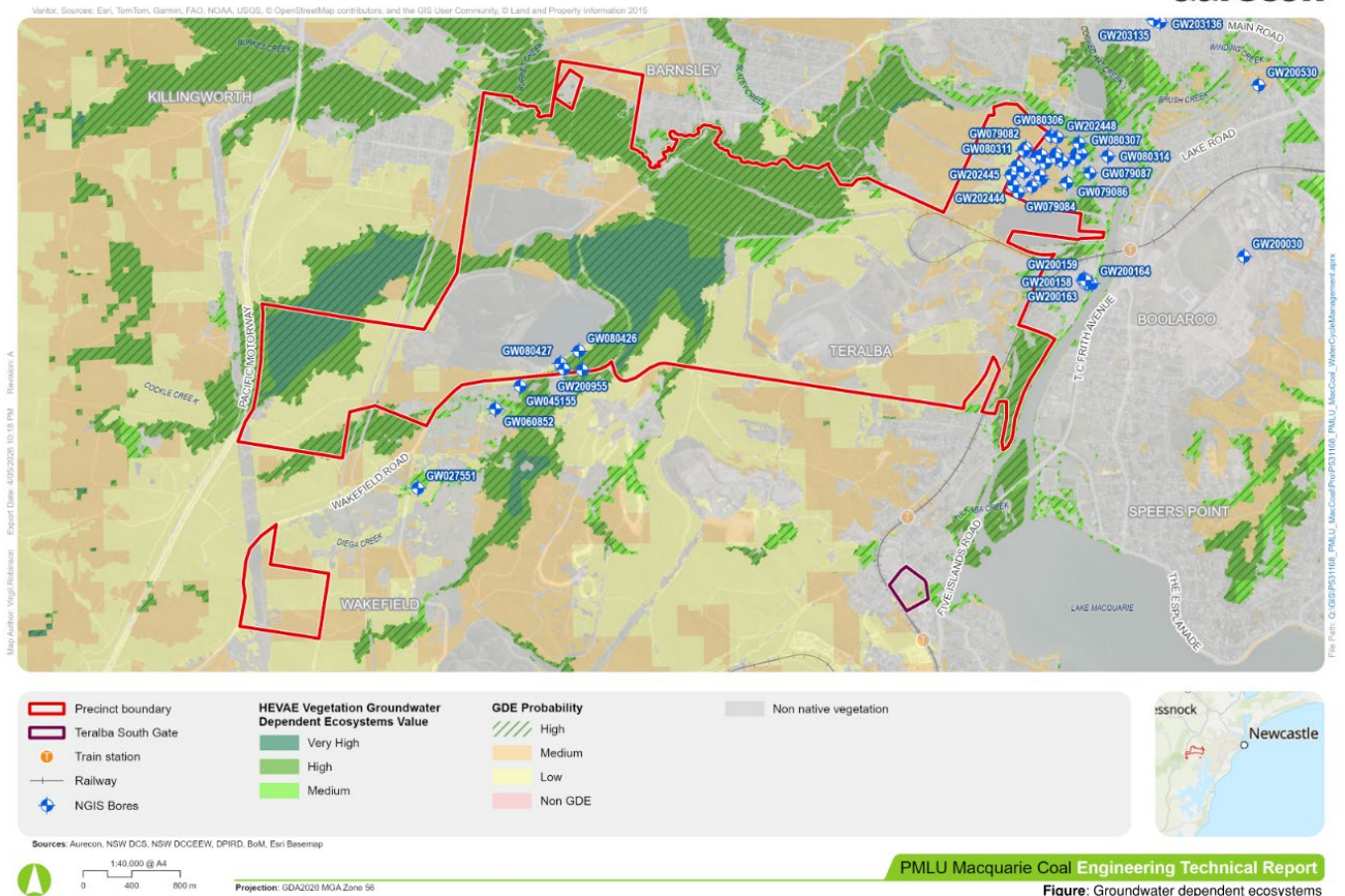


Figure 3-15 Groundwater dependant ecosystems

Key Fish Habitat

Key Fish Habitat within the Precinct includes Cockle Creek, Burkes Creek and Diega Creek. These three creeks are listed as Key Fish Habitat as they are Strahler > 3. Key Fish Habitat Areas are situated around the confluence of Cockle and Diega Creek in the centre of the Precinct. Key Fish Habitat mapping can be seen in Figure 3-16.

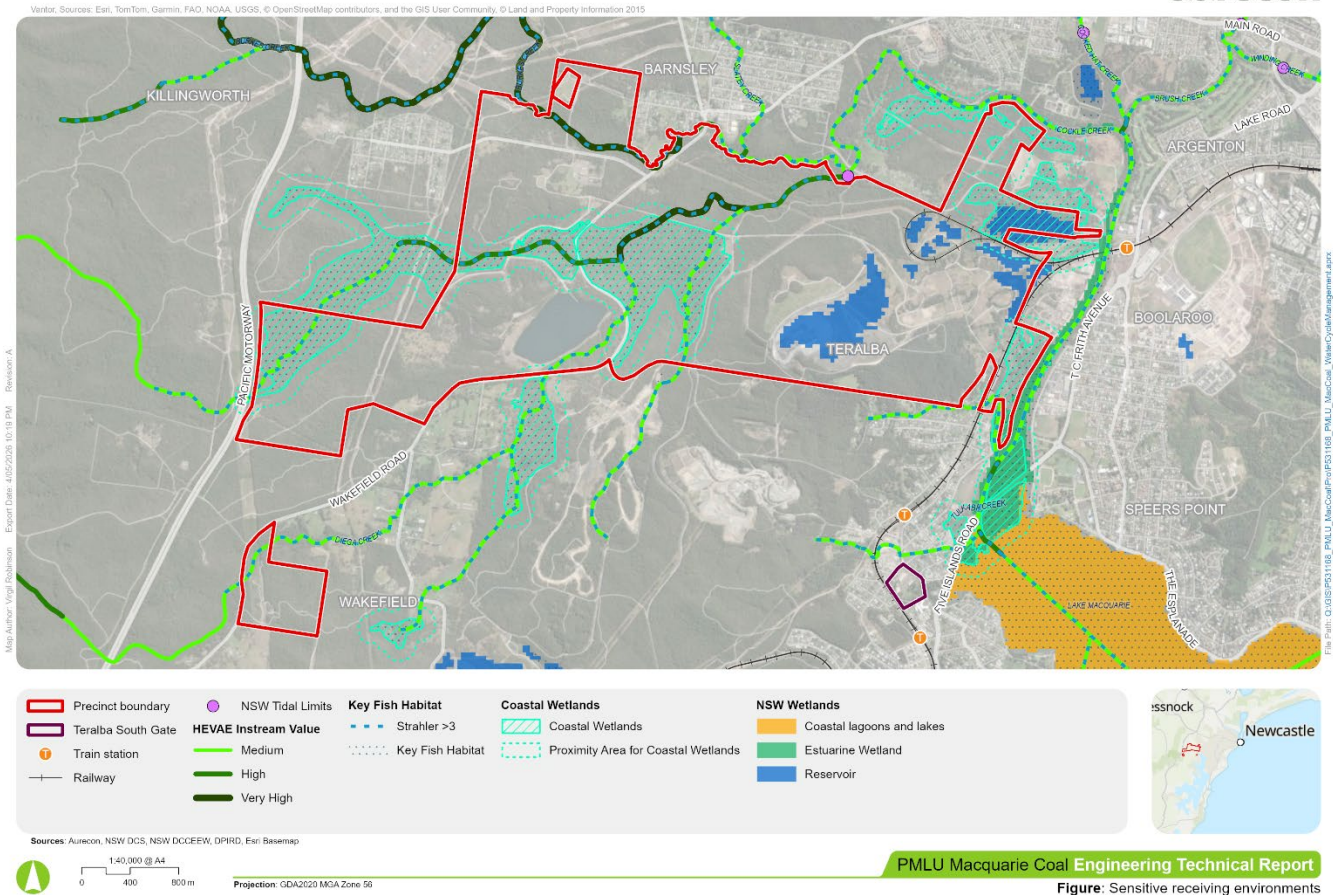


Figure 3-16 Sensitive receiving environments

3.2.9 Preliminary Riparian Corridor Extent

The riparian corridor mapping has not been completed for the Precinct and therefore there was not data available to show the relevant riparian corridors. A preliminary riparian corridor width was determined based on the widths in Table 3-6 but assumed specific channel widths depending on the stream order. These assumptions were as follows:

- Strahler 1: 22m buffer (11m either side of hydroline, assuming a 2 metre channel width)
- Strahler 2: 45m buffer (22.5m either side of hydroline, assuming a 5 metre channel width)
- Strahler 3: 70m buffer (35m either side of hydroline, assuming a 10 metre channel width)
- Strahler 4+: 100m buffer (50m either side of hydroline, assuming a 20 metre channel width)

Figure 3-17 spatially maps the preliminary vegetated riparian zone (VRZ) using the above assumptions.

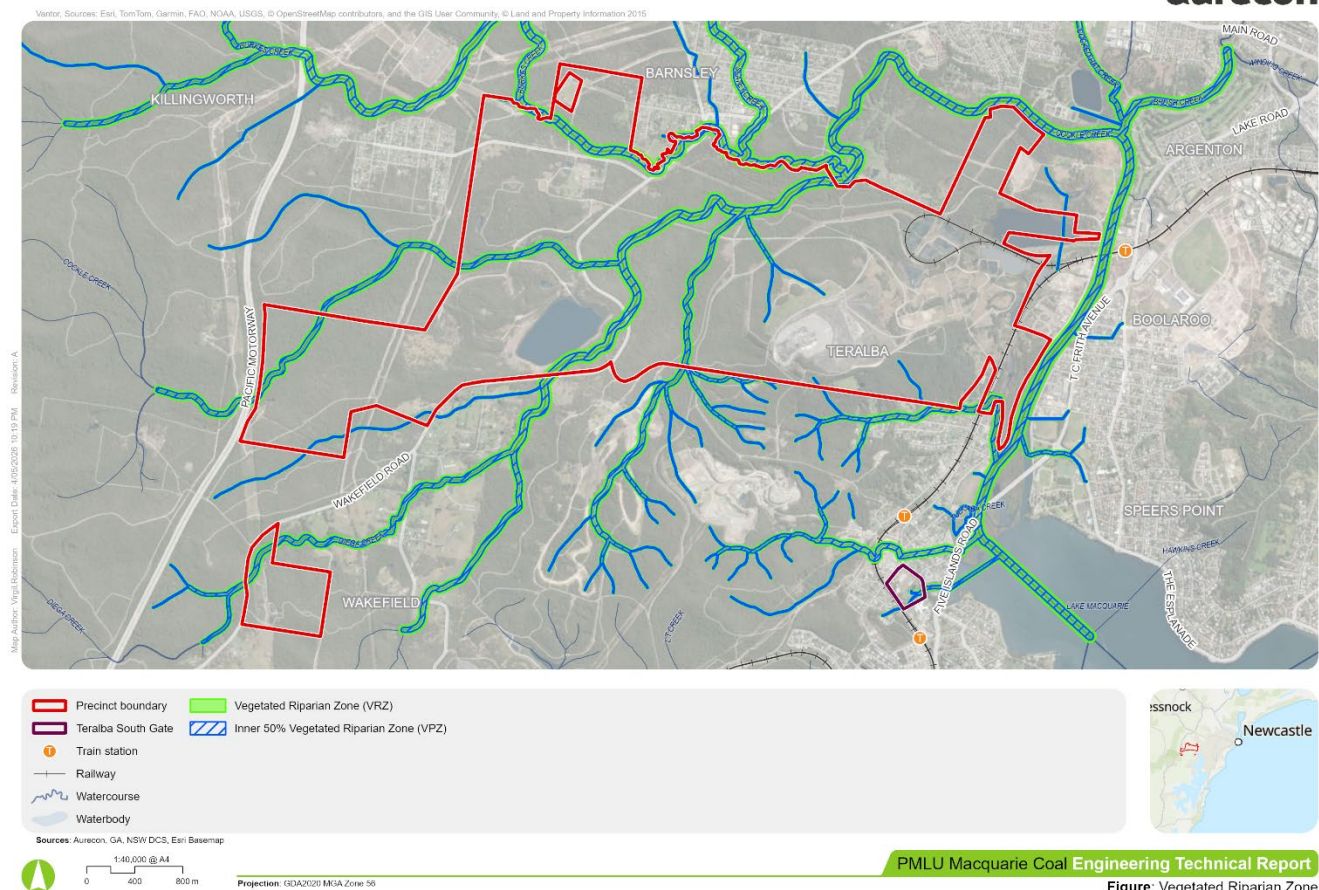


Figure 3-17 Preliminary vegetated riparian zone

3.2.10 Indigenous Values of Lake Macquarie

Lake Macquarie is the Traditional Country of the Awabakal people and Lake Macquarie City has the second largest population of people who identify as being Aboriginal and Torres Strait Islander in NSW. 'Awaba' is the word meaning place of flat or plain surface and is used to describe what is now known as Lake Macquarie (Awabakal Language Program, 2026). There are traditional art and shell middens which can be found around Lake Macquarie, indicating a culture who thrived on the area's rich shores and woodlands (Lake Macquarie City Council, 2026). Kerrie Brauer an Awabakal Traditional Owner summarises the relationship of the Awabakal people to the water and lands of Lake Macquarie (Lake Mac Libraries, n.d.):

"Whilst a diversity of attachment and experience is recognised, it is also recognised that the landscape, vegetation and watercourses of the Awabakal Cultural Boundary form part of an Aboriginal cultural landscape of traditional and contemporary cultural and spiritual value to many Aboriginal people."

The Awabakal people also have a continuing, contemporary history of trying to protect and preserve their Cultural Heritage within the LGA Regions within their Traditional Country. ... Aboriginal lore requires that the Aboriginal cultural landscape (which includes Aboriginal heritage sites, landscape features of cultural value, the plants, animals and water) are cared for, so that it will survive for future generations of our Peoples."

3.3 Existing Drainage and Water Management Infrastructure

Existing stormwater management within the Precinct is through natural flow paths within the undeveloped areas and via water management infrastructure across the disturbed site. Water management infrastructure on site includes catchment dams, sediment dams, storage dams, spillways and drainage systems. Clean water runoff is currently diverted around the site where possible to minimise contamination. Disturbed area runoff is managed by catch drains and delivered to sediment basins for treatment and reuse on site prior to being directed to the licensed

discharge points at Cockle Creek. The existing water management infrastructure across the site is shown in Figure 3-18.

Existing water management at the Macquarie Coal Preparation Plant (MCP) is managed in accordance with the MCP water management plan, the MCP erosion and sediment management plan and the Oceanic Coal Australia Limited (OCAL) Mine Closure Plan. Runoff generally drains towards the north east of this portion of the Precinct and is captured in a series of dams. Due to the mine closure, there is no current reuse of this water and captured water is discharged following settlement via a licensed discharge point (EPL 1360) at Pond 3 which discharges to Cockle Creek. During the site closure works, the maturation ponds were filled and water was redirected to the downstream pond 3. The tailings facility remains uncapped and effectively operates as a dam, holding captured rainfall and runoff.

At the West Wallsend Colliery site, surface water is managed via an Environmental Management System (EMS) and erosion and sediment controls have been constructed in alignment with relevant guidelines. The surface water management includes a series of dams where water is captured and held prior to discharge at the licensed discharge point (EPL 1360). The site includes dirty water management of all process locations. This site holds a water access license under the Water Act 1912 for extraction of groundwater from the hard/fractured rock aquifers (license 20BL169793). This licence is understood to be still valid following site closure.

The water management for the rest of the site is largely via natural flows paths. Culverts are located under roads where they cross waterways including the Rhondra Road crossing of Diega Creek and Wakefield Road crossing of Cockle Creek. It is understood that there is largely no other formalised drainage infrastructure across the site.

Groundwater management has included monitoring groundwater levels and quality at some discrete locations around Diega Creek, but large scale groundwater monitoring has not been undertaken within the Precinct.

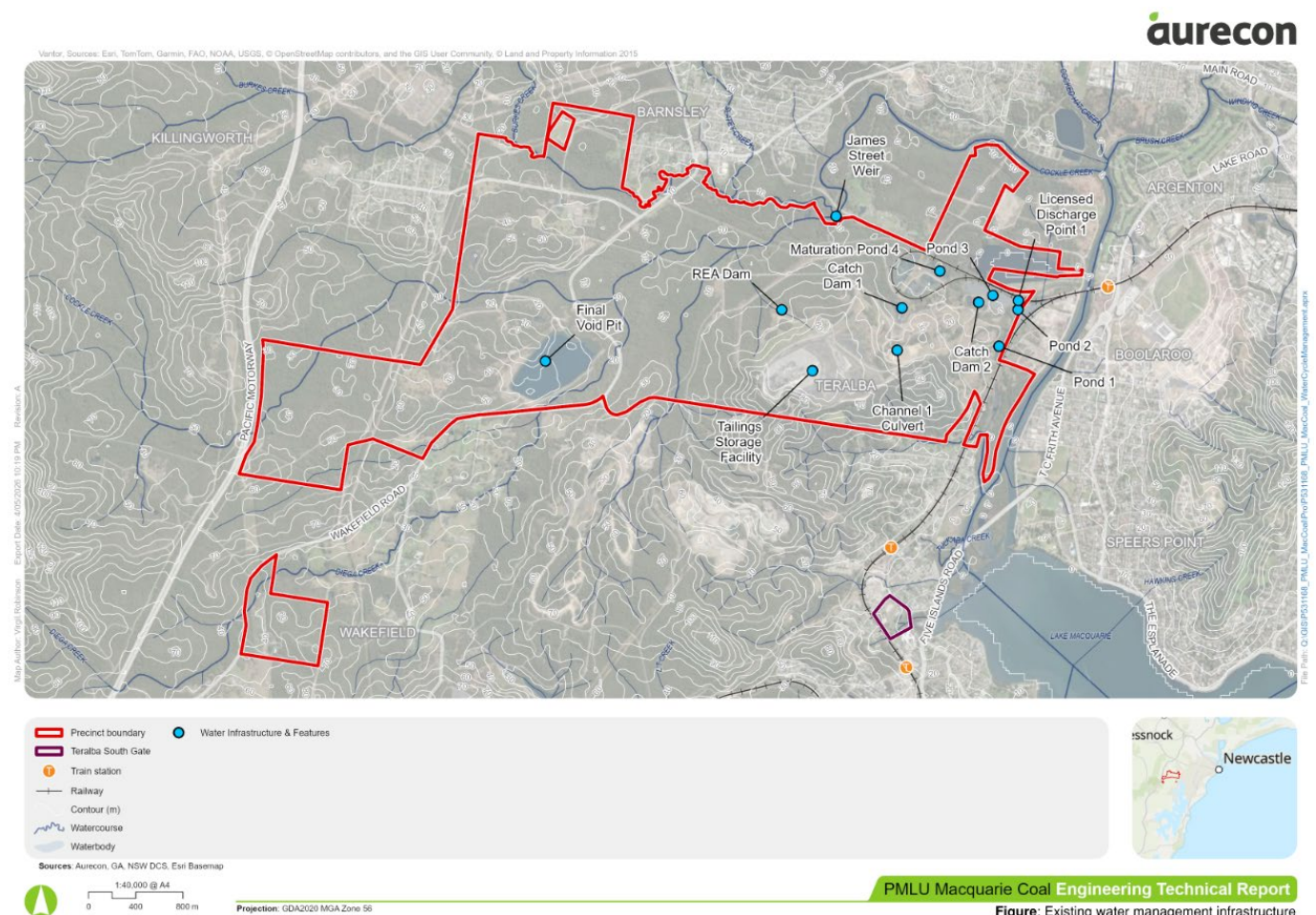


Figure 3-18 Existing water management infrastructure

3.4 Mine Rehabilitation and Post-Mining Water Context

The Precinct is located within a post mining landscape where there is existing infrastructure including water storage, water treatment and water diversions on site. The landscape within the Macquarie Coal Preparation Plant area has been rehabilitated to some degree which has included capping of existing water management infrastructure and rerouting of flows to other locations. There are a number of retained ponds through this site and the West Wallsend Colliery site which continue to be used under existing conditions to provide treatment or settlement prior to discharge at the licensed locations. Most of the water treatment infrastructure is located within the 100year ARI flood extent and tidal inundation zones from Cockle Creek. The water quality and sediment quality of these basins is largely unknown but is anticipated to be impacted from the historical mining usage and is therefore expected to have some contamination associated. The tailings dam also remains uncapped at this stage and retains runoff. There is a dam failure area associated with the tailings dam. Additionally, the pit lake void remains a large water feature of the site, and includes an overflow to Cockle Creek. Comprehensive water quality analysis of Pit Lake Void has been undertaken over the last two years, although this data was not available during the time of reporting and hence has not been reviewed for this report. The water quality of the Tailings Dam is unknown at this stage. Further detail about the potential presence of contamination is contained in the Contamination Technical Assessment (CTA).

The EPL License 1360 is still current for the two discharge locations at Burkes Creek and Cockle Creek. It is understood that there is a water access license held as part of the West Wallsend Colliery site (license 20BL169793) which is understood to be valid following mine closure.

The opportunities to reuse water management infrastructure in the future will be limited by:

- Suitability of size of infrastructure. This includes suitable size of infrastructure for conveyance, detention and treatment.
- Location and flooding/tidal inundation. Repurposing of these areas may be limited by inundation levels and whether this is acceptable for the proposed infrastructure (e.g. detention basins or treatment infrastructure).
- Final landform and levels of infrastructure. Levels will dictate what infrastructure has the potential to be repurposed for future development
- Contamination and safety. Further understanding of potential contamination and safety of these assets will further identify their reuse potential

Outside of the disturbed sites, the landform is fairly natural and is anticipated to flow through natural flow paths towards Cockle Creek and its tributaries. Through the undisturbed landscape, new water management infrastructure including conveyance pipes/channels, culverts and treatment infrastructure will be required.

Development across the site will also need to consider protected areas, sensitive environments and riparian corridors with appropriate buffers and distances provided between these locations and water management infrastructure.

4 Existing Flood Risk

This section presents a strategic assessment of flooding within the Precinct and its implications for future land use planning, development feasibility and infrastructure provision. The assessment considers the principal flood mechanisms affecting the Precinct, the distribution of flood hazard and associated land use constraints, the implications of extreme flood events, and the interaction between flooding and future access, movement networks and emergency management considerations.

The assessment is primarily informed by the best available regional flood investigations undertaken by Lake Macquarie City Council, including the Upper Cockle Creek Floodplain Risk Management Study and Plan (2019), Winding Creek and Lower Cockle Creek Flood Study (2017) and Lake Macquarie Waterway Flood Risk Management Study and Plan (2012). These studies provide the adopted understanding of flood behaviour across the Cockle Creek catchment and Lake Macquarie waterway and form the primary evidence base for identifying flood-related constraints and opportunities relevant to the Master Plan. However, the studies do not reflect the

latest guidance from Australian Rainfall and Runoff, Version 4.2. The key difference with the latest edition is the update to the climate change factors, which may have an impact on the flood risk areas. This would particularly be the case with the PMF flooding, which historically had not included climate change, but is now recommended.

Consistent with the strategic nature of the Master Plan, the assessment focuses on identifying flood risk and floodplain constraints that influence land use allocation, developable land, infrastructure planning and future staging, rather than undertaking detailed flood modelling or design. Where appropriate, recommendations are provided identifying matters requiring further investigation during subsequent planning, subdivision and development assessment stages.

4.1 Flood Mechanisms and Flood Behaviour

Flooding within the Precinct is influenced by a combination of catchment and downstream hydraulic processes. The Precinct experiences flooding from three primary mechanisms, which can be broadly defined as:

- **Regional Flooding** – flooding associated with Cockle Creek and its major tributaries during significant catchment flood events;
- **Local Flooding** - flooding resulting from rainfall runoff generated within the local catchments and conveyed through the local overland flow and drainage network; and
- **Tidal Inundation** - elevated water levels associated with tidal fluctuations within Lake Macquarie and their influence on downstream reaches of Cockle Creek.

These mechanisms operate at different spatial scales and influence flood behaviour across the Precinct to varying degrees. Regional flooding from Cockle Creek represents the dominant flood mechanism affecting the Precinct and has therefore informed the adopted flood planning levels and strategic flood risk assessment. Local flooding may occur independently of regional flooding where drainage capacity is exceeded during intense rainfall events, while tidal inundation primarily influences downstream water levels and tailwater conditions within the lower reaches of Cockle Creek.

An understanding of each flooding mechanism is important for informing future land use planning, drainage design and flood risk management within the Precinct. The characteristics and implications of each mechanism are discussed in the following sections.

4.1.1 Regional Flooding

As noted, flooding associated with Cockle Creek is considered regional flooding and represents the dominant flood risk affecting the site as it crosses the site from a southwest to northeast direction, draining to Lake Macquarie. The regional flood risk varies in impact on the floodplain to varying degrees and can be generally classed into the following categories (note that these are not related to Bureau of Meteorology flood warning categories):

- **Minor Flooding:** Cockle Creek water levels may remain largely within channel; however, elevated lake water levels can restrict discharge, resulting in increased tailwater effects and localised inundation adjacent to the creek corridor.
- **Moderate Flooding:** Overbank flow from Cockle Creek occurs, with floodwaters extending across the adjacent floodplain. Flooding in these events is typically shallow to moderate in depth but widespread, affecting low-lying areas contiguous with the creek.
- **Major Flooding:** During major flood events, extensive overbank flooding occurs along Cockle Creek and main tributaries, with floodwaters spreading across the broader floodplain and connecting downstream with inundation from Lake Macquarie. Under these conditions, flood extents expand significantly and flood depths increase, resulting in hazardous flooding conditions.

The existing flood information indicates that parts of the site are subject to inundation during the 1% AEP event, with more extensive and higher-hazard flooding occurring under PMF. These flood risk areas are mapped in Figure 4-1. The interaction between Cockle Creek flood flows and elevated lake water levels further impact inundation duration and reduced drainage capacity during regional flood events.

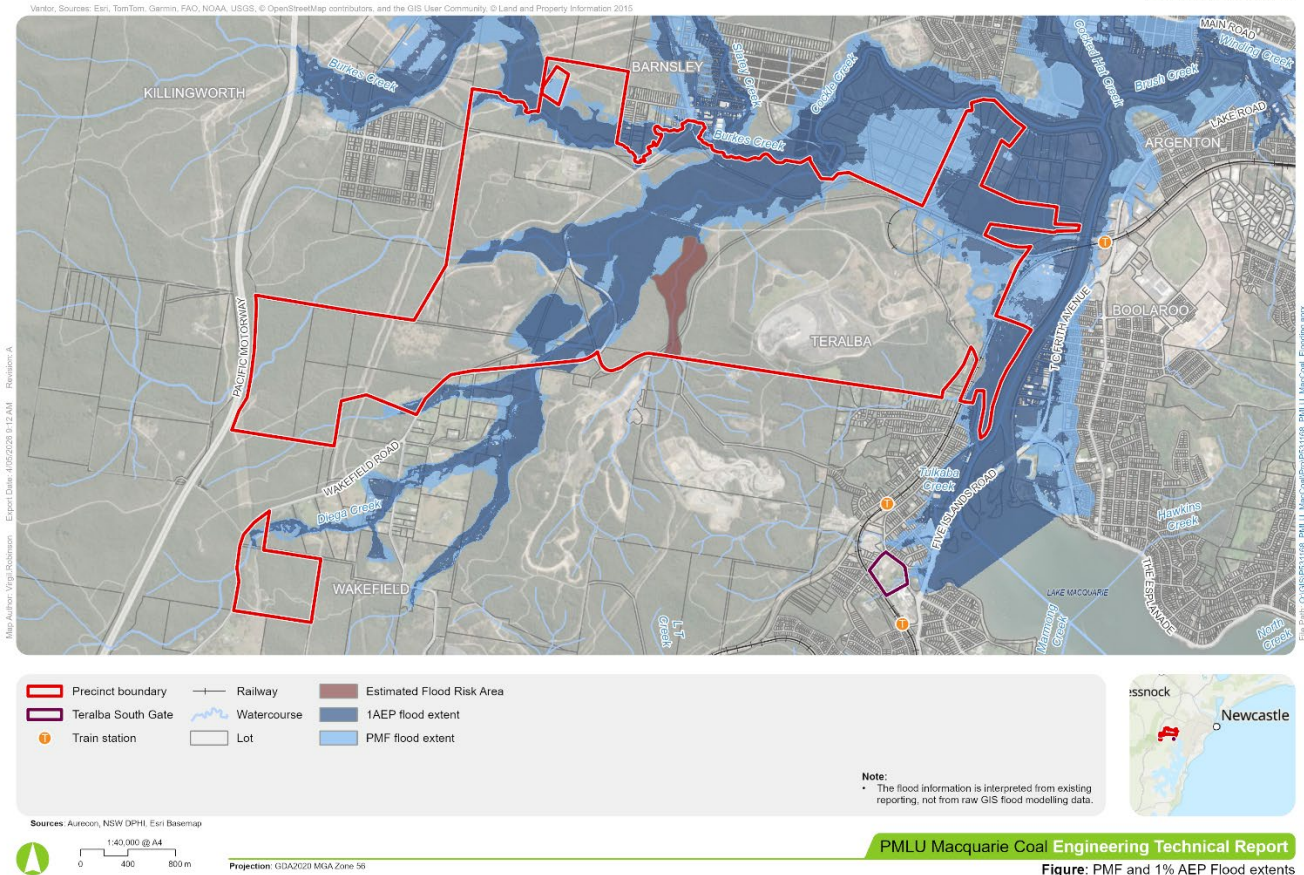


Figure 4-1 PMF and 1% AEP Flood extents map

4.1.2 Local Flooding

Local flooding within the Precinct is associated with intense rainfall falling directly on the local catchments and resulting in overland flow, minor watercourse flooding and temporary ponding in low-lying areas. Unlike regional flooding from Cockle Creek and its main tributaries, local flooding is generally driven by the capacity of the local drainage system and the topographic characteristics of the surrounding land rather than large-scale catchment inflows.

The Precinct contains a number of natural drainage depressions and ephemeral flow paths that convey runoff towards Cockle Creek and its tributaries. During high intensity rainfall events, these drainage pathways may exceed their conveyance capacity, resulting in shallow overland flow and temporary inundation of adjacent low-lying land. The final developed Precinct landform and local topographic variations may further influence the direction and extent of local runoff and drainage behaviour.

At the Master Plan level, detailed local drainage behaviour has not been assessed and will be dependent on future landform modification, infrastructure provision and development layout. Consequently, the extent and severity of local flooding should be confirmed through detailed drainage modelling and design in accordance with Council requirements at subsequent planning and development stages.

Future development should maintain existing overland flow paths where practicable, avoid creating adverse drainage impacts on adjoining land and provide local stormwater infrastructure designed to safely convey runoff while accounting for downstream tailwater conditions and the potential interaction with regional flood events. The detailed design of future drainage infrastructure should demonstrate that appropriate levels of service are achieved and that local flooding risks are managed without increasing flood impacts elsewhere within or downstream of the Precinct.

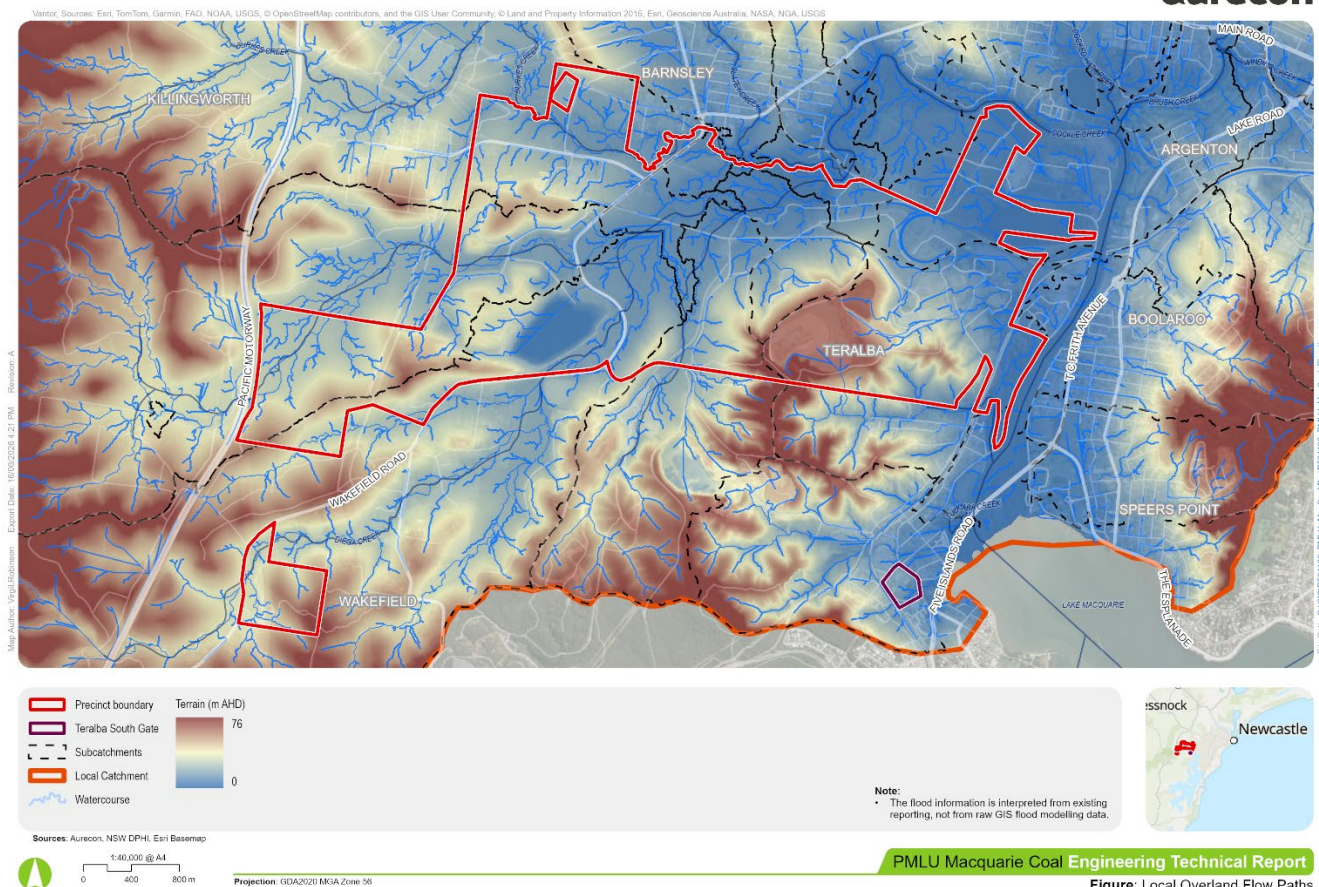


Figure 4-2 Local overland flow paths

4.1.3 Tidal Inundation

Tidal inundation within the Precinct is associated with elevated water levels in Lake Macquarie and the resulting hydraulic influence on the lower reaches of Cockle Creek. Cockle Creek discharges to Lake Macquarie, which in turn drains to the Pacific Ocean via the narrow and shallow 4 km long Swansea Channel. Under normal conditions, the lake level is approximately 0.1 m AHD and tidal fluctuations are relatively small, typically in the order of ± 0.05 m due to the hydraulic restriction imposed by Swansea Channel. However, elevated ocean levels associated with high tides, storm surge and wave setup, together with inflows from the surrounding catchments during major rainfall events, can significantly increase lake water levels and the tailwater conditions experienced within lower Cockle Creek. Barnsley Weir is known as the tidal limit of Cockle Creek however there is limited information to confirm details of the hydraulic control. Based on the flood mapping of ocean dominated events, the weir appears to be overtopped indicating that its influence is limited to frequent (eg. daily tide) events.

The Lake Macquarie Waterway Flood Risk Management Study and Plan (2012) recognised that peak lake water levels are governed by the coincidence of elevated ocean conditions and catchment runoff. During historical events, including the February 1990 and June 2007 floods, peak lake levels reached approximately 1.0 m AHD, contributing to elevated water levels within Cockle Creek and its lower tributaries.

For flood planning purposes, the adopted design coastal inundation level for the 1% AEP event under the 2100 sea level rise scenario (+0.9 m) is approximately 2.32 m AHD. This level represents a joint probability event, combining an elevated ocean condition with a coincident catchment rainfall event, consistent with the methodology adopted in the Lake Macquarie Waterway Flood Risk Management Study (2012). Consequently, the adopted design level reflects the combined influence of coastal processes and catchment runoff rather than a coastal storm occurring in isolation.

Figure 4-3 illustrates both the projected permanent inundation extent under the 2100 sea level rise scenario and the 1% AEP coastal inundation extent based on a peak lake level of 2.32 m AHD. These figures demonstrate that

the eastern portions of the Precinct may be influenced by future coastal inundation, particularly in low-lying areas adjacent to Cockle Creek and associated drainage corridors.

Areas subject to permanent inundation represent a significant development constraint. Development within these areas would require substantial engineering intervention, such as filling to achieve suitable finished surface levels or the construction of elevated structures supported above projected future water levels. Such measures are likely to increase development costs, require careful consideration of flood storage and hydraulic impacts, and may introduce additional environmental and planning constraints. Consequently, avoidance of permanently inundated land is considered the preferred planning outcome where practicable.

Similarly, development within areas affected by the adopted 1% AEP (2100) coastal inundation level should demonstrate that future land uses can safely accommodate the combined effects of elevated lake levels and coincident catchment flooding. Future planning and detailed design should consider the implications of elevated tailwater conditions on flood behaviour, stormwater drainage performance and finished surface levels to ensure that development remains resilient to the projected effects of climate change and coastal inundation.

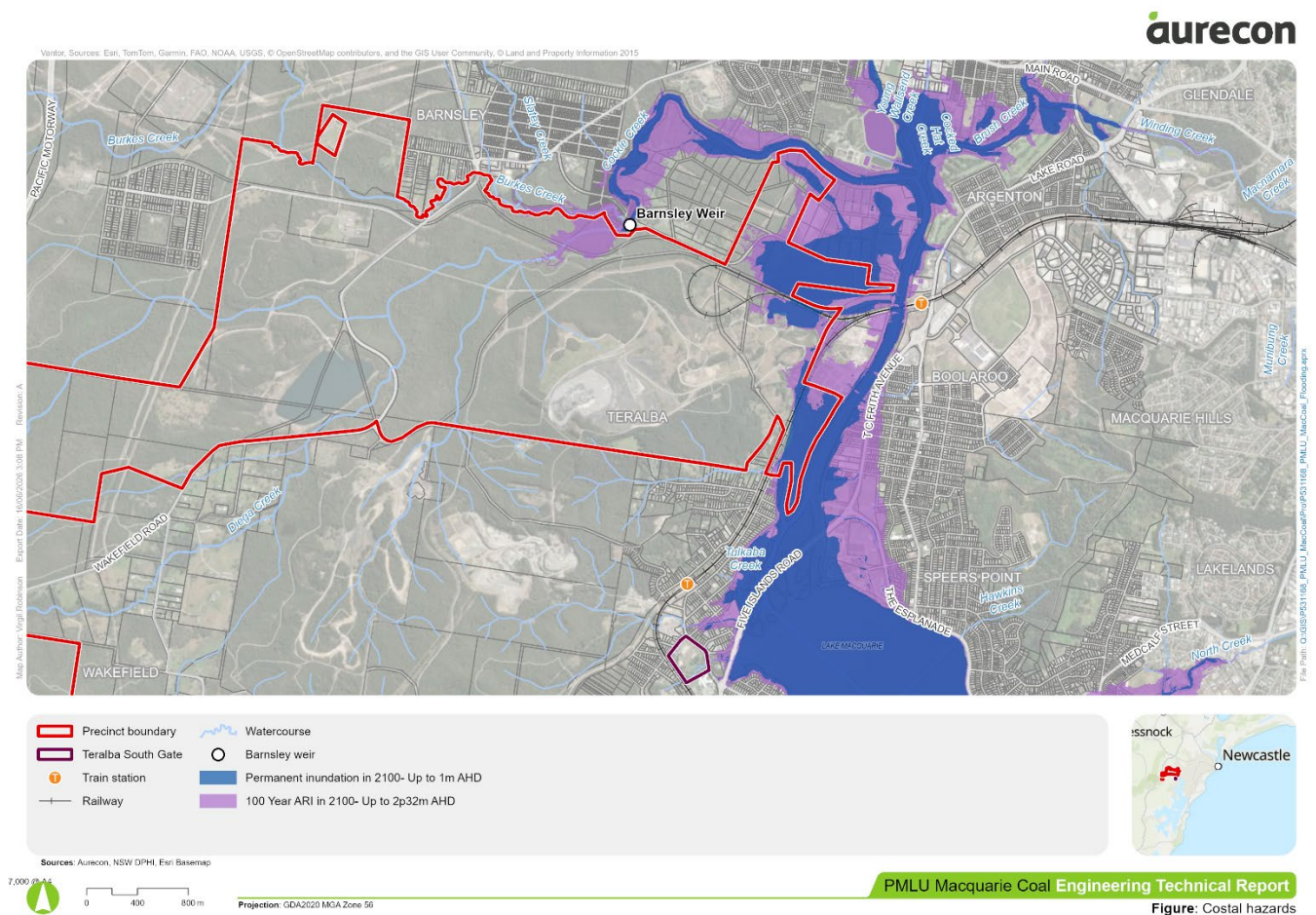


Figure 4-3 Coastal hazards

4.2 Hydraulic Categorisation (Flood Function)

In addition to flood extent and hazard, the hydraulic function of the floodplain is an important consideration for future land use planning and development. The Upper Cockle Creek Flood Study (WMAwater, 2019) categorises the floodplain into Floodway, Flood Storage and Flood Fringe areas based on the hydraulic behaviour of floodwaters.

- **Floodways-** are areas that convey a significant proportion of flood flows and play a critical role in transporting floodwaters through the floodplain. Development within floodways has the greatest potential to increase upstream flood levels and should generally be avoided where practicable.

- **Flood Storage-** areas provide temporary storage of floodwaters during flood events and assist in attenuating flood levels. The loss of flood storage through filling or development may increase flood levels unless appropriate compensatory measures are implemented.
- **Flood Fringe-** areas comprise the remaining flood-prone land outside of floodways and flood storage areas and generally have a lower hydraulic significance. Subject to detailed assessment and appropriate mitigation, these areas may provide greater opportunities for future development than floodways or flood storage areas.

Hydraulic categorisation provides an additional layer of information to support the flood constraint framework and refinement of the Structure Plan. As a general planning principle, future development should avoid floodways where practicable, minimise encroachment into flood storage areas and direct development towards lower constraint areas of the floodplain.

Recognising the strategic nature of the Precinct, there may be opportunities for targeted encroachment into floodway or flood storage areas where supported by detailed hydrologic and hydraulic modelling. In these circumstances, the proposed development and associated mitigation measures should demonstrate that changes to flood behaviour, including flood levels, flow velocities, flood hazard and flood conveyance, remain acceptable and do not result in unacceptable impacts on surrounding land or infrastructure. This approach provides flexibility for future precinct planning while maintaining the primary objective of preserving floodplain function and managing flood risk to an acceptable level.

4.3 Flood Hazard, Life Safety and Land Use Compatibility

4.3.1 Flood Hazard and Life Safety

Flood hazard provides an indication of the potential risk to people, vehicles and property during a flood event and is influenced by the depth, velocity and rate of rise of floodwaters. Areas subject to deeper and faster flowing floodwaters present a greater life safety risk and may significantly restrict emergency access, evacuation and future development opportunities.

Figure 4-4 presents the PMF flood hazard covering the Precinct. The hazard is categorised as either Low or High hazard and has been simplified from the hazard classifications in accordance with Australian Rainfall and runoff 2019 (Version 4.2). The high hazard recognises that higher hazard areas represent an increasing risk to people and vehicles and may become unsafe during flood events. In addition to mapped hazard classifications, floodways and areas characterised by deep or fast-flowing floodwaters should be recognised as key flood constraints due to their role in conveying flood flows and the associated safety risks.

At the strategic planning level, these areas should be considered when refining the Master Plan and locating future development, transport infrastructure and essential services. Development within areas subject to significant flood hazard may require substantial mitigation measures and should demonstrate that flood risk to occupants and emergency responders can be appropriately managed.

Accordingly, areas subject to significant flood hazard represent important planning constraints and should generally be avoided for land uses that increase population exposure or rely on safe evacuation during flood events.

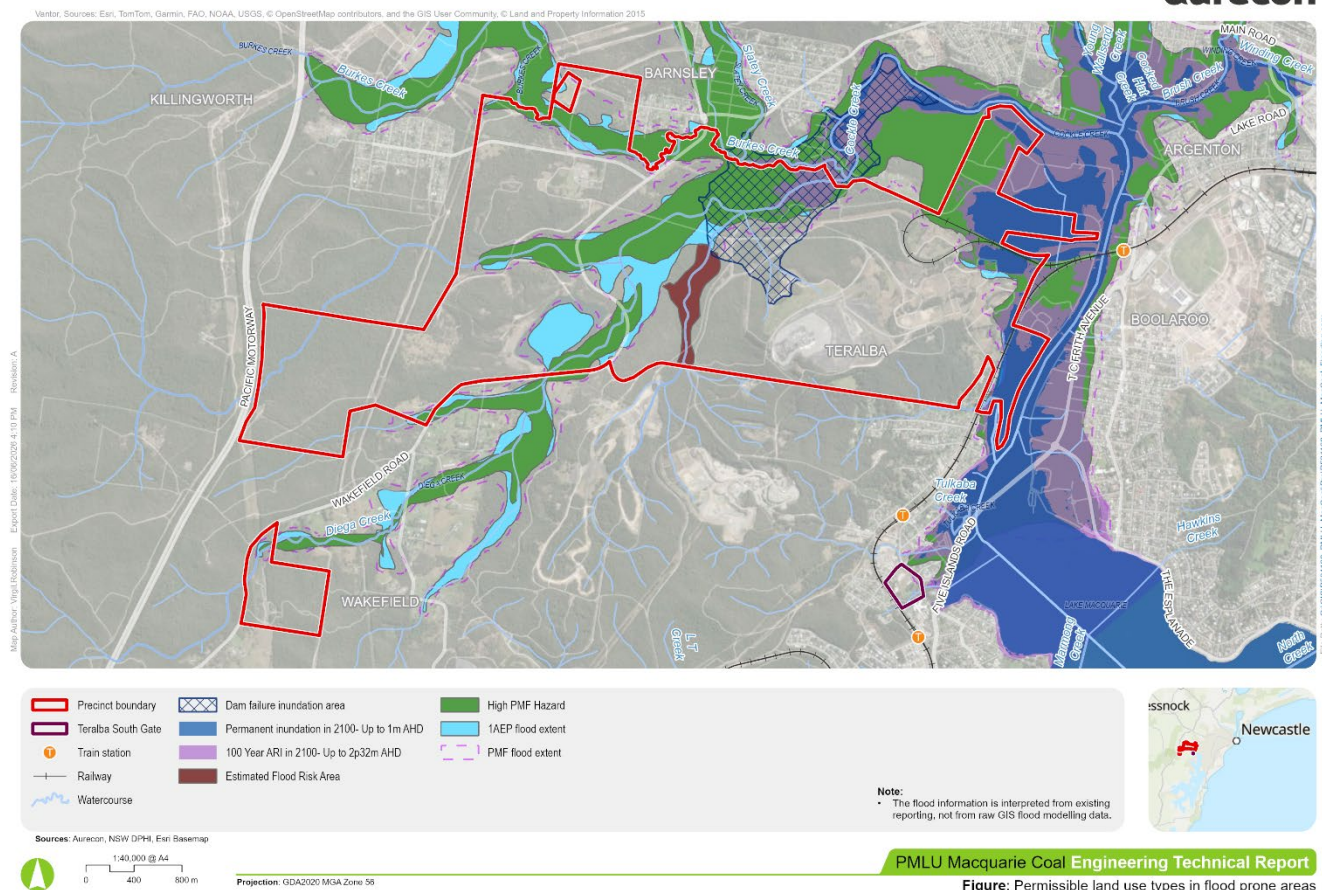
4.3.2 Land Use Compatibility

Land subject to frequent inundation, significant flood hazard or important flood conveyance and storage functions is generally more suited to compatible land uses such as environmental conservation, passive recreation, drainage corridors, agriculture and other uses that can tolerate periodic flooding with minimal risk to people or property.

Land subject to lower levels of flood hazard may be suitable for industrial, commercial and other low-occupancy land uses, provided detailed assessment demonstrates that flood risk can be appropriately managed and that emergency access, drainage and flood resilience objectives can be achieved.

Conversely, land subject to significant flood hazard is generally unsuitable for critical infrastructure, schools, hospitals, aged care facilities, emergency services infrastructure and other evacuation-sensitive land uses where

The Master Plan should therefore seek to direct future development towards areas of comparatively lower flood risk while recognising flood-prone land as an important constraint that influences developable land boundaries and future staging. Land use planning should avoid introducing additional population or evacuation-sensitive development into areas subject to significant flood hazard unless detailed assessment demonstrates that an acceptable level of flood risk can be achieved and appropriate mitigation measures implemented.



4.4 Extreme Event Risk

4.4.1 Probable Maximum Flood (PMF) Risk

The current understanding of PMF flood behaviour within the Precinct (refer Figure 4-5) is based on the Upper Cockle Creek Floodplain Risk Management Study and Plan (2019) and the Winding Creek and Lower Cockle Creek Flood Study (2017). These studies provide the best available information for strategic planning; however, they predate the latest guidance contained within Australian Rainfall and Runoff (ARR2019 Version 4.2), which recognises that Probable Maximum Precipitation (PMP) and consequently PMF flood events may also be influenced by climate change. Future updates to the flood modelling incorporating current climate change guidance

may therefore increase the extent and severity of PMF inundation, potentially affecting areas currently identified as having lower flood risk. Accordingly, the extent of PMF flooding should be reviewed during subsequent planning and design stages as updated modelling becomes available.

There is limited planning guidance that defines development suitability solely based on PMF inundation. Instead, PMF mapping should be used to inform life safety considerations, emergency management planning, infrastructure resilience and the suitability of critical or evacuation-sensitive land uses.

From a strategic planning perspective, the following principles should be considered:

- Development that is evacuation-sensitive, critical or accommodates large numbers of people should generally be avoided within areas subject to significant PMF flood hazard unless detailed assessment demonstrates that an acceptable level of life safety risk can be achieved.
- Future land use planning should consider the potential for rapid onset flooding, flood isolation and restricted emergency access during extreme flood events when determining the suitability of development within the PMF extent.
- Critical infrastructure and essential community services located within the PMF extent should be assessed to ensure that failure during an extreme flood event would not result in unacceptable risks to surrounding communities or emergency response capability.
- Buildings and infrastructure that are required to remain operational during or immediately following extreme flood events should be designed to withstand the hydraulic and debris loading associated with PMF conditions, including flood forces, buoyancy effects and debris impact where appropriate.
- Essential utilities and services, including transport infrastructure, power, communications, water and sewer infrastructure, should consider PMF flood impacts where service disruption could adversely affect surrounding communities or emergency management operations.
- The resilience of key transport corridors should be considered where PMF flooding has the potential to restrict emergency access, evacuation or post-event recovery. Future upgrades to strategic transport infrastructure should balance the required level of service with the engineering complexity and floodplain impacts associated with achieving higher flood immunity standards.

Given the strategic nature of the Master Plan, the PMF assessment should be regarded as a high-level planning tool that informs land use compatibility, infrastructure resilience and emergency management considerations. Detailed PMF assessment, incorporating the latest flood modelling, climate change guidance and proposed development layouts, should be undertaken during subsequent planning and design stages where development or infrastructure is proposed within or adjacent to the mapped PMF extent.

4.4.2 Dam Failure Risk

The Precinct will inherit a Tailing Storage Facility (TSF) that is in the process of being capped at the time of this study. The TSF is a declared dam under the Dam Safety Act 2015 and the Dam Safety Regulation 2019. The potential for failure can result in downstream flooding, presenting a new flood risk to downstream areas.

Dam failure assessments have been undertaken to understand the potential flood inundation area, the population at risk (PAR) and potential loss of life (PLL) with that area. This was captured in a Dam break failure impact assessment by ATC Williams (2023). Further details on the dam failure risk are presented in the Dam Engineering Report.

The failure inundation area is shown overlaid on the PMF flood hazard mapping in Figure 4-5. The PMF flood is more extensive and considered the dominant flood risk mechanism for the area, except for a small area of inundation between the failure location at the spillway to the extent of the PMF flood. This area would be considered as a 'High Flood Constraint Area' (refer Section 4.3.1) with sensitive or high occupancy development avoided in the short to medium term.

The TSF will continue rehabilitation works overtime. As the works reduce the failure risk, the objective would be to de-declare the TSF. However, until such time, dam break assessments shall be undertaken progressively to map out changes in the failure flood inundation extents. It is expected that this extent will reduce and consequently

release areas of previously at-risk land for suitable development. Option studies for land use can be undertaken with an ongoing dam management program while the dam is still declared.

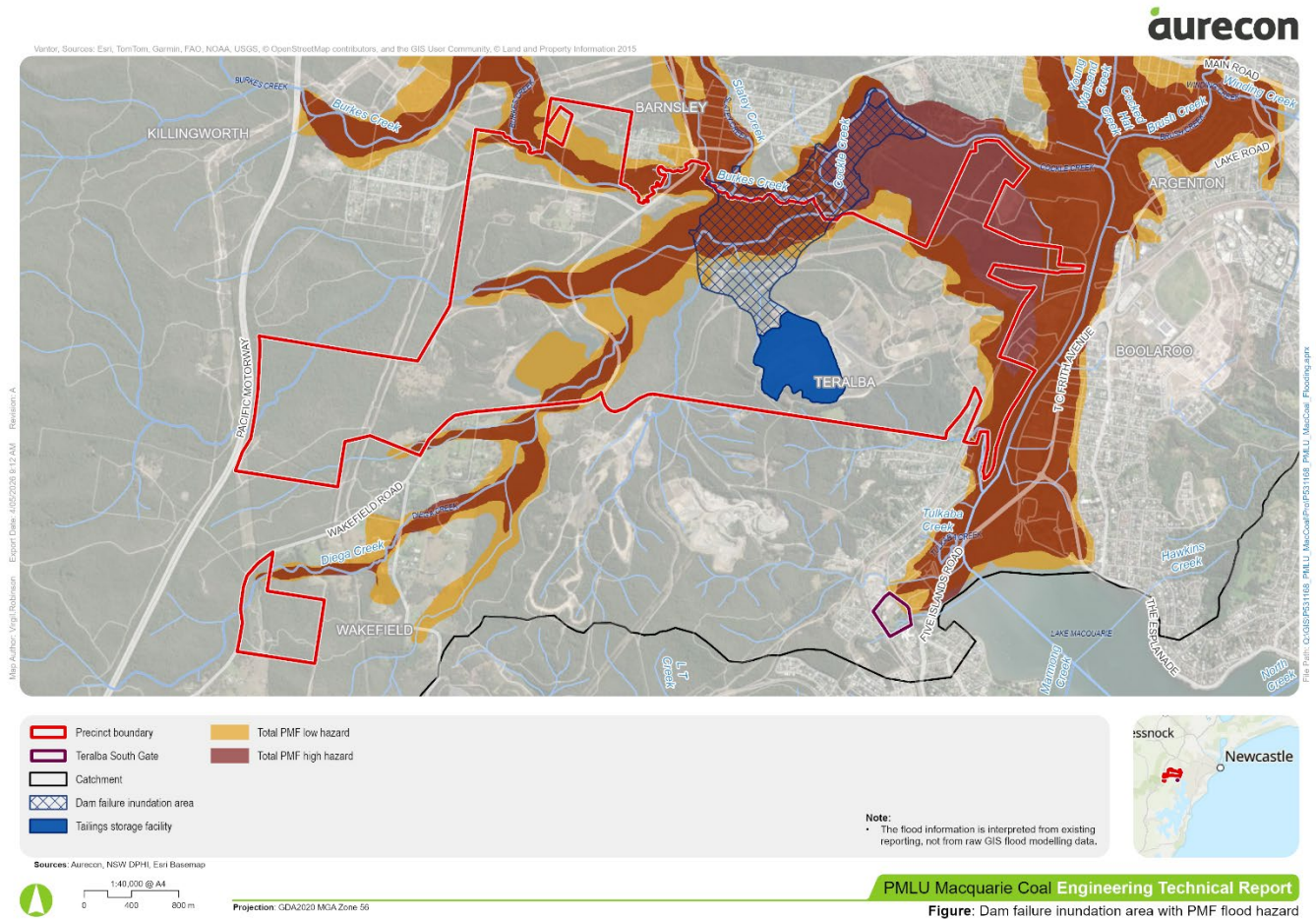


Figure 4-5: Dam failure inundation area with PMF flood hazard

4.5 Emergency Management

4.5.1 Emergency Management Strategy

Flood emergency management should be an important consideration in the future planning and development of the Precinct. As the site is largely greenfield, there is an opportunity to incorporate emergency access and evacuation considerations into the Master Plan to improve community safety and reduce future flood risk.

Regional flooding from Cockle Creek is the dominant flood mechanism affecting the Precinct. The Upper Cockle Creek Flood Study (2019) identified the 36-hour storm as the critical duration for the 1% AEP event and the 6-hour storm as the critical duration for the PMF event. While these durations indicate the storm events that generate the highest flood levels, the available information does not clearly define the available warning time before flooding occurs. In addition, there is currently no dedicated flood warning system for the Cockle Creek catchment.

Accordingly, evacuation should be adopted as the preferred emergency management strategy for future development. Future planning should seek to maintain access to higher ground and provide transport networks that remain functional for as long as practicable during flood events.

At this stage, shelter-in-place is not considered necessary, as the preliminary flood emergency response assessment indicates that evacuation is generally achievable (refer to Section 4.4.2). Furthermore, shelter-in-place is unlikely to be appropriate within areas subject to significant PMF flood hazard due to the potential life safety risks associated with extreme flooding.

There is currently limited guidance for emergency management of non-residential development exposed to regional flooding. Consequently, future development proposals should demonstrate that appropriate evacuation arrangements can be achieved and that emergency access is maintained where required. The emergency management strategy should be reviewed and refined to account for local flooding as development layouts, transport networks and flood modelling are updated during subsequent planning and design stages.

4.5.2 Flood Emergency Response Classification

A high-level flood emergency response classification has been undertaken to provide an initial understanding of the emergency management characteristics of the Precinct under extreme flooding conditions. The assessment is intended to support strategic land use planning and identify areas where future development may require more detailed emergency management consideration. It is not intended to replace a detailed flood emergency response assessment or Operational Flood Emergency Management Strategy, which should be prepared as future development proposals and infrastructure layouts are refined.

The emergency response classification has been determined using the decision flowchart presented in Figure 17 of the NSW Flood Risk Management Manual – Support for Emergency Management Planning (EM01) and applied to the Probable Maximum Flood (PMF) extent across the Precinct. The PMF extent adopted for this assessment was derived from the Upper Cockle Creek Flood Study (WMAwater, 2019) hydraulic model outputs and interpreted from the available flood mapping.

The Figure 17 framework classifies flood-affected areas through a sequential assessment of:

- whether the land is inundated during the PMF event;
- whether pedestrian or vehicle evacuation is practical before evacuation routes become flooded; and
- whether isolated land remains above flood level or becomes fully submerged.

The resulting emergency response classification mapping for the Precinct is presented in Figure 4-6.

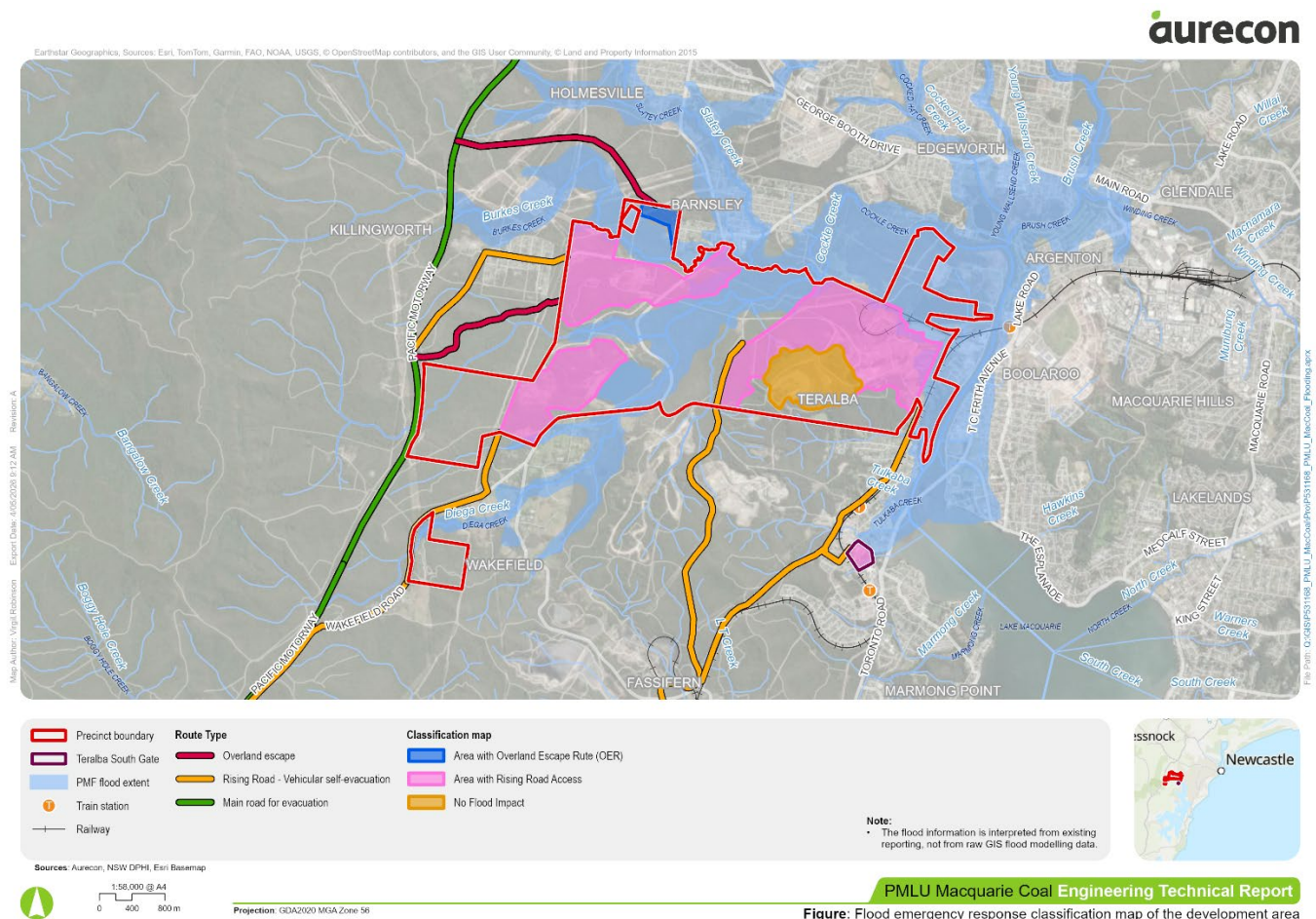


Figure 4-6 Flood emergency response classification map of the development area

The following assumptions have been adopted in undertaking the preliminary assessment:

- Existing unsealed roads have been assumed to remain trafficable for emergency egress, or access for emergency services, where they provide rising access and are not directly inundated by floodwaters.
- The rising road connection to the Pacific Highway is assumed to provide an emergency access point, notwithstanding that direct public access to the highway may not currently exist. It is assumed that emergency services would be able to access this location if required.
- Existing roads located within the mapped PMF extent may remain above the PMF water surface for evacuation; however, these roads have not been adopted as evacuation routes due to the potential for embankment instability or isolation during extreme flooding. Preference has therefore been given to routes located outside the mapped PMF extent.
- The emergency response classification should be reviewed as development layouts, road infrastructure and flood mitigation measures are refined to ensure that appropriate evacuation opportunities and emergency response arrangements can be achieved for future land uses.

The emergency response classification has been undertaken at a strategic planning level using the currently available PMF mapping and topographic information and should be regarded as indicative only. The classification has been derived from visual interpretation of the PMF mapping and is intended to inform the Master Plan by identifying broad emergency management constraints and opportunities. As future land use planning, infrastructure design and development staging are refined, the emergency response classifications should be reviewed and confirmed using the underlying hydraulic model outputs from the Upper Cockle Creek Flood Study (WMAwater, 2019), or any subsequent update, together with detailed terrain data, proposed road networks, finished surface levels and flood mitigation measures.

4.5.3 Flood Sensitive Transport Links

The transport network servicing the Precinct has an important role in supporting future development, emergency access and evacuation during flood events. As such, flood-sensitive transport links represent a key planning consideration and should be considered during the refinement of the Master Plan and future infrastructure planning.

Future development should demonstrate that suitable access to higher ground can be achieved, that the transport network is resilient to flood isolation where practicable, and that emergency vehicle access and evacuation arrangements are appropriate for the proposed land uses. The required flood immunity of future transport infrastructure should be commensurate with its intended function and level of service.

The proposed crossing of Cockle Creek and the Diega Creek floodplain, together with the proposed Lower Hunter Freight Corridor, will require detailed flood assessment to determine appropriate alignment, levels and structural arrangements. These crossings have the potential to influence flood behaviour and will require careful consideration to balance transport connectivity, flood risk and constructability.

Existing transport links within the Precinct are also affected by flooding to varying degrees. Wakefield Road is mapped as being overtopped during the 1% AEP flood event; however, the flooding appears to be largely confined to the creek crossing locations based on the Upper Cockle Creek Flood Study (2019). Consequently, relatively localised road raising and bridge or culvert upgrades may provide a practical means of improving flood immunity across this corridor.

The Weir Road traverses flood-prone land near the existing rail loop before crossing eastward towards lower Cockle Creek. The Winding Creek and Lower Cockle Creek Flood Study (2017) indicates that sections of this road are inundated during the 1% AEP flood event, reducing its reliability as an evacuation or emergency access route during major floods. Proposed road corridors should see to improve this with connectivity south through the site.

The existing rail loop is generally above the 1% AEP flood level but lies partially within the PMF extent. While this indicates a relatively high level of flood immunity for more frequent events, consideration should be given to the potential impacts of extreme flooding on network resilience and emergency response planning.

Any future road or rail crossings that traverse the Cockle Creek floodplain should consider the required level of service and corresponding flood immunity appropriate to their intended function. Higher immunity standards may require increased structure heights, longer bridge spans or greater embankment raising, which may increase construction complexity, cost and potential impacts on flood behaviour. Accordingly, flood immunity objectives should be balanced against floodplain impacts and the operational requirements of the transport network during the detailed planning and design stages.

5 Drainage Feasibility and Existing Constraints

The feasibility of future development within the Precinct is influenced not only by flooding constraints but also by the ability to effectively collect, convey and discharge stormwater runoff. At the Master Plan stage, the assessment has been limited to a strategic review of topography, drainage pathways, receiving waterways and flood behaviour to identify areas where drainage may influence future land use planning, infrastructure provision and development staging. Detailed stormwater modelling and drainage design have not been undertaken and should be completed during subsequent planning and development phases.

Drainage feasibility across the Precinct is expected to vary according to local gradients, available discharge locations and downstream hydraulic conditions. In general, flatter terrain presents greater drainage challenges due to reduced hydraulic gradients, requiring larger drainage infrastructure or open channels to convey runoff. Conversely, steeper areas generally provide improved gravity drainage opportunities but may require additional drainage structures, erosion protection and energy dissipation measures to safely manage runoff velocities. In addition, elevated downstream water levels associated with regional flooding and tidal influences may reduce drainage capacity and should be considered during future infrastructure planning.

5.1 Drainage Constraints and Development Implications

The Precinct contains a number of physical characteristics that have the potential to influence future drainage feasibility and the cost and complexity of development. These include:

- extensive low-lying floodplain areas;
- high groundwater table within the lower floodplain areas;
- interaction with Cockle Creek flood levels;
- downstream tailwater influences;
- shallow drainage gradients; and
- local depressions and natural storage areas.

Collectively, these characteristics may limit conventional gravity drainage opportunities in some parts of the Precinct and require more extensive drainage infrastructure than would otherwise be required on unconstrained land.

Table 5-1 Drainage implications based on slope constraints

Slope (%)	Equivalent Slope (°)	Drainage Characteristics
0–0.5%	0.0–0.3°	Very flat terrain: Conventional gravity drainage can be difficult to achieve without significant excavation depths. Open channels, swales, broad drainage corridors or other low-gradient conveyance systems may be required, with an increased potential for local ponding.
0.5–7%	0.3–4.0°	Ideal slope: Generally, well suited to conventional gravity stormwater drainage with efficient runoff conveyance and practical pipe grades. Typically represents the most favourable terrain for urban development and servicing.
7–20%	4.0–11.3°	Moderate slopes: Gravity drainage remains feasible; however, increased runoff velocities require greater consideration of erosion protection, energy dissipation measures, scour management and dedicated stormwater infrastructure. Earthworks and retaining structures may become more significant depending on development layout.
>20%	>11.3°	Steep terrain: Drainage is generally achievable but requires careful design to manage runoff velocities, erosion, slope stability and construction constraints. Significant earthworks, retaining structures and specialised drainage measures may be necessary to facilitate development.

Areas characterised by shallow gradients typically require larger drainage conduits or open drainage channels to achieve acceptable hydraulic performance while maintaining practical pipe grades. Conversely, areas with steeper grades generally provide more efficient drainage but may require additional pits, erosion protection measures and energy dissipation structures to manage increased flow velocities and minimise scour at drainage outlets.

Drainage feasibility is also influenced by downstream flood levels within Cockle Creek and associated waterways. During regional flood events or periods of elevated lake levels, tailwater conditions may reduce the capacity of gravity drainage systems, surcharge outlets and increase local ponding durations. Future drainage infrastructure should therefore be designed having regard to both local runoff conditions and downstream hydraulic controls.

The combination of topographic constraints, flood interactions and drainage servicing requirements is likely to influence the developability and staging of future development areas. Higher land with favourable drainage gradients and unconstrained discharge opportunities is expected to provide more cost-effective development opportunities and may be more suitable for early activation. Conversely, low-lying land subject to flood interaction or constrained drainage conditions may require substantial earthworks, drainage infrastructure augmentation or other engineering measures before development can proceed.

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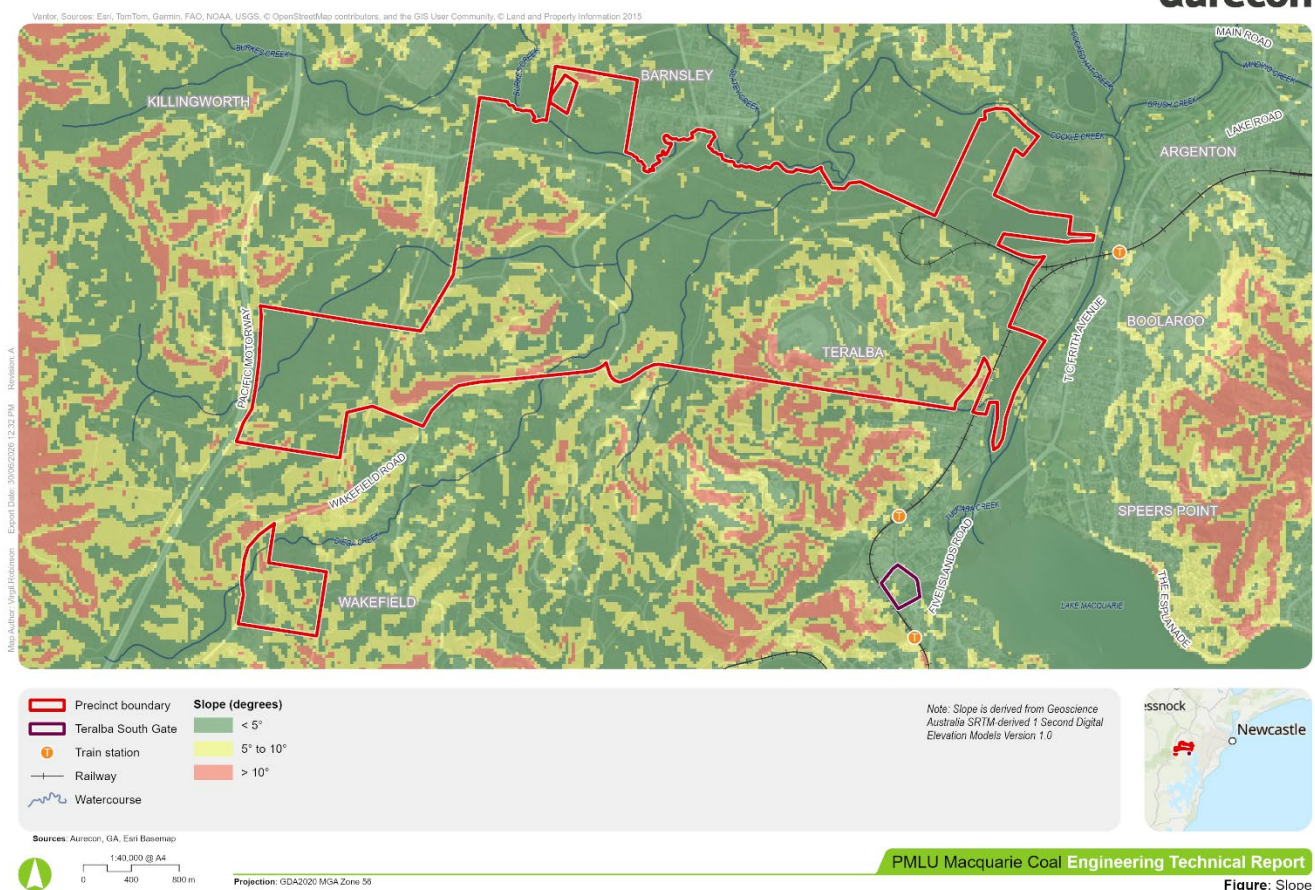


Figure 5-1: Indicative Drainage Constraints

5.2 Overland Flow Paths and Outlet Points

The existing landform contains numerous natural drainage pathways that convey runoff towards Cockle Creek, Burkes Creek and associated tributaries. These overland flow paths perform an important hydraulic function and should be recognised during future land use planning to avoid creating adverse flooding or drainage impacts.

Future development should maintain the function of existing drainage pathways and provide suitable conveyance corridors where modification is unavoidable. The obstruction or redirection of established overland flow paths has the potential to increase local flooding, alter flow distribution and transfer drainage impacts to adjacent land.

The location and capacity of future drainage outlets will require detailed assessment during subsequent planning and design stages. Particular consideration should be given to outlets discharging to flood-affected waterways where elevated downstream water levels may reduce drainage performance during major rainfall events. In these locations, additional drainage infrastructure, outlet controls or local storage may be required to achieve acceptable levels of service.

The drainage network should be planned to accommodate both minor and major storm events while preserving existing flow paths and maintaining compatibility with the surrounding floodplain. This approach will assist in minimising flood risk, reducing infrastructure costs and support staging of future development across the Precinct.

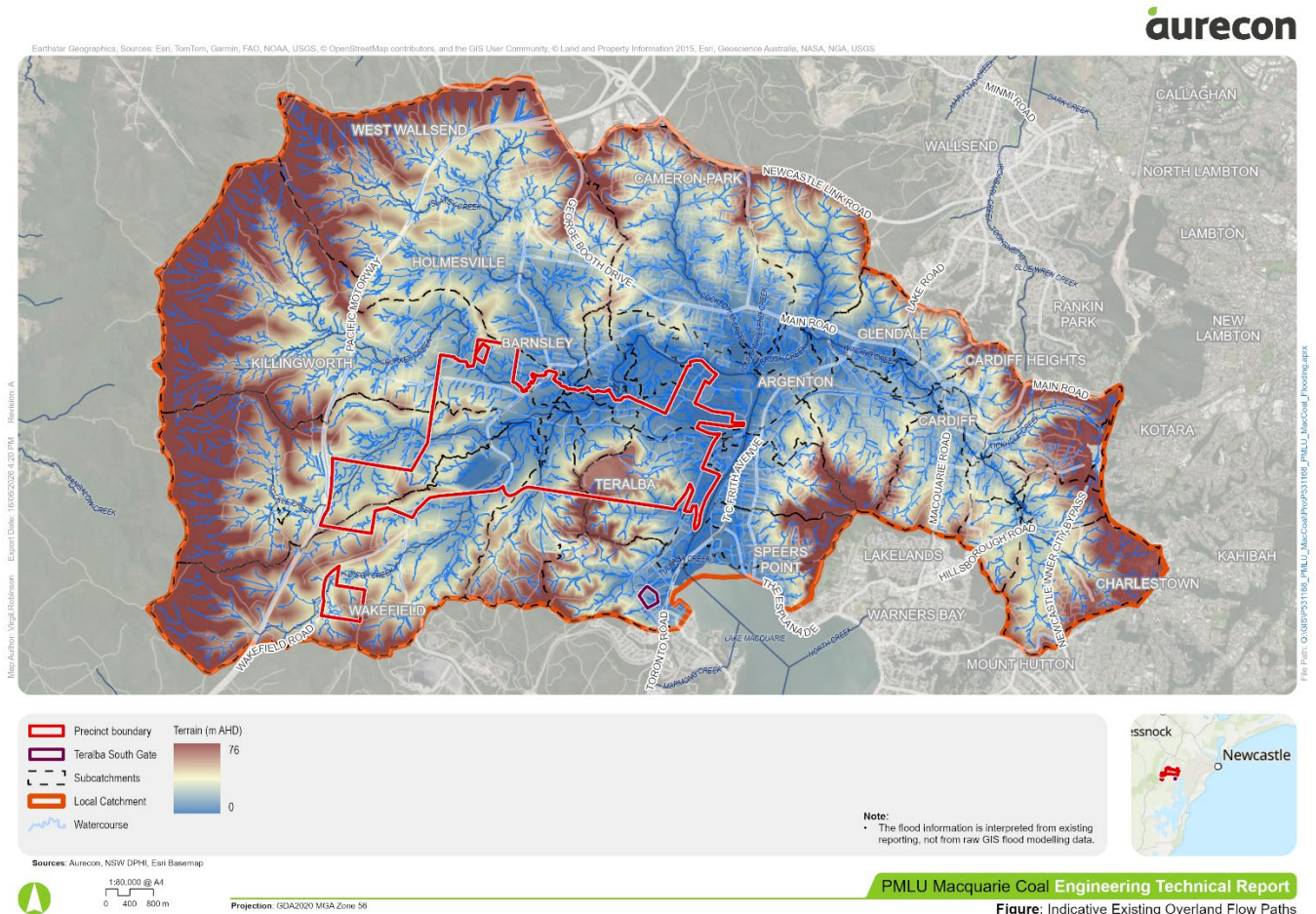


Figure 5-2: Indicative Existing Overland Flow Paths

5.3 Groundwater, ASS and Salinity Considerations

There is limited, sparse information about the hydrogeology within the Precinct, however some information suggests shallow groundwater around Cockle Creek and tributaries in low lying areas. Groundwater recharge rates are unknown but based on hydrological soil type are anticipated to be slow.

Groundwater considerations that are relevant to the project include:

- Shallow groundwater – groundwater levels pose a constraint due to possible inundation risk and development feasibility particularly in low lying areas near the waterways. Groundwater levels also pose a risk to stormwater management infrastructure including detention basins and bioretention basins which need to be designed to account for groundwater levels.
- Groundwater aquifer sustainability, ecosystems and recharge – there is a risk that development could impact infiltration and recharge of groundwater. Precinct planning should identify and protect critical groundwater areas and dependent ecosystems during the planning stage.

- Groundwater extraction or depletion particularly during construction may pose a threat to stability and increase subsidence risk.
- Acid sulphate soils are anticipated to occur around the site, largely situated within the low lying areas around waterways, in particular near Cockle Creek. These soils create acidic conditions which can pose environmental and structural risks.
- The overall salinity hazard assessment for the broader Lake Macquarie / Hunter region identifies the North Coast and Hunter LLS regions as areas of concern for salinity, with significant impacts on land and local infrastructure possible in high hazard areas. Salinity within soils and groundwater can cause corrosion of concrete, steel and other building materials. Areas where salinity is identified may require specific construction standards and procedures to address the potential impact.

Comprehensive hydrogeological investigation and risk assessment will be essential to identify groundwater constraints, inform design modifications, and establish long-term management protocols that protect both development viability and water resource sustainability. Additionally, detailed acid sulfate soil investigations are recommended to inform future development. Further detail about the presence of Acid Sulfate Soils and salinity risk is included in the Contamination Technical Assessment (CTA).

6 Water Cycle Management Assessment

6.1 Integrated Water Cycle Management Pathway

Water cycle management (WCM) is a key consideration for future development within the Precinct. WCM is defined as the holistic approach that addresses competing demands placed on a regions water resources whilst optimising social benefits and protecting environmental values of receiving waterways.

Diega Creek and Cockle Creek are the two key waterways located within the Precinct. Burkes Creek is also a waterway which is located at the northern side of the Precinct. Existing stormwater management within the Precinct is through natural flow paths within the undeveloped areas and via water management infrastructure across the disturbed site which includes catchment dams, sediment dams, storage dams, spillways and drainage systems. Cockle Creek is the main receiver of stormwater runoff from the Precinct, receiving flows from upstream tributaries, natural flow paths and water directed from the disturbed areas via two licenced locations.

Key variables and interactions associated with groundwater within the Precinct include presence of groundwater dependent ecosystems (GDEs), groundwater levels and flow paths, and groundwater quality and potential contamination. The Precinct is largely underlain by soils with slow infiltration rates which may indicate limited or slow recharge of groundwater aquifers within the Precinct.

Key sensitive receiving environments that have been considered include:

- Coastal wetlands, key fish habitat and groundwater dependent ecosystems (GDEs) within the Precinct, concentrated largely within the riparian corridor of Diega and Cockle Creeks.
- Downstream sensitive receiving environments including Lake Macquarie and Cockle Creek estuarine environments

Drainage of stormwater within the Precinct will aim to minimise changes to sub-catchments boundaries and direct flows via existing drainage paths where possible. Formalisation of drainage will occur to capture and direct water to stormwater treatment prior to discharge to the downstream receiving environment. Onsite stormwater detention (OSD) will be required within the Precinct and the stormwater treatment strategy will consider options for both on-lot and end-of-pipe treatment to allow for flexibility of development size and staging. Stormwater treatment opportunities include rainwater tanks, gross pollutant traps, bioretention, wetlands and street trees.

For water supply (potable and non-potable), refer to the Utilities report.

The adoption of an integrated approach to water cycle management prioritises capture, treatment and reuse where feasible and aims to minimise water quality and quantity impacts to sensitive receiving environments.

6.2 Water Cycle Management Opportunities

There are various water cycle management opportunities for the Precinct such as utilising existing infrastructure for stormwater detention and treatment uses, harvesting rainwater through water tanks and reuse on site and integrating stormwater management with landscape design.

6.2.1 Existing Infrastructure

The 'Discharge Water Management Strategy' for the Macquarie Coal Preparation Plant (SLR Consulting, 2018) indicated that dams at the plant have adequate capacity to contain runoff in accordance with the EPL 1360 and Blue Book requirements (Landcom, 2004) for sediment dams. Pending further investigation, there may be opportunities for reuse of existing infrastructure for stormwater detention and treatment uses. For example, the MCPP and West Wallsend Colliery ponds could be repurposed for onsite detention. Existing runoff diversions and catch drains may be reused depending on final landform and topography changes.

6.2.2 Harvesting and Reuse

There are opportunities for an integrated water cycle management such as harvesting stormwater and reuse on site. Stormwater can be harvested at an on-lot or precinct scale. At an on-lot scale stormwater from rainwater tanks can be used for non-potable re-use such as flushing of toilets and landscape irrigation. At a precinct scale stormwater can be used to supply water to local industry, irrigation or for other water uses. Urban development typically impacts volume, frequency and quality of stormwater runoff. Harvesting can reduce the volume of water flowing into the drainage system and so reduce stream erosion and minimise the impacts of urbanisation on aquatic ecosystems. Further, water quality can be improved through re-use by abstracting a proportion of polluted stormwater for re-use, trapping pollutants in on-line storages and returning treated stormwater to receiving waterways. Finally, stormwater harvesting can also substitute in part for existing mains water uses whilst balancing the need to provide flows for urban creeks and rivers (DEC, 2006).

6.2.3 Treatment

Water quality reduction targets can be achieved by using typical Water Sensitive Urban Design (WSUD) treatment trains using Stormwater Quality Improvement Devices (SQIDs). Depending on the scale of the development and planning agreements, SQIDs can potentially be provided at the lot, street, subdivision or precinct scales. Examples of typical treatment trains from the Lake Macquarie City Council SQID Guidelines 2013 (LMCC, 2013) are listed in Table 6-1.

The local constraints may prohibit the implementation of these typical treatment trains in some sections of the Precinct. These constraints are discussed in Section 4.5.3.

Table 6-1 Examples of treatment trains and scales

On lot / at source treatments (lot & street)	End of pipe – primary treatments (subdivision & precinct)	End of pipe secondary / tertiary treatments (subdivision & precinct)
■ Rainwater tank ■ Raingarden ■ Permeable paving ■ Infiltration trench ■ Vegetated filter strip ■ Grassed swale ■ Bioretention swale ■ Tree pit filter	■ Gross pollutant trap ■ Sediment basin	■ Constructed wetland ■ Bioretention basin ■ Infiltration basin

Furthermore, there are opportunities to integrate stormwater treatment into the landscape to create visual and recreational amenity of developments. Treatment devices such as raingardens, biofiltration basins, vegetated filter strips, street trees and constructed wetlands can be integrated into the landscape to provide stormwater, visual and recreational benefits.

6.2.4 Groundwater

There may be opportunity to replenish groundwater aquifers through WCM opportunities that promote infiltration of soils. This may be particularly beneficial if the Precinct is identified as having depleted aquifers. At this stage due to a lack of monitoring and sampling information, groundwater levels and infiltration rates of soils is unknown. As mentioned in Section 3.2.5, the soils across the Precinct are anticipated to have slow rates of infiltration due to their hydrological soil group. Thus, opportunities for groundwater recharge such as implementation of “leaky” treatment trains which utilise infiltration may be limited. Further ground investigations and analysis will determine whether groundwater recharge is suitable for the Precinct.

7 Structure Plan

7.1 Proposed Structure Plan

The proposed structure plan has been developed by GHD and is the principal spatial framework for the Master Plan. It identifies the arrangement of developable areas, environmentally constrained areas, recreational areas, movement networks and key constraints across the Precinct. The structure plan outlines the spatial logic for future development at a precinct scale, providing a framework within which detailed design, subdivision and development assessment will occur. The boundaries between land use categories shown on the structure plan are indicative and may be refined through detailed technical investigation and assessment at the development application stage.

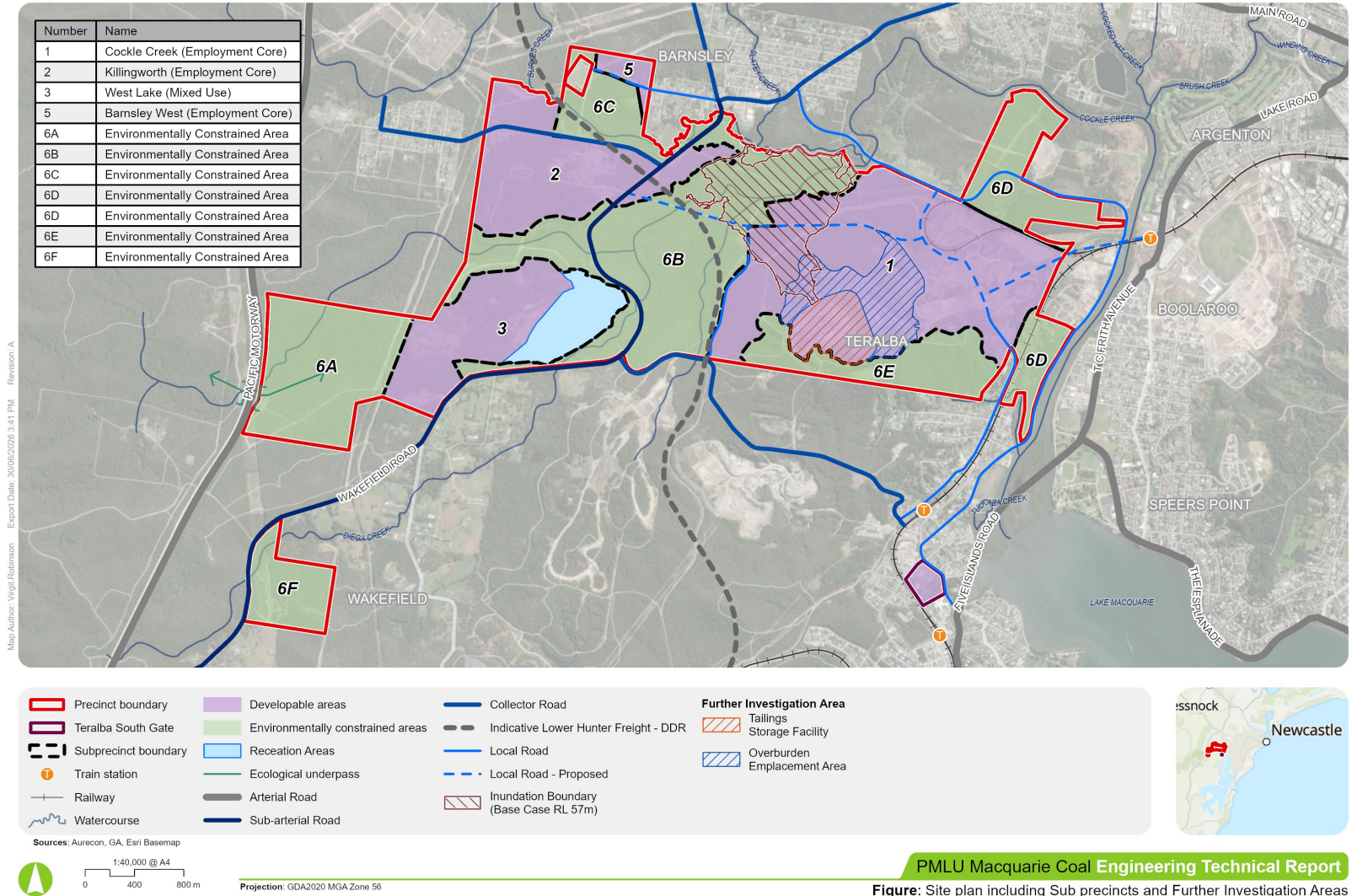
The structure plan has five developable sub-precincts with the last sub-precinct consisting of the remaining area made up of the environmentally constrained area.

The following categories are used to describe land within the structure plan as shown in Figure 7-1:

- **Developable areas** – Land assessed as most likely to be suitable for development, subject to further technical investigation and site-specific assessment at the development application stage. Developable areas are concentrated in locations with relative geotechnical stability, accessibility to existing or planned transport networks and proximity to utility infrastructure or feasible servicing solutions.
- **Environmentally constrained areas** – Land identified as having high biodiversity values or where natural features (watercourses, wetlands, steep slope, flooding) constrain development for employment uses. Detailed environmental investigations are required prior to development or disturbance of vegetation in these areas. Environmentally constrained areas form an interconnected network across the Precinct that underpins ecological connectivity, water cycle management and landscape identity.
- **Recreational areas** – Land suitable for active and passive recreation facilities, balanced with protecting biodiversity and cultural values. Recreational areas are primarily associated with the West Lake sub-precinct and the conservation corridor network, where the landscape and environmental values support nature-based tourism, walking and cycling trails and community recreational facilities.
- **Further investigation areas** – Land constrained in the short to medium term by geotechnical, dam safety, flood or biodiversity factors requiring resolution before development can proceed. Further investigation is required, as described in this Master Plan, prior to development. The release of this land for development is contingent upon the completion of relevant technical investigations and the achievement of applicable staging triggers. These areas have been mapped mainly in the Cockle Creek sub-precinct with small areas mapped in the eastern section of the Killingworth sub-precinct.

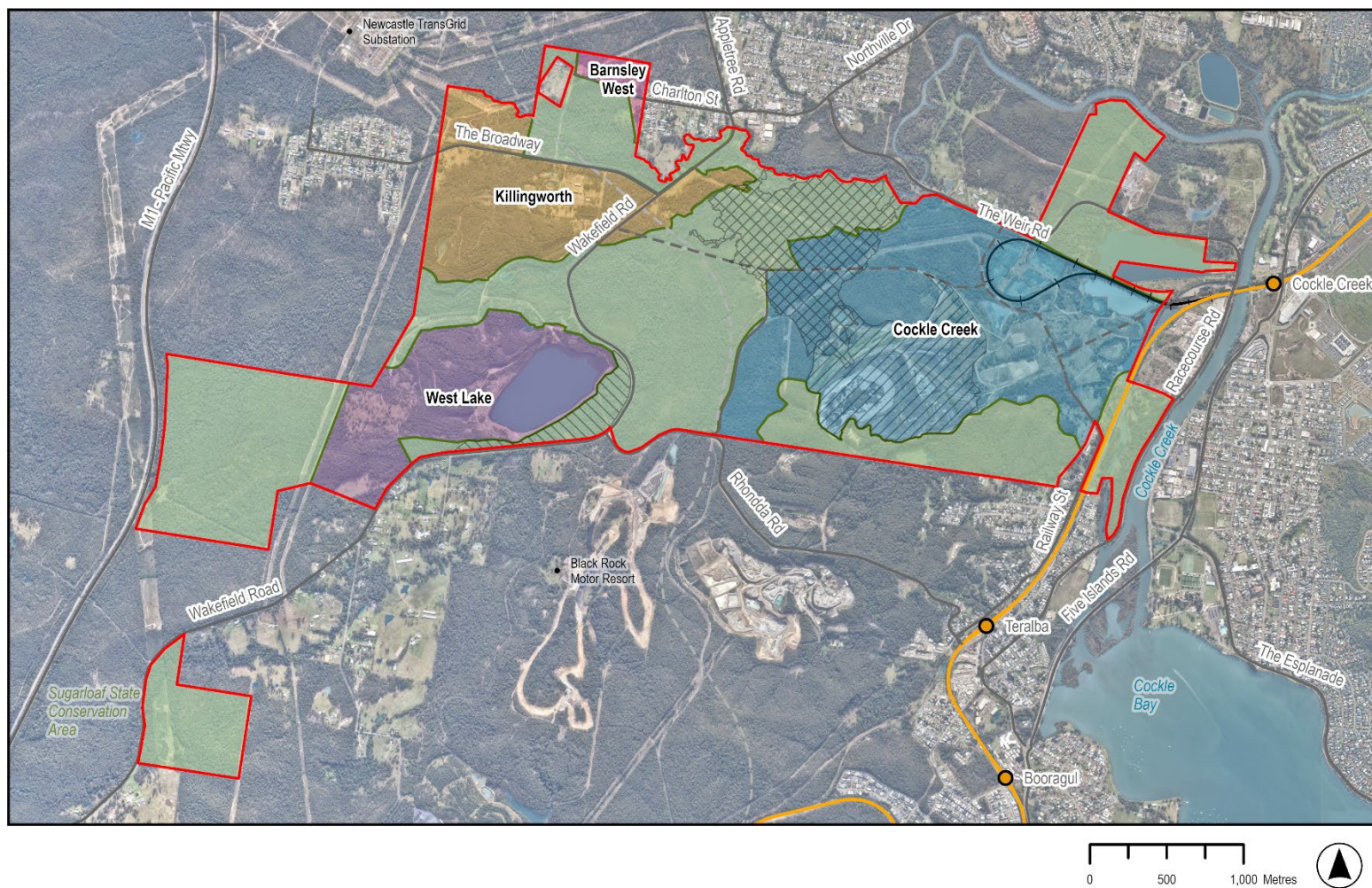
The rail loop to the northeast has been retained, along with most of the local roads within the Precinct. The six sub-precincts are shown in Figure 7-2 and the land use and areas summarised in Table 8-1

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Figure: Site plan including Sub precincts and Further Investigation Areas

Figure 7-1 Structure plan (GHD, June 2026)



Legend

▭ Precinct boundary

● Railway station

— Rail loop

— Railway line

— Road

- - - Road - potential

- - - Indicative Lower Hunter Freight Corridor

▭ Barnsley West (light Industrial)

▭ Cockle Creek (employment)

▭ Killingworth (employment)

▭ West Lake (mixed use)

▭ Environmentally constrained areas

▭ Further investigation area - dam break inundation zone

▭ Further investigation area - tailings storage facility

▭ Further investigation area - West Lake

Figure 7-2 Structure Plan with sub-precincts

Table 7-1 Land Use Analysis

Sub-precinct	Name	Land Use (Typology)	Total land area (ha)	Road Contingency	Allowance for open space
1	Cockle Creek	Circular Economy, Waste and Recycling; Industrial and Advanced Manufacturing; Data Centres and Digital Infrastructure; Freight Logistics, Distribution and Supply Chain Hub; Energy Transition & Renewable Infrastructure	299	15%	8%
2	Killingworth	Circular Economy, Waste and Recycling; Industrial and Advanced Manufacturing; Data Centres and Digital Infrastructure; Freight Logistics, Distribution and Supply Chain Hub; Energy Transition & Renewable Infrastructure	124	20%	8%
3	West Lake	Community, Recreation, Agriculture and Residential	126	25%	30%
4	Teralba Southgate	Residential mixed use	7	10%	0%
5	Barnsley West	Light industrial / commercial	11	10%	20%
6	Further investigation areas	Subject to Further Investigation	621	-	-
	Totals		1188		

8 Flooding and Water Cycle Management Strategy

8.1 Flood Management Strategy

Flooding presents a significant constraint to future development within the Precinct and should be considered as a key input to the Structure Plan. The preferred approach is to avoid development within the most flood-sensitive areas, including areas defined as floodways and important flood storage areas, wherever practicable.

Where avoidance is not practical, there may be opportunities for targeted development within lower constraint areas of the floodplain, supported by appropriate flood mitigation measures. In practice, a combination of land use planning, structural mitigation and non-structural management measures is likely to be required to balance development outcomes with acceptable flood risk. The suitability and effectiveness of these measures should be confirmed through detailed flood modelling and concept design during subsequent planning stages.

8.1.1 Regional Detention Basins

Regional detention basins may reduce downstream flood levels by temporarily storing runoff during major rainfall events and releasing flows at a controlled rate following the flood peak. This approach primarily can mitigate increases in peak flow runoff associated with future urban development (refer Section 8.2.1) or it may be used to manage the existing catchment flows from Cockle Creek and Diega Creek upstream of the Precinct. Either way, the approach manages the flood risk downstream to mitigate impacts of development or potentially unlock areas previously at risk of flooding.

The effectiveness of detention basins depends on their size and location within the catchment. Due to the dispersed nature of the proposed development areas and the varying topography, a distributed basin network may be more practical than a single regional basin when directly mitigating increases in peak flow from development (also refer Section 8.2.1). Where detention basins are used to offset and unlock downstream development, areas such as the Final void pit or along (offline) from the main creek lines provide opportunities.

8.1.2 Floodway Corridors

The provision of dedicated floodway corridors may improve flood conveyance through the Precinct and reduce the hydraulic impacts associated with future development for local and possibly regional scale flooding. Measures may include localised channels, diversion channels or the preservation of broad overland flow corridors through the development area.

Maintaining flood conveyance is particularly important where development may encroach into flood risk areas, as reducing the available flow area may increase flood levels in other areas. Future development planning stages should therefore consider retaining sufficient floodway corridors through the Precinct to preserve floodplain connectivity while balancing development outcomes (refer Figure 8-1).

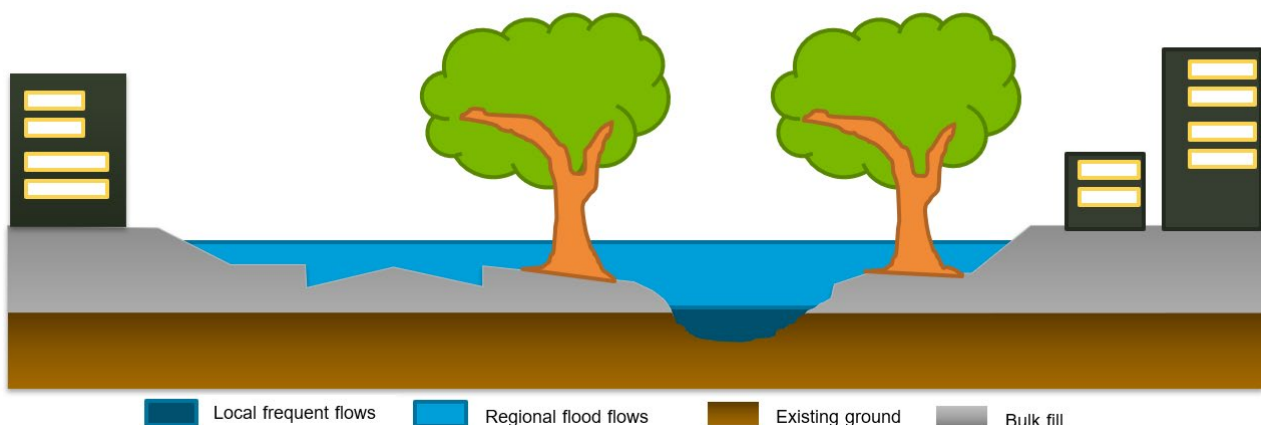


Figure 8-1: Typical floodway corridor example

8.1.3 Bulk Filling and Flood Storage Offsets

Bulk filling is the most common approach used to facilitate development within flood-prone land by raising finished surface levels above the adopted flood planning level. While this protects development from flooding, it also displaces floodwater by reducing the available floodplain storage.

The magnitude of flood impacts is generally related to the volume of fill placed within the floodplain, although local floodplain geometry and hydraulic behaviour also influence the response. Where flooding is predominantly characterised by flood storage, the impacts of bulk filling may be partially offset through the creation of compensatory flood storage elsewhere within the floodplain (refer Figure 8-2).

Compensatory flood storage is typically created by excavating an equivalent volume below the design flood level to offset the storage lost through filling. However, the effectiveness of this approach depends on the hydraulic characteristics of the floodplain and the location of the offset storage.

There may be limited opportunity to create meaningful compensatory storage where groundwater levels are shallow, as excavated areas may become permanently filled with groundwater and provide little additional flood storage during flood events. In these circumstances, alternative flood mitigation measures or strategic use of other floodplain areas may be required.

Where development encroaches into floodways, flood storage offsets alone are unlikely to mitigate flood impacts as these areas are primarily important for conveying flood flows rather than storing floodwater. In these locations, maintaining or replacing flood conveyance through dedicated floodway corridors or diversion channels may provide a more appropriate management approach.

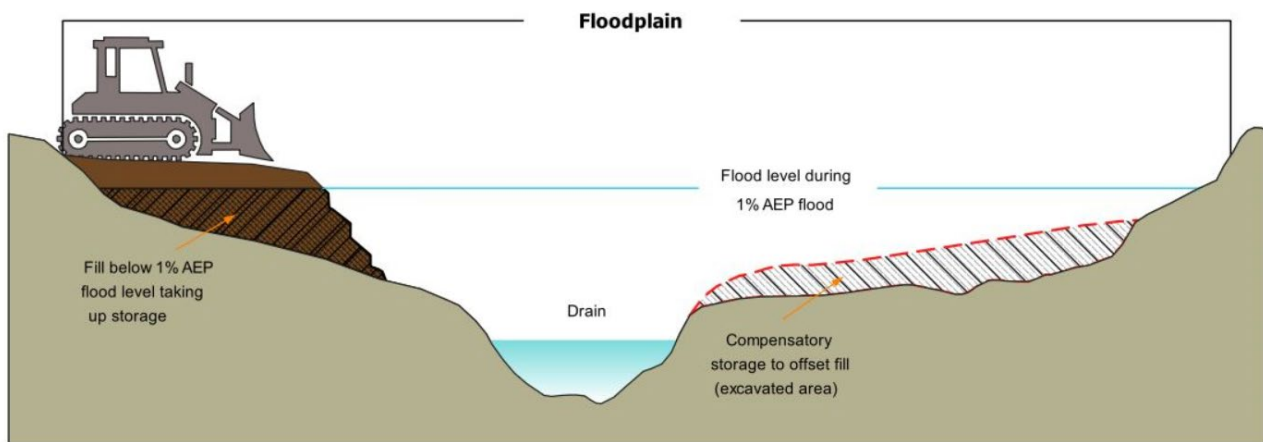


Figure 8-2: Conceptual Floodplain Filling and Flood Storage Offset Arrangement

8.1.4 Non-Structural Management Measures

In some circumstances, structural mitigation measures alone may not eliminate all flood impacts associated with future development. Non-structural measures may therefore form part of an integrated flood management strategy where residual impacts remain.

Such measures may include land use planning controls, restrictions on future development within flood-sensitive areas, emergency management planning, or, where justified, compensation arrangements where changes in flood behaviour cannot be reasonably avoided.

Overall, future development within the Precinct is likely to require a combination of flood storage management, preservation of flood conveyance, detention basins, careful optimisation of development footprints and appropriate non-structural management measures. The final mitigation strategy should be informed by detailed hydraulic

modelling to demonstrate that flood impacts can be managed to an acceptable level while maximising development opportunities.

8.2 Stormwater Management Strategy

8.2.1 Stormwater Detention

Stormwater detention will be required for development within the Precinct, as the increase in impervious surfaces will lead to increased runoff being generated under developed conditions.

As part of the proposed stormwater management approach, stormwater detention will be required to be incorporated at either an on-lot or sub-precinct scale to ensure post-development discharge rates do not exceed pre-development conditions. Stormwater detention prevents urban flash flooding, protects local infrastructure and reduces erosion. Opportunities for detention are to:

- Provide on-lot on site detention (OSD) through rainwater tanks (or storage tanks for larger allotments) to reduce runoff at the source, reducing peak flows entering the downstream network. This also benefits the council piped network as it reduces the pipe size shown against catchments. On lot OSD design must be in line with Council DCP guidelines outlined in Section 3.1.2. Stormwater detention of roads and public infrastructure or open space would be required to be managed separately by Council where on-lot detention is provided.
- Manage detention via a precinct or sub-precinct scale which would involve providing regional basins at the end of sub-catchments to provide detention for all upstream development. This type of system would manage cumulative catchment flows, providing a centralised detention function that supports efficient conveyance and reduces reliance on oversized pipe infrastructure. Refer to Section 8.1.1. Stormwater detention of roads and public infrastructure or open space would be incorporated into a regional basin detention strategy.

The advantages and disadvantages to each detention strategy should be considered when determining the best option for development within the Precinct.

- Rainwater tanks should be chosen when there is limited space available but moderate to low runoff attenuation required. Rainwater tanks offer the easiest opportunity for reuse on site and have fairly simple maintenance requirements, however they are normally sized for smaller events and may not effectively manage extreme storm conditions or large catchment areas.
- Storage tanks are usually designed for larger runoff volumes than rainwater tanks and can more effectively manage higher peak flow reduction. Tanks can be combined with a rainwater harvesting strategy to provide reuse on site. Tanks are usually suited to larger development allotments due to high design and cost requirements as well as complex maintenance regime.
- Regional detention basins offer an economical solution to large catchment areas which can achieve high peak flow reduction. Regional basins attenuate peak flows but do not largely reduce total runoff volume. Stormwater infrastructure also is required to be sized for higher flows up to the detention basin location.

The suitability of chosen detention infrastructure and management will be dependent on the type and staging of development as well as landform and number of discharge locations. It is anticipated that different detention options will be suitable for different sub-precincts. There may be alternate uses for water captured onsite such as reuse opportunities from rainwater tanks which will further reduce discharge to waterways and be aligned with Lake Macquarie City Council's principle to maximise water reuse.

Consideration should also be given to ownership and maintenance requirements. Onsite detention options are normally owned and maintained privately, whilst regional basins are usually public assets. Maintenance burden of these assets should be considered when determining the most suitable option.

Regardless of detention approach, stormwater quantity will be required to be managed prior to discharge to the receiving environment in conjunction with Council.

When quantifying stormwater detention volumes and identifying potential basin locations, the interaction with the mainstream floodplain should also be assessed. As shown in Figure 8-3, most of the 1% AEP flood extent is within

Developable Land subject to Further Investigation. Impervious surfaces within West Lake, Cockle Creek, Killingworth and Barnsley West sub-precincts encroach small areas of the 1% AEP extent and therefore no widespread impacts are expected.

Further assessment will be required for the West Lake sub-precinct, where an existing lake is located within the proposed development footprint. Decisions regarding whether the lake is retained, modified or filled will influence both the available floodplain storage and the Water Management Strategy for the site, including the location, volume and configuration of any required detention infrastructure.

The Killingworth sub-precinct also includes areas mapped as floodway, where development is generally discouraged due to the need to maintain flood conveyance. Future design stages should therefore seek to avoid or minimise encroachment into these areas where practicable.

No flood modelling has been conducted as part of this assessment, and earthworks volumes are not yet available. It is therefore not currently possible to quantify the volume of floodwater displaced or confirm the required size, location and configuration of detention basins. These matters should be assessed during subsequent design stages once more detailed development layouts, earthworks information and stormwater management strategies are available.

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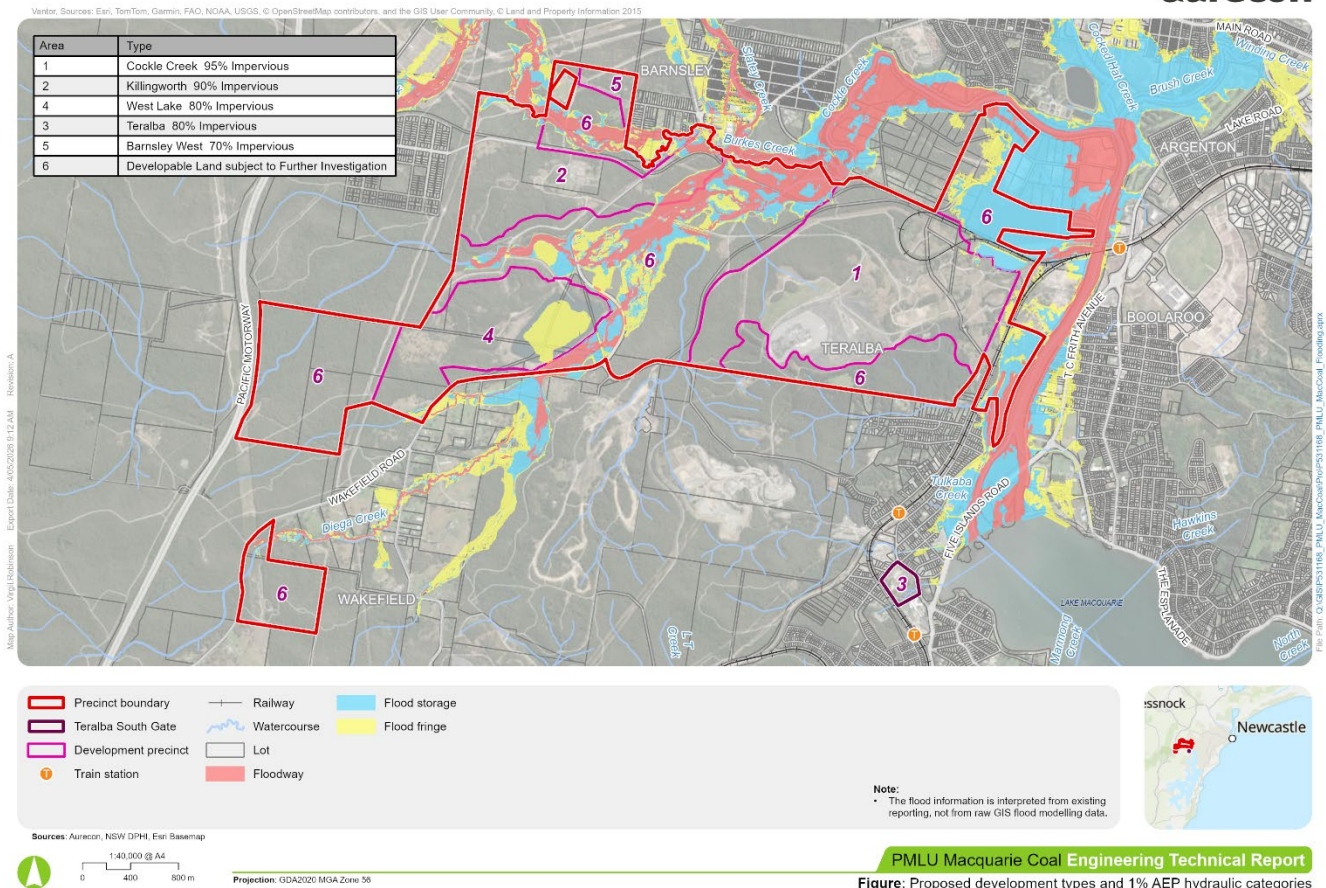


Figure 8-3 Percentage impervious surfaces of each sub-precinct and 1% AEP flood extent.

8.2.2 Stormwater Conveyance

The Stormwater Management Strategy will aim to ensure the protection and function of waterways. The Precinct will be designed to work with nature to use the existing landform to dictate the stormwater conveyance network.

The minor drainage system should be designed to capture and convey minor system (or nuisance flows) as per the LMCC DCP. The minor system flows are defined as localised flowing with an ARI of 2-5 years. The major drainage system is part of the public drainage system designed to convey the major storm flows.

Catchment runoff should be directed through a clearly defined pipe or channel network that conveys flows to designated treatment nodes before discharge to receiving environments. Major overland flow paths should be preserved if possible, ensuring they remain unobstructed and integrated into the Precinct layout to safely convey exceedance flows.

Drainage of stormwater within the Precinct will aim to minimise changes to sub-catchments boundaries and direct flows via existing drainage paths where possible to protect and maintain the hydrological regime of the receiving waterways. Formalisation of drainage within the Precinct will capture and direct water to stormwater treatment and detention prior to discharge to the natural water systems to mitigate impacts from increased runoff. Scour protection and erosion controls should be implemented upstream of discharge locations to natural waterways.

In addition, if no sub-precinct scale or end of pipe stormwater retention or treatment scheme is developed for a site, development should demonstrate that impervious areas are captured for reuse or managed on lot through infiltration and retention prior to discharge to the drainage system. Lake Macquarie City Council DCP indicates that only 10% of impervious areas on a site can be discharged directly to the receiving drainage system.

8.2.3 Natural Water Systems

The stormwater strategy recommends development considers natural water systems including those within the Precinct and those downstream. In alignment with Section 3.1.3 and the Lake Macquarie City Council DCP, natural water systems including waterways and estuaries should be protected and not adversely impacted by development. This will be done by retaining creeks and natural drainage pathways within their vegetated riparian zone (VRZ). Preliminary mapping of the VRZ for the Precinct is shown in Figure 3-17 and further discussed in Section 3.2.9.

Natural systems including all natural watercourses through the Precinct should be maintained in a natural state in alignment with the Controlled Activity Approvals on Waterfront Land. This will require buffer distances from waterways to be in place to maintain the riparian vegetation and retain ecological processes. Proactive rehabilitation of waterways is also recommended where development has the potential to exacerbate an issue or where additional vegetation buffers between development and the waterway is required.

8.3 Water Quality Management Strategy

The Water Management Strategy adopted will be dependent on what targets are required to be achieved based on existing/ historical land use. For brownfield development, the strategy will be required to meet best practice pollutant load reduction targets. For greenfield development, the strategy will be required to meet no net increase to pollutant loads (subsequently referred to as requiring assessment of Neutral or Beneficial Effects (NorBE)). Water quality targets are stipulated by the Lake Macquarie City Council DCP and estuary health nutrient loading targets outlined in Section 3.1.1. Simple modelling using eWater MUSIC X software was conducted on a number of 1 ha MUSIC models to understand preliminary spatial requirements for specific WSUD measures. A more detailed MUSIC model and treatment train should be developed for the Precinct to further refine and determine the strategy.

8.3.1 Water Quality Targets

Around half of the developable area within the proposed structure plan is stipulated by the Lake Macquarie City Council DCP pollutant reduction targets outlined in the second column of Table 3-4. However, there are sections of Cockle Creek, Killingworth, West Lake and Barnsley West sub-precincts which are zoned as conservation and rural and thus NorBE is required. These areas are marked up in Figure 8-4 and developable areas from the structure plan which consist of rural and environmentally constrained land are summarised in Table 8-1.

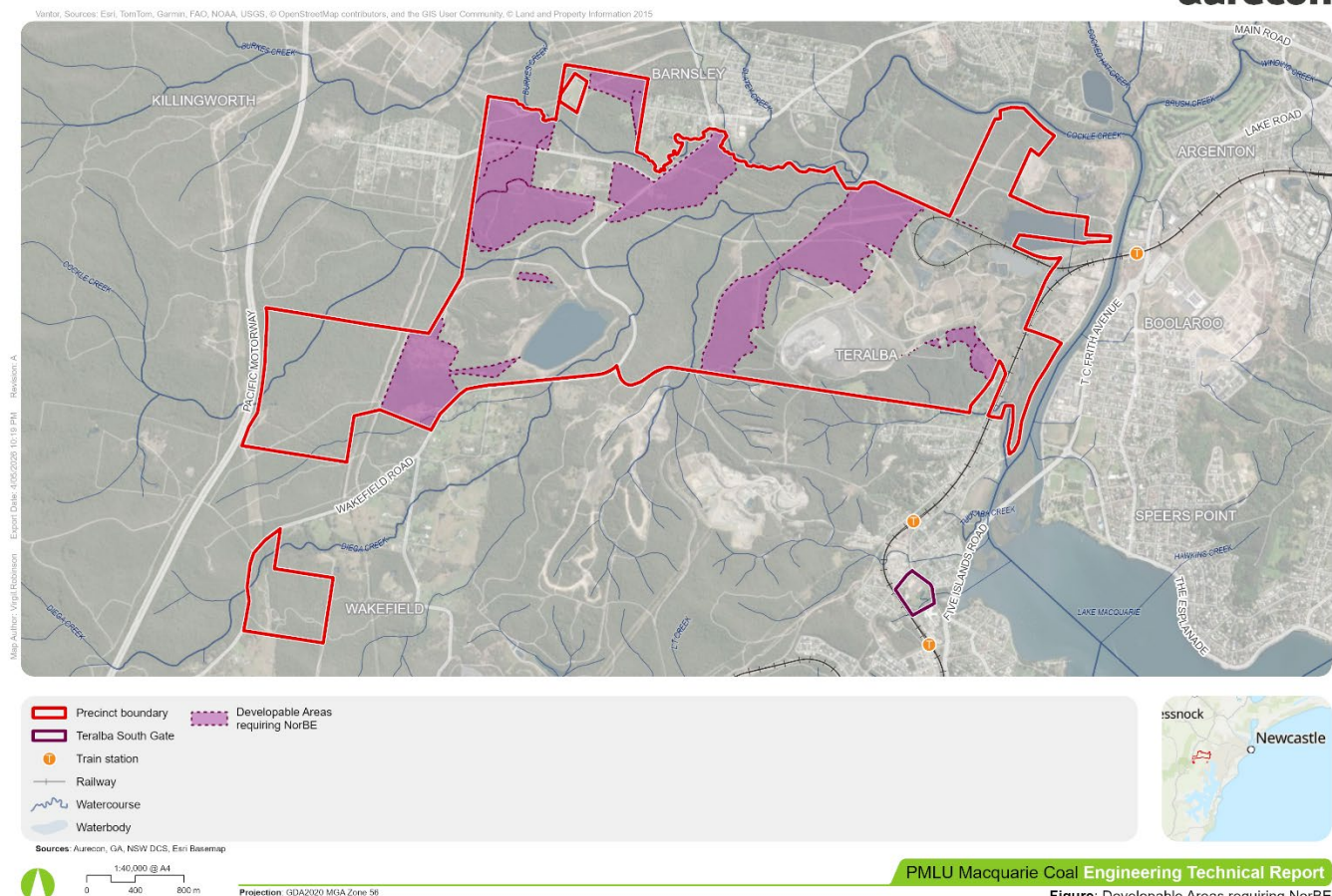


Figure 8-4 Developable areas requiring NorBE

Table 8-1 Developable area

Land use code	Description of land use	Developable Areas (ha)	% of developable area
C2	Environmental Land use	198	40%
C3	Environmental Management	33	7%
RU2	Rural Landscape	14	3%

Basic MUSIC modelling was conducted to estimate the current site performance and Table 8-2 summarises pollutant loads running off the catchment for loamy sand and silty clay loam catchments. This modelled performance provides an estimated target for urban (residential, commercial, industrial areas) and dense rural residential development. It is important to note that ground investigations are needed to confirm exact soil type of Precinct and further detailed modelling of catchments is required to determine accurate targets within greenfield areas.

Table 8-2 Preliminary pre-development pollutant loads for 1hectare greenfield areas based on two example soil types

	Loamy sand	Silty clay loam
Total Suspended Solids (kg/yr)	24.5	49.7
Total Phosphorus (kg/yr)	0.1	0.1
Total Nitrogen (kg/yr)	0.9	1.2
Gross Pollutants (kg/yr)	0	0

8.3.2 Water Sensitive Urban Design Treatment Train

A combination of on-lot and precinct scale SQIDs have been proposed to support the Master Plan and deliver water quality targets adopted to protect the highly valued environment downstream of the Precinct and respond to the local constraints. The key proposed WCM measures are summarised in Table 8-3 and further discussed in Sections 8.3.3 to 8.3.5. An indicative treatment train is shown in Figure 8-5. This treatment train has been considered suitable for most circumstances but would need to be refined and modelled in detail in subsequent planning and design stages.

Table 8-3 WSUD Measures proposed

WSUD Measure	Description
Rainwater tanks	Rainwater tanks to collect water from roofs and supply non potable water demand such as landscape irrigation.
Gross Pollutant Trap (GPT)	A GPT is proposed for larger allotments to remove litter and sediment from stormwater runoff prior to discharge to the street drainage system. A GPT may also be provided at the end of pipe and may be more suitable for upstream residential development or development of smaller allotments where on-lot application is not feasible. For larger allotments and industrial and commercial development, a GPT should be proposed on lot to manage litter prior to discharge to the drainage system.
Biofiltration basins	Biofiltration basins are proposed as on-lot and end-of-pipe treatment measures as the preferred treatment option. These basins are able to meet the water quality targets within a smaller footprint when compared with constructed wetlands. Additionally, Lake Macquarie City Council has indicated a preference towards biofiltration basins in recent years.
Wetlands	In some locations, constructed wetlands may be proposed in place of biofiltration to achieve water quality targets. Constructed wetlands are more suitable for end-of pipe treatment and can be utilised within integrated landscape planning as a visual amenity, linking public open space with stormwater management infrastructure.
Street Trees	Street trees should be considered where biofiltration basins or other infrastructure is unable to be integrated into the design. Street trees allow for passive irrigation of trees through rainwater captured and diverted to the tree pits along roads or other impervious surfaces. Due to maintenance requirements for individual trees, preference should be given to consolidated biofiltration in place of street trees where possible.

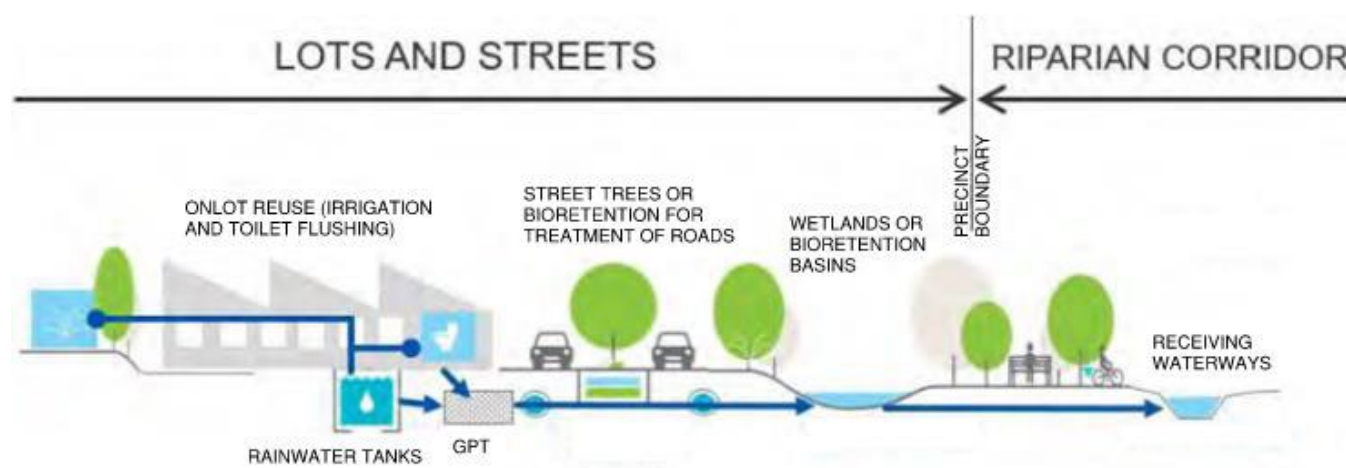


Figure 8-5 Indicative WSUD treatment train proposed for the Precinct

8.3.3 On-lot Treatment

Rainwater Tanks

For the proposed water quality management strategy roof runoff was assumed to be captured in rainwater harvesting tanks and reused for non-potable uses such as landscape irrigation and/or toilet flushing.

Preliminary MUSIC modelling was conducted to model a typical treatment train with rainwater tanks paired with biofiltration basins for treatment, see Figure 8-5. The roof area was considered 60% of the developable area which was routed to the rainwater tanks with the remaining area going through a GPT. Ultimately, rainwater tank sizing would be further informed on an individual lot basis, dependent on the downstream infrastructure and achievement of water quality and detention targets for the site and Precinct. Further development of the potential onsite reuse will also determine sizing and capture rates for the rainwater tanks. The tanks could be used for irrigation and other non-potable uses such as toilet flushing.

Gross Pollutant Traps

On lot gross pollutant traps (GPTs) are proposed to be incorporated to provide source-control treatment for individual larger allotments. These devices capture coarse sediments, litter, organic debris, and other gross pollutants before runoff enters the downstream drainage network, reducing pollutant loads conveyed to precinct-scale treatment assets. This helps maintain performance and longevity of downstream end of pipe infrastructure. On-lot GPTs also support compliance with water quality objectives by distributing treatment across the catchment rather than relying solely on end-of-pipe systems. Their placement within accessible locations ensures ongoing maintenance can be undertaken efficiently, preserving treatment effectiveness over the life of the development.

A nominal GPT unit was adopted for the basic modelling conducted for the removal of gross pollutants, sediment and some nutrients. If other units are proposed at a later stage, then analysis to confirm equivalent performance would be required. While on-lot measures support flexible delivery time frames, there is a high risk of on-lot measures failing due to poor management and maintenance practices in private ownership.

8.3.4 Road Treatment

If no end of pipe treatment is available and water quality is being managed on lot, public roads should have an independent treatment management approach. The approach to on street treatment would be via bioretention basins where size and routing of flows to larger downstream locations is possible. Where space is not available or for smaller portions of roads, street trees could be considered for implementation along public roads in the Precinct. The replacement of natural vegetated landscape with impervious areas increases urban temperatures and stormwater runoff. Street trees will be passively irrigated by directing stormwater into their tree pits, allowing them to naturally absorb moisture and helping to create a cooler, greener streetscape. Figure 8-7 is an example of a biofiltration street tree.

Preliminary MUSIC modelling assumed biofiltration street trees were assumed to be incorporated at a ratio of approximately 16 trees per hectare of development.



Figure 8-6 Example of bioretention street tree (Blacktown City Council, 2026)

8.3.5 End of Pipe Treatment

The preferred approach to achieving the required pollutant reduction targets is through an end-of-pipe treatment train incorporating a biofiltration basin and GPT or a constructed wetland and GPT. The suitability of these systems will be dependent on catchment characteristics, potential of inundation from flooding or tidal flows, available land and integration with landscape design. As part of the proposed strategy, wetlands and larger biofiltration basins will be located within public open space areas adjacent to the riparian corridors and positioned at local catchment low points. These end-of-pipe systems provide robust treatment of sediments, nutrients, and gross pollutants while also offering opportunities for habitat creation and landscape amenity. Where these assets are intended to become Council-owned infrastructure, appropriate drainage and access easements will be required to ensure long-term maintenance and operational functionality.

Preliminary MUSIC model has been undertaken to size bioretention basins and wetlands. To achieve council pollutant reduction targets and nutrient reduction targets the bioretention basins need to be at least 1% per hectare of development and wetlands will need to be 5% per hectare of development footprint when paired with end of pipe GPTs. These GPTs work in the same way as on lot GPTs but are bigger for higher flows.

9 Integrated Spatial Planning Implications

9.1 Flooding Constraint Classification Framework

The Precinct has been classified into high, moderate and lower flood constraint areas based on the strategic flood assessment undertaken for the Precinct. The classification provides a framework for identifying areas where flooding is likely to constrain future development and where additional investigation may be required.

The flood constraint classification is summarised in Table 9-1 and spatially presented in Figure 9-1. The framework supports an avoidance-first approach, directing future development towards lower constraint areas while recognising that targeted encroachment into parts of the floodplain may be feasible where supported by appropriate flood mitigation measures and detailed assessment. This approach seeks to balance development opportunities with the need to preserve floodplain function and manage flood risk.

The characteristics and planning implications of each flood constraint category are discussed in the following sections.

Table 9-1: Constraint and developability levels

Constraint level	Developability	Description
High Flood Constraint Areas	Constrained	Major creek and drainage lines, high hazard PMF flooding, 1% AEP floodways. Avoid sensitive and high occupant development in these areas
Moderate Flood Constraint Areas	Potentially Developable	Low hazard flooding (PMF and 1% AEP). Flood fringe and storage areas. Potential reclamation areas. Further flooding and drainage assessment required
Low Flood Constraint Areas	Developable	Outside identified regional flooding. Typically, standard engineering management of local flooding and drainage. Development less constrained but still required assessment

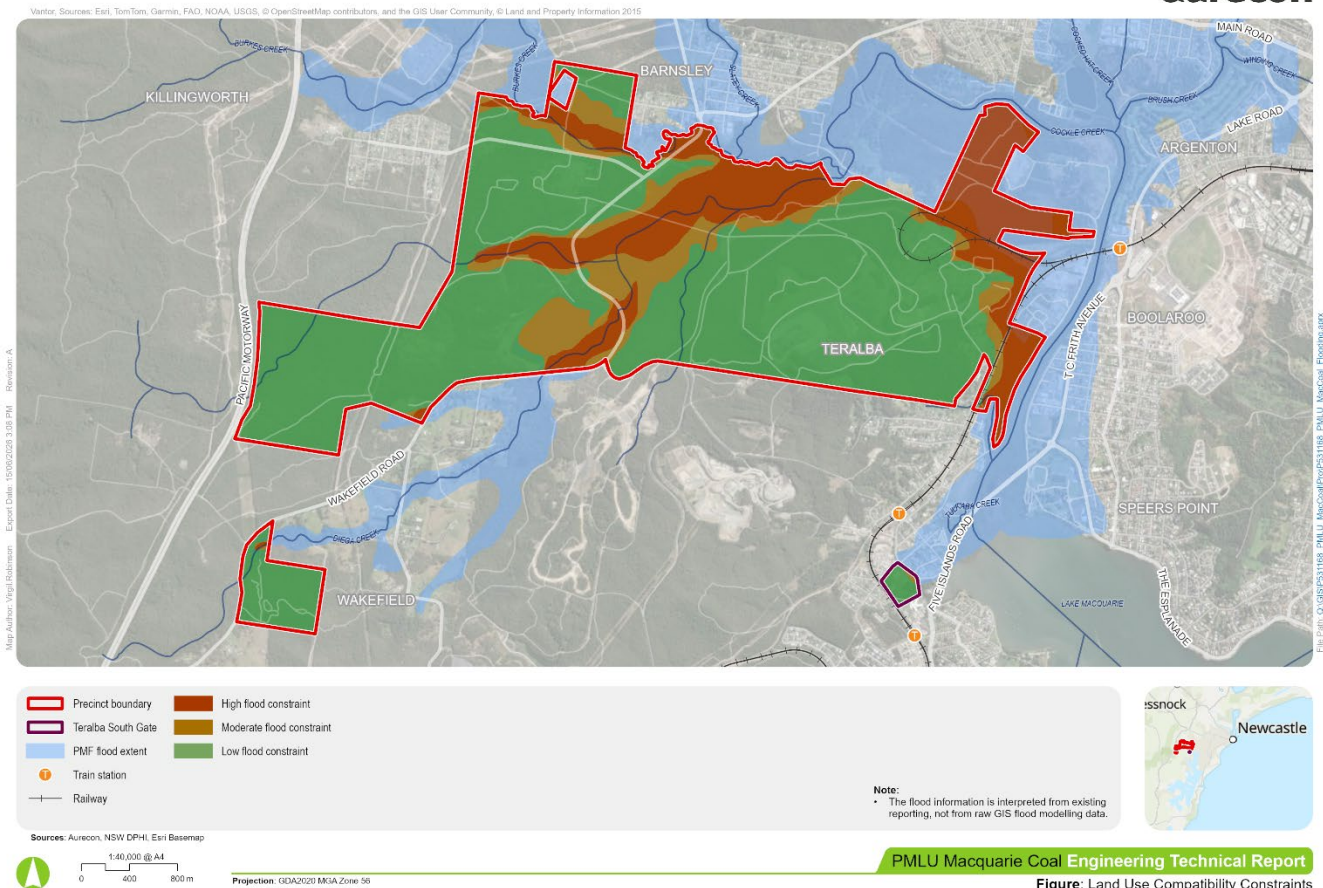


Figure 9-1 Land Use Compatibility Constraints

9.1.1 High Flood Constraint Areas

The Cockle Creek, Burkes Creek and Diega Creek floodplain corridors represent the primary flood constraints within the Precinct. These waterways and their associated floodplains perform important flood conveyance and flood storage functions and accommodate significant flood flows during regional flood events.

Development within these corridors has the potential to:

- increase flood levels and flood risk to surrounding land;
- reduce available flood storage;
- obstruct flood conveyance pathways;
- increase flood isolation and evacuation risk; and
- require substantial flood mitigation measures and associated infrastructure upgrades.

The confluence of Cockle Creek, Burkes Creek and Diega Creek is of particular importance, as it represents a hydraulically complex area where multiple flood mechanisms interact and floodplain connectivity is critical to the overall performance of the catchment. Development within these areas should be minimised unless supported by detailed flood investigations demonstrating that flood behaviour and flood risk can be appropriately managed.

Accordingly, these areas are generally better suited to land uses that are compatible with periodic inundation, including drainage functions, open space, environmental conservation, utility corridors and other low-intensity uses that do not significantly alter floodplain behaviour.

9.1.2 Moderate Flood Constraint Areas (Further Investigation Areas)

Moderate constraint areas generally include land adjacent to high hazard floodplain corridors, consist of low hazard flooding, low-lying drainage depressions, and areas with poor drainage gradients.

While these areas may ultimately be suitable for development, they are likely to require further investigation such as detailed flood assessment, drainage investigations and potentially additional engineering measures to demonstrate that flood risk and drainage constraints can be appropriately managed. Development within these areas should carefully consider flood immunity, drainage feasibility, emergency access and the potential impacts of earthworks on flood behaviour.

9.1.3 Lower Constraint Areas

Areas located outside the mapped flood extent and away from major drainage corridors generally represent lower flood-related constraints and potentially are subject to more local flooding and drainage issues and may provide greater opportunities for future development. These areas are likely to require less extensive flood mitigation infrastructure and may be more suitable for early activation within the Master Plan.

Notwithstanding their comparatively lower flood risk, future development should still be supported by detailed flood assessment, drainage feasibility assessment and access and evacuation assessment to confirm development suitability. The identification of lower constraint land through the Master Plan should therefore be regarded as indicative only and does not remove the requirement for subsequent flood and drainage investigations during future planning and development approval processes.

9.1.4 Master Plan Implications

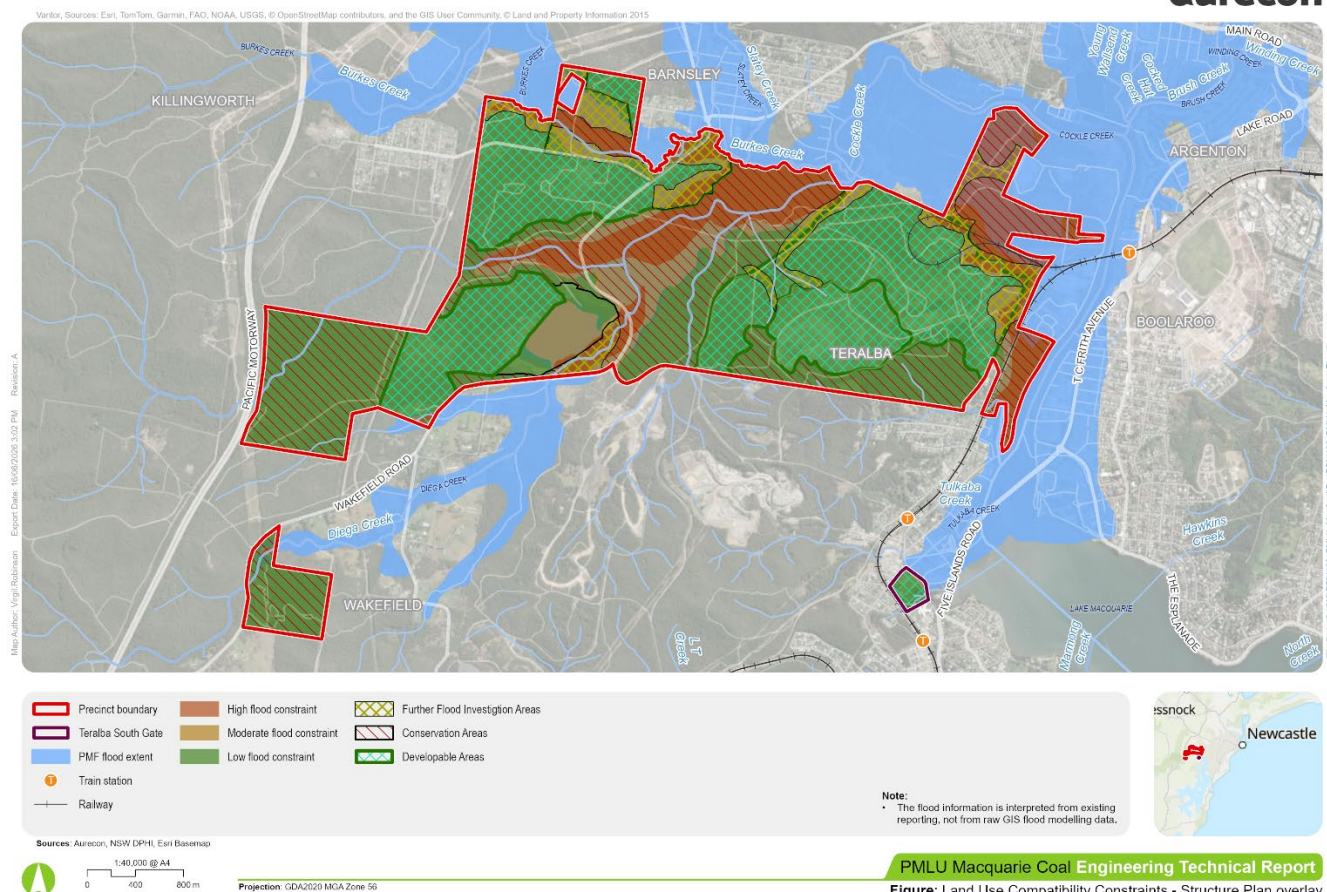
The flood constraint classification has been compared with the draft Master Plan to identify areas where flooding may influence future development and where additional investigations are required. The assessment is intended to guide strategic planning and should not be interpreted as confirming or preventing development within any particular area.

The review indicates that some development areas in the Master Plan extend into land identified as having moderate or high flood constraints. These areas will require further flood assessment to demonstrate that flood risk, drainage, emergency access and evacuation can be appropriately managed.

Conversely, some flood-constrained land may have the potential to support future development where appropriate flood mitigation measures are implemented. Measures such as detention basins, floodway corridors, channel improvements and flood storage management may reduce flood impacts and increase the amount of developable land, subject to detailed investigation.

One key aspect of the areas identified for further investigation is the encroachment into the dam failure inundation area. The concept is that the existing failure flood inundation area is based on current landform. Given the low risk of failure there is opportunity to control the failure pathway. This allows for the opportunity to reclaim a portion of the at risk area through an engineering mechanism.

Figure 9-2 presents the flood constraint classification overlaid on the Master Plan and identifies areas where further flood investigations may either support future development opportunities or confirm that existing flood constraints should be retained within the final Master Plan.



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 Figure: Land Use Compatibility Constraints - Structure Plan overlay

Figure 9-2 Land Use Compatibility Constraints – Master Plan overlay

9.2 Integrated Water Management Strategy

9.2.1 Impervious Assumption

With consideration of the low permeability of the site area (see Section 3.2.5) and the impervious percentages from the DCP in Table 9-2, a summary of impervious percentages was determined. These were based on the land use from the Master Plan for each sub-precinct. The adopted impervious % for the assessment is summarised in Table 9-3.

Table 9-2 Impervious % from DCP

Type of development	Impervious %
Single Allotment and Small Residential Subdivision	40% to 60%
Medium and High-Density Residential Development	60% to 80%
Large Greenfield Subdivisions	40% to 60%
Commercial and Industrial	80% to 100%
Infrastructure	50% to 100%

Table 9-3 Impervious % based on land use

Sub Precinct	Name	Industry Type	Land Area (ha)	Impervious %
1	Cockle Creek [Employment Core]	Heavy industrial	168.4	95%
2	Killingworth [Employment Core]	Light industrial	112.2	90%
3	West Lake [Mixed Use]	Tourism / resort	89.4	80%
4	Teralba Southgate [Mixed Use]	Residential (greenfield)	7	80%
5	Barnsley West [Industrial]	Light industrial / commercial	11	70%

9.2.2 Proposed Strategy

The proposed integrated Water Management Strategy has been developed to coordinate surface water across the entire Precinct and is shown in Table 9-4. The strategy adopts a whole-of-catchment approach, ensuring that each sub-precinct contributes to shared water quality, water quantity, and liveability objectives while responding to its specific land-use, topography, and hydrological characteristics.

Table 9-4 Integrated Water Management Strategy for each sub- precinct

Sub-precinct	Name	Land use type	Integrated water management	Further Investigations
1	Cockle Creek	Heavy industrial	- Sub-precinct drains in a north easterly direction towards Cockle Creek at Weir Road. - Combination of on lot and end of pipe WSUD measures to be considered. WSUD measures to be positioned at local catchment low points. - Small allowance for open space (Table 7-1) and thus preference for bioretention basins for treatment due to anticipated space restrictions. Bioretention basins estimated to be approximately 1.7 ha for the catchment size. However, this is dependent on entire treatment train. - Onsite detention provided and managed either on lot or sub-precinct depending on development staging. - There may be opportunity to reuse existing water management infrastructure including ponds in the north east for stormwater detention or treatment functions.	- Sub-precinct contains soil landscape classified as 'disturbed terrain'. Behaviour of this soil type is unknown. Further geotechnical investigation is needed. "Leaky" WSUD measures not proposed for aquifer recharge as there is uncertainty on groundwater levels. Further groundwater investigation is needed to determine whether infiltration could be provided to increase groundwater recharge opportunities. - Further investigation of stability, potential contamination and redesign opportunities for reuse of existing water infrastructure within this sub-precinct
2	Killingworth	Light industrial	- Sub-precinct drains towards the northwest towards Burkes Creek. - Combination of on lot and end of pipe WSUD measures to be considered. WSUD measures to be positioned at local catchment low points.	"Leaky" WSUD measures not proposed for aquifer recharge as there is uncertainty on groundwater levels. Further groundwater investigation is needed. - Further investigation of stability, potential

Sub-precinct	Name	Land use type	Integrated water management	Further Investigations
			- Small allowance for open space (Table 7-1) and thus preference for bioretention basins for treatment due to anticipated space restrictions. Bioretention basins estimated to be approximately 1.1 ha for the catchment size. However, this is dependent on entire treatment train. - Onsite detention provided and managed either on lot or sub-precinct depending on development staging. - There may be opportunity to reuse or repurpose existing water management infrastructure within the existing West Wallsend colliery for stormwater detention or treatment functions.	contamination and redesign opportunities for reuse of existing water infrastructure within this sub-precinct
3	West Lake	Tourism / resort	- Sub-precinct drains towards Cockle Creek in the north. - End of pipe treatment proposed. Location of WSUD measures to be positioned at local catchment low points. - Largest allowance for open space (Table 7-1). Opportunity for wetlands to be integrated as a key landscape feature. Wetlands estimated to be approximately 4.5 ha for the catchment size. However, this is dependent on entire treatment train. - The Pit Void Lake has already been identified for recreational use, and there is potential for it to serve a dual-purpose role for stormwater detention and social value.	"Leaky" WSUD measures not proposed for aquifer recharge as there is uncertainty on groundwater levels. Further groundwater investigation is needed. - Further investigation of stability, potential contamination and reuse of Pit Lake Void for stormwater detention purposes.
4	Teralba southgate	Residential (greenfield)	- Sub-precinct drains towards the southeastern drainage line of Cockle Creek. - No allowance for open space (Table 7-1) and developed as a standalone location and thus only on-lot measures are feasible. - Onsite detention managed on lot.	
5	Barnsley West	Light industrial / commercial	- Sub-precinct drains south towards Burkes Creek. - Combination of on lot and end of pipe WSUD measures to be considered. WSUD measures to be positioned at local catchment low points. - Moderate allowance for open space (Table 7-1) and thus preference for bioretention basins for treatment due to space restrictions. Bioretention basins estimated to be approximately 0.11 ha for the catchment size. However, this is dependent on entire treatment train.	"Leaky" WSUD measures not proposed for aquifer recharge as there is uncertainty on groundwater levels. Further groundwater investigation is needed.

Sub-precinct	Name	Land use type	Integrated water management	Further Investigations
6	Developable Land subject to Further Investigation	Subject to investigation	No WCM strategy proposed at this stage. To be developed following further investigation.	

9.3 Interface With Other Disciplines

The flooding, drainage and WCM assessment has several interfaces with other technical disciplines and planning activities that should be considered during refinement of the Master Plan and future development stages. Coordination between disciplines will be important to provide appropriate flood risk management and water cycle management while achieving the desired land use and infrastructure outcomes.

- **Transport and Staging** – The location and flood immunity of future road networks, bridge crossings and access routes should be coordinated with the flood assessment to ensure suitable access, evacuation and emergency response can be achieved throughout each stage of development.
- **Contamination and Rehabilitation** – Earthworks associated with bulk filling, flood storage offsets or drainage infrastructure should consider any contamination or rehabilitation requirements associated with former mining land or other disturbed areas.
- **Geotechnical** – Ground investigations should be aligned with geotechnical investigations for a coordinated approach to determining ground conditions, hydrogeology and soils.
- **Biodiversity and Open Space** – Floodways, flood storage areas and drainage corridors may also provide opportunities for open space and ecological connectivity. The preservation of these areas should be coordinated with biodiversity objectives and the broader open space network.
- **Utilities and Servicing** – The location and design of utilities, drainage infrastructure and essential services should consider flood risk, flood immunity and access requirements for long-term resilience and service continuity during flood events. Where existing utilities such as the Ampol fuel pipeline impact drainage and potential flood risk management strategies, lowering or adjustment of the fuel pipeline would unlikely be feasible. As such future drainage and potential flood mitigation would require consideration of this constraint and be designed accordingly.
- **Statutory Planning** – Future planning controls, development staging and land use zoning should be informed by the identified flood constraints and consistent with applicable flood planning policies and Council requirements.

The flood and WCM assessment should continue to be refined alongside these disciplines as the Master Plan progresses. In particular, the final Master Plan, transport network, development staging and flood mitigation strategy should be developed in an integrated manner so that flood risk, infrastructure planning and land use outcomes are appropriately balanced.

10 Master Plan Objectives and Performance Criteria

The objectives and performance criteria in this section have been developed to guide future land use planning, infrastructure provision and development within the Precinct. They are informed by the flooding and water cycle management investigations and seek to ensure that future development responds appropriately to flood risk, drainage constraints, water quality considerations and broader water management outcomes. The objectives establish the desired strategic outcomes for the Precinct, while the performance criteria provide a framework for assessing future development proposals and identifying where further investigation or management measures may be required.

10.1 Proposed Objectives

The proposed objectives establish the desired outcomes for flooding and water cycle management within the Precinct, having regard to the opportunities and constraints identified through the technical investigations.

- Direct future development towards areas with comparatively lower flood risk and drainage constraints.
- Protect the floodway and flood storage functions of the waterways and associated floodplain areas.
- Avoid intensification of development within areas subject to significant flood hazard or life safety risk.
- Support a movement network that considers flood resilience to future climate risk, emergency access and flood isolation risks.
- Ensure future development is compatible with the flood risk (hazard, function) and does not increase flood risk to surrounding land, infrastructure or communities.
- Establish a framework for future detailed flood risk, drainage and emergency management assessments to demonstrate any management and mitigation measures are suitable for developing in areas of the floodplain.
- Recognise drainage feasibility as a key consideration in determining developable land and future staging.
- Ensure future development incorporates drainage infrastructure that is compatible with downstream hydraulic controls, including flood and tidal influences.
- Implement Water Sensitive Urban Design (WSUD) measures to achieve urban water cycle management objectives
- Minimise water quality and quantity impacts from surface water runoff to waterways and sensitive receiving environments including GDEs, sensitive wetlands and downstream estuarine environments
- Ensure compliance with Lake Macquarie Council Development Control Plan including OSD and stormwater pollution reduction targets
- Ensure compliance with NSW Estuary Health Risk Dataset which defines nutrient load targets for estuary catchments
- Preserve groundwater recharge where feasible and minimise impacts to groundwater levels and quality
- Implement proactive construction phase sediment and erosion control measures are implemented to protect downstream sensitive receptors.

10.2 Proposed Performance Criteria

The proposed performance criteria provide guidance for future planning and development assessment and identify the matters that should be addressed to achieve the intended outcomes for the Precinct.

- Flood planning levels (including freeboard) are to be as per Council requirements and include allowance Climate Change to year 2100.

- Development is to demonstrate that it will not unreasonably increase flood levels, flood extents, flow velocities or flood hazard on surrounding land outside of the precinct.
- Development within flood-affected or drainage-constrained land is to demonstrate that flood risk and drainage constraints are acceptable for the development type or have been mitigated to achieve an acceptable level of impact.
- Future road, access and movement network planning is to consider flood risk, flood immunity to align with level of service required and flood isolation risks.
- Flood-constrained land identified through the Master Plan is generally not intended for intensive and/or sensitive development, evacuation-sensitive land uses or critical infrastructure unless supported by a detailed assessment and suitable management measures are implemented.
- Future development applications undertake detailed flood, drainage and emergency management investigations.
- Development must achieve applicable stormwater quality targets including pollutant load reduction and nutrient load limits relevant to the Lake Macquarie catchment.
- High-priority groundwater dependent ecosystems must be protected in accordance with the applicable policy framework.
- Development must demonstrate an integrated water cycle and WSUD response that minimises hydrological and water quality impacts on downstream sensitive receivers.
- WSUD measures must be designed to treat frequent storm events, including flows up to and including the three-month ARI event
- Development must maintain, to the extent practicable, natural flow paths, sub-catchment function and flow direction and must not rely solely on piped or heavily engineered stormwater systems where this would increase erosion, scour or downstream impacts.
- Public domain, landscape and street design must integrate WSUD assets to support urban cooling, biodiversity, amenity and place identity.
- Construction-phase sediment and erosion controls must be implemented in accordance with the Blue Book

11 Staging Triggers and Later-Stage Investigations

The flooding and water cycle management investigations undertaken for the Master Plan have identified a range of opportunities, constraints and areas requiring further investigation. Given the strategic nature of the Master Plan, detailed engineering investigations have not been undertaken at this stage.

Future planning, subdivision and development proposals will require progressively more detailed investigations to demonstrate that flooding, drainage, water quality, groundwater and related environmental constraints have been appropriately considered. The staging triggers outlined below identify the circumstances under which additional investigations, modelling, design development and consultation with relevant authorities may be required.

11.1 Triggers Table

The Master Plan has identified a number of flooding and water cycle management matters that will require further investigation as planning and development progresses. The trigger framework presented in Table 11-1 identifies the circumstances under which additional technical assessment, modelling, design development or authority consultation may be required. The framework is intended to guide future planning and development assessment processes and ensure that investigations are undertaken at a level of detail commensurate with the scale, location and risk profile of the proposed development.

Table 11-1 Trigger framework

Trigger Condition	Affected Area	Flooding Requirement	Water Cycle Management Requirement	Timing	Responsible Discipline	Master Plan Consequence if Unresolved
Development proposed within or adjacent to flood-affected land	Floodplain areas associated with Cockle Creek, Burkes Creek, Diega Creek, and tributaries	Detailed flood impact assessment and confirmation of applicable flood planning levels	Demonstrate drainage feasibility and stormwater management approach	Development Application	Flooding / Civil	Development footprint, land use or staging may require revision
Significant earthworks, filling or landform modification proposed	Areas subject to regrading or landform change	Assessment of impacts on flood behaviour, flood storage and flood conveyance	Assessment of impacts on drainage patterns and runoff characteristics	Concept Design / Development Application	Flooding / Civil	Earthworks extent may require modification
Development requiring new flood-sensitive road connections or crossings	Transport corridors and access routes	Access, flood immunity and flood isolation assessment	Assessment of drainage impacts on transport infrastructure	Transport Planning / Development Application	Flooding / Transport / Civil	Road alignment or staging may require revision
Development involving evacuation-sensitive land uses or critical infrastructure	Sensitive land uses and critical facilities	Flood emergency management and access/egress assessment	Demonstrate continuity of essential services	DA	Flooding / Emergency Management	Land use suitability may require reassessment
Subdivision or precinct-scale development proposals	New development precincts	Confirmation of flood planning levels and flood risk management measures	Detailed stormwater servicing strategy and drainage concept design	Subdivision Application	Flooding / Civil	Development yield and staging may be affected
Development affecting waterways, drainage corridors or floodplain areas	Waterway corridors and drainage reserves	Assessment of floodplain impacts and hydraulic functionality	Water quality assessment and receiving water protection measures	DA	Flooding / Water Resources	Development footprint may require modification
Development requiring discharge to external waterways or drainage systems	Precinct drainage outlets	Confirmation of flood impacts at discharge locations	Stormwater quantity and quality assessment	DA	Civil / Water Resources	Additional infrastructure or treatment measures may be required

Trigger Condition	Affected Area	Flooding Requirement	Water Cycle Management Requirement	Timing	Responsible Discipline	Master Plan Consequence if Unresolved
Areas identified as having potential groundwater constraints	Groundwater-sensitive locations	Assessment where groundwater conditions may influence flood mitigation or drainage infrastructure	Groundwater investigation and management measures	DA	Hydrogeology	Development design may require modification
Areas identified as having potential ASS or salinity constraints	ASS or salinity risk areas	Assessment where flood mitigation works interact with ASS-affected soils	ASS and/or salinity investigation and management plan	DA	Geotechnical / Environmental	Earthworks methodology may require modification
Development requiring works within regulated waterways, drainage infrastructure or flood-prone land	Areas subject to external authority jurisdiction	Confirmation of flood risk management outcomes	Obtain relevant authority approvals and licences	Prior to approval	Relevant Discipline and Proponent	Development design may require modification

11.2 Exclusions and Deferred Items

The purpose of this Master Plan assessment is to identify flooding and water cycle management constraints relevant to strategic land use planning. The assessment is not intended to replace the detailed investigations and design processes that will be required to support future development applications and infrastructure delivery.

Accordingly, the following matters have been deferred to subsequent planning, design and approval stages unless specifically required to resolve a critical Master Plan decision:

- Catchment-wide hydrologic and hydraulic model development or major updates to existing flood models.
- Two-dimensional hydraulic modelling of flood risk.
- Flood impact assessments associated with final landform designs, filling strategies or infrastructure layouts.
- Stormwater network design, including pipe sizing, drainage layouts and operational design.
- Stormwater detention and water quality treatment system design.
- Detailed water sensitive urban design (WSUD) sizing and performance modelling.
- Detailed receiving water quality modelling and compliance assessment.
- Emergency management planning and evacuation procedures.
- Authority consultation and approval processes associated with future development applications.

These investigations should be undertaken progressively as future land use layouts, infrastructure requirements and development staging are refined. The need for any investigation should be commensurate with the scale, location and risk profile of the proposed development and informed by the staging triggers identified in Section 9.1.

12 Conclusions and Recommendations

The flooding and drainage assessment has identified that flood risk is a key consideration for future planning and development within the Precinct. Regional flooding from Cockle Creek, local overland flow paths and downstream tailwater influences create varying levels of flood constraint across the site, influencing land use suitability, infrastructure planning and development staging.

While significant flood-prone areas exist, the assessment indicates that opportunities may be available to facilitate future development through appropriate land use planning, flood mitigation measures and detailed flood and drainage investigations. The findings provide a strategic framework to guide refinement of the Master Plan and support future planning decisions.

Water cycle management is a key consideration for future development within the Precinct. Diega Creek, Cockle Creek and Burkes Creek are the key waterways located within the Precinct. Existing stormwater management within the Precinct is through natural flow paths within the undeveloped areas and via water management infrastructure across the disturbed site which includes catchment dams, sediment dams, storage dams, spillways and drainage systems. The adoption of an integrated approach to water cycle management prioritises capture, treatment and reuse where feasible and aims to minimise water quality and quantity impacts to sensitive receiving environments.

12.1 Summary of Findings

The following lists the key findings of the assessment:

- Flooding within the Precinct is primarily influenced by regional flooding from Cockle Creek, together with local flooding, drainage and downstream tidal influences.
- Flood constraints vary across the Precinct and have been classified into high, moderate and lower constraint areas to assist future planning and Master Plan refinement.
- Floodways and major flood storage areas should generally be retained to preserve floodplain function and minimise increases in flood risk.
- Several proposed development areas intersect flood-constrained land and will require further flood and drainage investigations to demonstrate development feasibility and identify appropriate mitigation measures.
- Emergency management should adopt evacuation as the preferred strategy, with future transport planning considering flood resilience and flood isolation risks.
- Flood mitigation measures, including detention basins, floodway corridors, flood storage management and development footprint optimisation, may provide opportunities to increase developable land, subject to detailed investigation and hydraulic modelling.
- The assessment has been undertaken at a strategic Master Plan level using the latest available flood studies and supporting information at a desktop level and therefore provides a suitable level of confidence for identifying broad flooding constraints and opportunities across the Precinct.
- The assessment is not intended to confirm development suitability or mitigation feasibility at individual sites. Detailed flood modelling, drainage investigations and emergency management assessments will be required during subsequent planning and design stages to refine flood constraints, assess climate change impacts and confirm the feasibility of proposed development and mitigation measures.
- Diega Creek, Cockle Creek and Burkes Creek are the key waterways located within the Precinct. Lake Macquarie is a sensitive receiving waterway for the project.
- Stormwater quality targets from the DCP and nutrient load limits (kg/ha) outlined in the NSW Estuary Health Risk Dataset (NSW Department of Planning, Industry and Environment, 2019) for Lake Macquarie Estuary are the key targets for water quality.

- The Water Management Strategy adopted will be dependent on what targets are required to be achieved based on existing/ historical land use. For brownfield development, the strategy will be required to meet best practice pollutant load reduction targets. For greenfield development, the strategy will be required to meet no net increase to pollutant loads (considered NorBE).

12.2 Recommendations for the Master Plan

A site-wide integrated Flood and Stormwater Management Strategy should be prepared following exhibition and prior to finalisation of development footprints, staging and infrastructure planning. The strategy should assess the cumulative impact of the proposed development across the wider catchment in greater detail, including changes to impervious area, runoff volumes and peak flows, floodplain storage, flood conveyance, flood levels and emergency access. This should be supported by detailed flood modelling, hydraulic modelling and MUSIC modelling undertaken post-exhibition, including testing of future development layouts, earthworks assumptions and stormwater management options.

The current assessment is considered suitable to support exhibition of the Master Plan, noting that further detailed assessment will be required to inform subsequent planning, design and development application stages. The following recommendations are therefore provided to guide the next stages of planning and development of the Precinct:

- Use the flood constraint mapping to guide refinement of the Master Plan and future development staging.
- Direct future development towards areas of comparatively lower flood risk where practicable.
- Preserve key flood conveyance corridors and flood storage areas within future land use planning.
- Undertake detailed flood modelling, drainage investigations and emergency management assessments for areas proposed for future development within or adjacent to flood-prone land.
- Investigate flood mitigation measures, including detention basins, floodway corridors, flood storage offsets and development footprint optimisation, to maximise development opportunities while maintaining acceptable flood risk outcomes.
- Review flood planning assumptions against the latest Australian Rainfall and Runoff (ARR2019) climate change guidance during subsequent planning and design stages.
- Ensure future development demonstrates appropriate flood immunity, drainage performance, emergency access and evacuation arrangements consistent with Council requirements and contemporary flood risk management practice.
- Utilise a whole of catchment approach coordinating surface water management across all sub-precincts while responding to local topography, land use, and hydrological characteristics.
- Adopt a Stormwater Management Strategy which considers stormwater detention, stormwater conveyance and natural water systems.
- Adopt a water quality management strategy that achieves water quality targets and WSUD measures (on-lot, end of pipe and/or road treatment).
- A surface water quality monitoring program should be developed to further understand and quantify the water quality within and downstream of the Precinct.
- Comprehensive hydrogeological, ASS and salinity investigation and risk assessment will be essential to identify groundwater constraints, inform design modifications, and establish long-term management protocols that protect both development viability and water resource sustainability.

13 References

- ANZECC (2000), 'Australian and New Zealand Guidelines for Fresh and Marine Water Quality'. Vol. 1 and 2. (Australian and New Zealand Environment and Conservation Council and Agriculture and Resource Management Council of Australia and New Zealand: Canberra).
- Awabakal Language Program, 2026. About Awabakal Language & Culture. Available at: <https://awabakallanguage.org.au/awabakal-language>
- Ball J, Babister M, Nathan R, Weeks W, Weinmann E, Retallick M, Testoni I, (Editors), 2019, Australian Rainfall and Runoff: A Guide to Flood Estimation, Commonwealth of Australia
- Blacktown City Council, 2026. Bioretention/ biofilter tree pits – planting. Available at: <https://www.blacktown.nsw.gov.au/Plan-build/Stage-2-plans-and-guidelines/Stormwater-management-and-water-sensitive-urban-design-WSUD>
- Bureau of Meteorology (2025a), Climate Data Online. Available At: <http://www.bom.gov.au/climate/data/>
- Bureau of Meteorology. (2025b). Climate classification maps. Retrieved from: <http://www.bom.gov.au/climate/maps/averages/climate-classification/>
- Department of Climate Change, Energy, the Environment and Water (DCCEEW), 2026. Hydrological Soil Groups (HSG) of NSW. Available at: <https://data.nsw.gov.au/data/dataset/hydrologic-groups-of-soils-in-nsw7f9e8>. Accessed 27th May 2026.
- Department of Environment and Conservation NSW, 2006. Managing urban stormwater: harvesting and reuse. Available at: <https://www.environment.nsw.gov.au/sites/default/files/managing-urban-stormwater-harvesting-reuse-060137.pdf>
- Department of Planning and Environment (DPE) 2022b. Controlled activities – Guidelines for riparian corridors on waterfront land. Fact sheet. Available at: https://publications.water.nsw.gov.au/watergroupjspui/retrieve/731c761f-bccf-4a6d-b9d5-0fedf2f4e9fa/Controlled_activities_-_Guidelines_for_riparian_corridors_on_waterfront_land.pdf
- Department of Planning and Environment (DPE), 2022a, Fact Sheet – Determining stream order. Available at: <https://www.water.nsw.gov.au/sites/default/files/2025-07/determining-strahler-stream-order-fact-sheet.pdf>
- Department of Planning, Industry and Environment (DPIE) 2026. Soil landscapes of NSW. Available at: <https://espade.environment.nsw.gov.au/> Accessed 1st June 2026.
- Lake Mac Libraries, n.d. Ala! Welcome to Awabakal country. Available at: <https://history.lakemac.com.au/narrative/5504>
- Lake Macquarie City Council (LMCC), 2013 Stormwater Improvement Devices Guidelines
- Lake Macquarie City Council, 2014. Lake Macquarie Development Control Plan (LMDCP). Available at: <https://www.lakemac.com.au/Development/Planning-controls-and-development-guidelines/Local-Planning-Controls/Lake-Macquarie-Development-Control-Plan>
- Lake Macquarie City Council, 2026. Our history. Available at: <https://www.lakemac.com.au/Our-Council/About-us/Our-history>
- Landcom (2004). **Managing Urban Stormwater: Soils and Construction**, 4th Edition (Volume 1)
- NSW Aquifer Interference Policy 2012. NSW Government policy for the licensing and assessment of aquifer interference activities
- NSW Government SES, 2020. Upper Cockle Creek Floodplain Risk Management Study and Plan. Available at: <https://floodddata.ses.nsw.gov.au/related-dataset/main-report>

NSW Department of Planning, Industry and Environment, 2019. NSW Estuary Health Risk Dataset: A first-pass risk assessment to help with the prioritisation of catchment management actions. Available at:

<https://data.nsw.gov.au/data/dataset/nsw-estuary-health-risk>

Office of Environment and Heritage, 2026. Health of our estuaries: Lake Macquarie. Available at:

<https://www.environment.nsw.gov.au/topics/water/estuaries/estuaries-of-nsw/lake-macquarie>

WaterNSW, 2025 WaterNSW Water Information Hub. Available at:

<https://realtimedata.watarnsw.com.au/water.stm>

WMAwater 2017, Winding Creek and Lower Cockle Creek Flood Study: Final Report, prepared for Lake Macquarie City Council, WMAwater, Sydney.

WMAwater 2019, Upper Cockle Creek Flood Study: Final Report, prepared for Lake Macquarie City Council, WMAwater, Sydney.